ऊर्जा दक्षता ब्यूरो
(भारत सरकार, विद्युत मंत्रालय)
BUREAU OF ENERGY EFFICIENCY
(Government of India, Ministry of Power)

आज़ादी का अमृत महोत्सव

No. 45/02/NMEEE/Energy Efficiency/2022-CCTS

18th September, 2026

**Sub: Seeking Inputs on identified methodologies for adoption in Offset Mechanism under Carbon Credit Trading Scheme (CCTS) – Reg.**

महोदय / महोदया जी,

उपरोक्त विषय पर ऊर्जा दक्षता ब्यूरो के पत्र सं. 45/02/NMEEE/Energy Efficiency/2022-CCTS की प्रति सूचना एवं आगे की आवश्यक कार्रवाई हेतु संलग्न है।

भवदीय,

(सौरभ डिड्डी)
निदेशक

**संलग्नक:** यथोपरि



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**Sub: Seeking Inputs on identified methodologies for adoption in Offset Mechanism under Carbon Credit Trading Scheme (CCTS) – Reg.**

Dear Sir/Madam,

This is with reference to the Carbon Credit Trading Scheme (CCTS), 2023 notified by the Ministry of Power, in which a national framework has been established with an objective to reduce the greenhouse gases emissions from the Indian economy by pricing the GHG emission through trading of the carbon credit certificates.

Further, in accordance with the Detailed Procedure for Offset Mechanism, the Bureau has identified and revised the following methodologies and associated tools for adoption under the offset mechanism, and thus seeking comments on the same.

I. BM AG04.XXX – Improved Agriculture Land Management
II. BM AG04.XXX – Production and use of Biochar
III. Tools:
    a. BM-T-0XXX: Soil Organic Carbon Model requirements and guidelines
    b. BM-T-0XX: Soil Organic Carbon Activity Tool for Cover Crops
    c. BM-T-0XX: Soil Organic Carbon Activity Tool for Increasing Soil Carbon Through Improved Tillage Practices
    d. BM-T-0XX: Soil Organic Carbon Activity Tool - Biostimulants for Soil Revitalisation
    e. BM-T-0XX: Soil Organic Carbon Activity Module- Application of Organic Soil Improvers from Pulp and Paper Mill Sludges
    f. BM-T-0XX: Soil Organic Carbon Activity Tool for Zero Tillage

The methodologies and tools are attached in the pdf format.

In view of the above, you are requested to provide inputs and suggestions on the methodologies in the required word format latest by 19th October, 2026 by email to sdiddi@beeindia.gov.in. and meenakshi@beeindia.gov.in.

**Yours sincerely,**

**(Saurabh Diddi)**
**Director, BEE**

Encl: As above



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विद्युत मंत्रालय
MINISTRY OF
POWER

**DRAFT TOOLS AND GUIDELINES**

# BM AGXX.XXXX

# Soil Organic Carbon Model Requirements and Guidelines

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Publication Date:

Version 1.0

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# 1. Introduction

1. The Soil Organic Carbon (SOC) Model Guidelines provide a comprehensive methodological framework for quantifying SOC stock changes in agricultural soils through modelling. These guidelines are designed to support non-obligated entities and accredited carbon verification agency (ACVA) and are complementary to the SOC Framework Methodology (SOC FM).

2. The guidelines do not prescribe specific SOC models but instead focus on a standardized procedure applicable to a range of models. Models are treated as "black boxes"— emphasis is placed on outputs, inputs and validation, rather than internal mechanics. The document outlines a seven-step workflow for applying SOC models: defining modelling objectives, selecting a model, collecting necessary data, calibrating the model, validating the model, predicting and estimating uncertainty and verifying model predictions over time.

3. The most important advantage of the Guideline is a procedure to gradually minimizing measurements using statistically secured model assistance. Key modelling objectives include:

    - Estimating SOC stocks with ground-based measurements
    - Estimating SOC stock variance to inform sampling strategies.
    - Supporting model-assisted SOC estimation for improved accuracy.
    - Enabling model-based estimation for calculating total SOC stock changes.

4. The project context including land use, soil types, climate and management practices must be clearly defined to guide model selection and ensure relevance. Calibration and validation require robust, context-representative data, ideally derived from within the project boundaries.

5. The guidelines provide detailed requirements for data quality, representativeness, statistical independence, and spatial/temporal consistency. Model validation is crucial when models are used for direct quantification (model-based estimation) and must meet specific performance metrics: bias, root mean square error (RMSE) and $R^2$ score.

6. Validation datasets must be independent from calibration datasets, and visual and statistical documentation is required for verification. Models must be revalidated as new field data becomes available.

7. The guidelines further emphasize uncertainty quantification and the need for transparency and reproducibility. They promote good scientific practices, including clear documentation of modelling decisions, data sources, assumptions, and limitations.

8. In essence, these SOC Model Guidelines aim to standardize and enhance the credibility of SOC modelling efforts across diverse agricultural contexts, enabling reliable carbon quantification that supports climate mitigation and sustainable land management practices.

## 1.1 Structure of SOC Model Guidelines

9. These guidelines are primarily directed at SOC sequestration non-obligated entity to help them in the process of using models in their SOC sequestration projects. Additionally, there

are some pointers for accredited carbon verification agency (ACVA) on how to properly verify the use of the model. Sections 4 to 6 of these guidelines outline the 5 typical sequences of steps required when applying a model in a SOC sequestration project.

10. These sections are generally agnostic to particular methodologies but do include specific explanations of how to apply SOC models in the context of the SOC FM.

11. As a framework methodology, the SOC FM was developed with a modular approach, allowing for extensions with activity modules that describe the exact activities to which it applies.

12. The SOC FM explicitly names “improved agricultural practices” as its core focus. Accordingly, these guidelines focus on SOC models dedicated to quantifying SOC stock (changes) on agricultural lands with improved (in the context of SOC sequestration) agricultural practices.

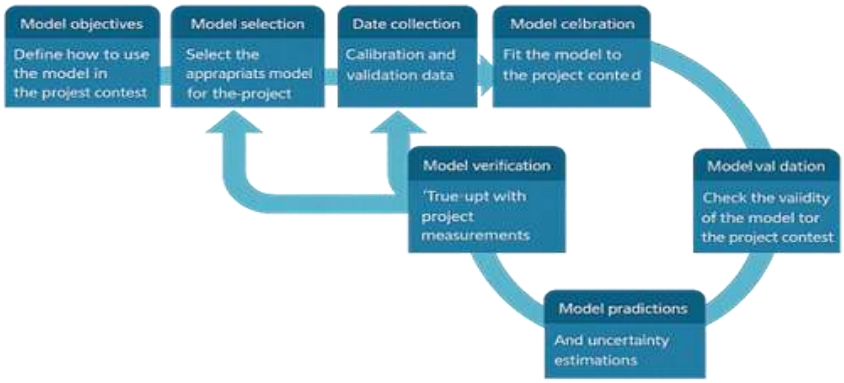



**Figure 1: Modelling workflow in the context of a SOC sequestration project**

13. The typical steps necessary to quantify SOC stock (changes) in a SOC sequestration project using any SOC model are ordered and presented in the following sections:

    - Define the Modelling Objectives
    - Select an Appropriate Model
    - Collect the Required Data
    - Calibrate the Model
    - Validate the Model
    - Make Predictions and Estimate the Uncertainty of the Predictions
    - Verify the Model Predictions

14. As shown in Figure 1, after determining the modelling objectives, selecting an appropriate model and collecting sufficient data for model calibration and validation, the subsequent process of model calibration, validation, prediction, and verification follows an iterative cycle.

This cycle has repeated recalibration and validation prior to making predictions and reverifying these predictions with ground-measured observations. Occasionally, a verification event can highlight the need for additional data collection or even the selection of a different model.

15. Each section of these guidelines has roughly the same structure with the following subsections:

| Subsection | Question addressed |
|---|---|
| • Rationale | Why is this needed? |
| • Requirements | What is required? |
| • How-to | How can the requirements be met? |
| • Required outputs | For the final outputs that need to be submitted to the ICM Administrator or the designated ACVA, what shall they look like? |

16. In terms of required outputs in general:

- The modelling objectives shall be communicated to the ICM administrator before the start of the respective calculation period (i.e., before SOC stocks are quantified for the start of the calculation period).
- No model validation is required when a model is used exclusively for variance estimation, which allows for efficient design-based SOC stock (change) estimation based on ground-measured samples.
- When a model requires validation (i.e., when the model is used for Quantification Approach 2 in the SOC FM, the reproducible model calibration and validation report shall be submitted to the ICM Administrator before the start of the calculation period. Upon request from the responsible ACVA, the full calibration and validation dataset shall be made available to the ACVA for the purpose of verifying the reproducibility of the results.
- When a model is used for direct quantification (prediction) of mean or total SOC stocks (change) (model-based estimation) as part of Quantification Approach 2 in the SOC FM at any point in the project, the mean/total SOC stock (change) predictions from the model shall be made available by the non-obligated entity to the ICM Administrator and ACVA before soil sampling for model validation takes place. Official model validation and verification by the ACVA can be done only with ground-measured observations that were taken after the model predictions were submitted.

# 2. Definitions

17. The model guidelines use the following definitions:

a. **Baseline Scenario** - The scenario that reasonably represents the anthropogenic emissions by sources of GHGs that would occur in the absence of the project activity. Continuation of the historical land management practices that are being followed in the last five years before the project start date.

b. **Baseline SOC stock** - Sum of SOC stocks in the project's spatial boundaries. For the first calculation period, the SOC stock at the beginning of the calculation period within the project's spatial boundaries is equal to the baseline SOC stock.

   $SOC_{BL,y}$ = SOC stock before the project start in stratum y [tCha-1].

c. **Diachronic data** - Data measured from the same location unit (point, plot aggregate, field-aggregate) at multiple points (dates) in time to establish a time series of paired data.

d. **Model calibration/Model training** - The process of adapting any (hyper)parameters of a model to a specific project context.

   For machine learning models, "model training" is a more commonly used term than "model calibration." Since process based models are already established in the SOC domain and machine learning models present an innovation in the industry, the term "model calibration" is used for both types of models throughout these guidelines.

e. **Model (hyper)parameters –** Model internal mechanics that might be calibrated but not changed after calibration from model run to model run in contrast to model input data.

**f.** **Model input data –** All data that is fed into a model to run a simulation or prediction. In statistical modelling, this is often referred to as the independent variable(s) or covariate(s) and in machine learning as feature(s) or predictor(s). In the process-based modelling domain, this refers to any measured or assumed input data ranging from initial SOC measurements to climate and management data.

g. **Model initialisation -** Part of model input data but specific to process-based modelling: setting initial conditions for process-based models to run, including the initial SOC measurements.

h. **Model selection** - The process of selecting an adequate model (version) for the declared modelling objectives and context.

i. **Model validation** – The demonstration of the selected SOC model's suitability to achieve the declared modelling objectives. This shall be demonstrated by satisfying certain statistics thresholds when comparing the model's predictions against ground-measured observations.

   A demonstration that a model within its domain of applicability possesses a satisfactory range of accuracy consistent with the intended application of the model. (Rykiel, 1996)

**j.** **Model verification** - Evaluating the actual accuracy and uncertainty that the model achieved in the project area, using measured reference data collected in the project area at regular intervals after the project start.

**k.** **Modelling unit** - Refer to the "Definitions and References" section in the SOC FM.

l. **Prediction unit -** The smallest spatial unit on which a model produces predictions. Examples: individual points (in geographical space), pixels, parcels, farms.

**m.** **Process-based model-** A model that encodes the relationship between inputs and outputs purely or mostly based on scientific assumptions on the real-world processes that govern this relationship (antonym: statistical model). Examples: RothC [Rothamsted

Carbon], DayCent [Daily Century]. These often include several levels of process detail, e.g., plant organ > plant > canopy or cell > plant organ > plant.

n. **Project scenario**- The scenario that will exist once the project activity is implemented and operational.

o. **Sampling unit**- The spatial level of granularity at which SOC reference values are sampled. Examples: individual points (in geographical space), pixels, fields, farms.

   For model validation and use, this shall be hierarchically consistent with the prediction unit, i.e., the physical sampling unit shall be small enough so that it can be spatially contained by the prediction unit. For example, the prediction units point, pixel, parcel, or farm can contain the sampling unit point, but a sampling unit field (e.g., composite field sample) cannot be used for model validation with a prediction unit point or pixel. In this case, the prediction unit shall become the field as well and model predictions shall be aggregated to that prediction unit, as described in Section 4.4.2 of these guidelines.

p. **Snapshot SOC stock**- The SOC stock at a single particular point in time.

**q. Statistical model**- A model that fits ("learns") a relationship between input data and the target variable purely or mostly based on mathematical relationships of measured data, with little or no hard-coded scientific assumptions on the real-world processes that govern this relationship (antonym: process-based model). Examples: generalised linear models, random forest models, neural networks.

r. **Target variable-** The entity under investigation that shall be predicted/simulated by the model. In SOC sequestration projects, this is typically the SOC stock at one point in time or the SOC stock change over a period of time. In statistical modelling, the term's response or dependent variable is also commonly used.

# 3. Scope & applicability

## 3.1. Scope

18. These guidelines provide methods to calculate project and/or leakage emissions resulting from the combustion of fossil fuels.

## 3.2. Applicability

19. These guidelines can be applied where carbon dioxide (CO2), methane (CH4), and nitrous oxide (N2O) emissions from fossil fuel combustion are calculated based on the quantity and properties such as chemical composition or net calorific value (NCV) of the fuel combusted.

20. Methodologies and other methodological tools applying these guidelines shall specify the particular combustion process to which they are being applied.

## 3.3. Model Guidelines Approval Date

21. The date of adoption of this document shall be effective from XX/XX/2026

# 4. Methodology Procedure

## 4.1. Project boundary

22. Refer to IALM methodology for the definition of the project boundary

## 4.2. Defining Modelling

23. First, the goals for SOC modelling shall be identified, and the specific (spatial and temporal) context within which the SOC modelling goals shall be attained. These steps are critical for selecting an appropriate SOC model and collecting adequate validation data.

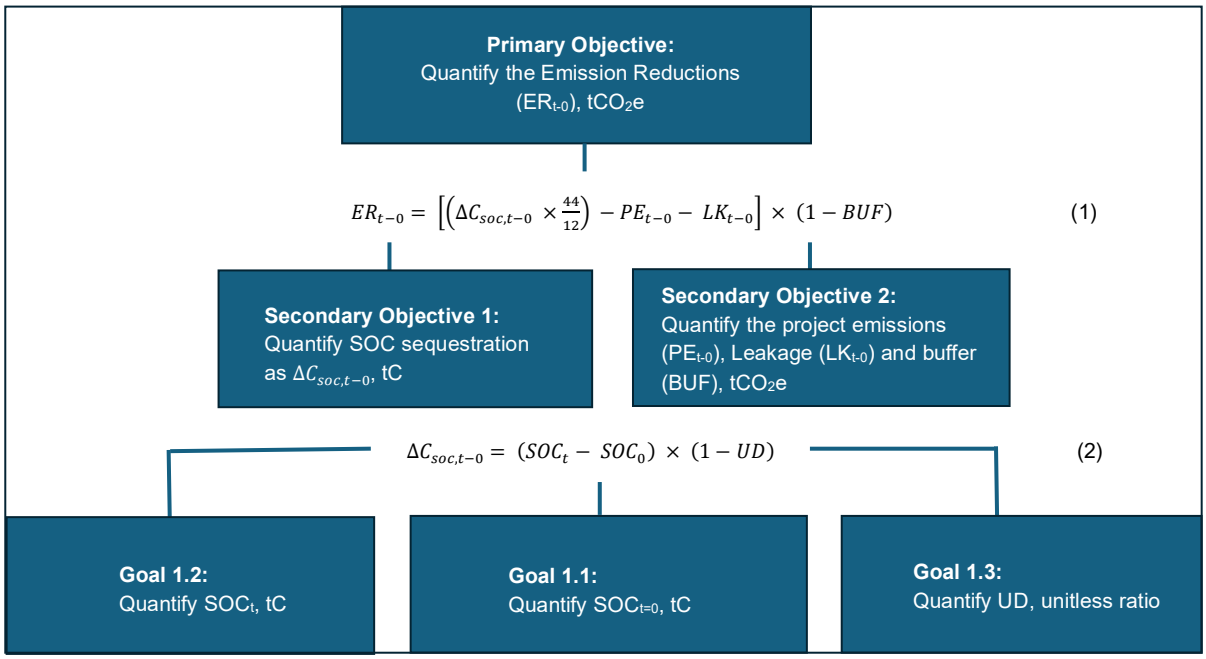



**Figure 2: Quantification objectives and goals from the SOC FM with the corresponding Equations 1 and 2**

### 4.2.1 The Potential SOC Modelling Goals

| **Box 1. Synopsis of modelling goals for projects with the SOC FM** |
|---|
| The ultimate goal of modelling within a SOC project is to quantify the SOC stock changes ΔCSOC, t-0 between two points in time, with t = end of calculation period, 0 = start of calculation period (the project). This change can be modelled directly or derived as the difference between the SOC stock snapshot at t=t - SOC stock snapshot at t=0. (See Box 3 in these guidelines.) SOC models, especially process based models, have traditionally been developed, calibrated, and validated at small plot scales in research settings. Whether they can be applied to larger-scale farming systems needs to be validated and verified as these projects take place. This means that soil sampling and measurement are still required, at least in the first calculation periods of the project, to ensure that the selected SOC model is adequately calibrated and validated to quantify the SOC stock changes (via direct modelling of change or modelling of snapshots in time). |

This means that over the project life cycle, three different strategies can be employed for quantifying SOC stock (changes), with different roles for modelling:

> **1. Design-based estimation:** Quantification of SOC stock via soil sampling and measurements. Modelling can be used prior to the sampling campaign to quantify the SOC stock variance over the project's spatial boundaries to delineate strata (modelling units) and determine the required sample size.
>
> **2. Model-assisted estimation** quantification of SOC stock via modelling SOC stock, in addition to ground-measured samples, to increase the precision of a design-based estimate.
>
> **3. Model-based estimation** quantification of mean and total SOC stocks over the project's spatial boundaries via a SOC model.

24. With any SOC sequestration project, the primary objective is to quantify the removed emissions in tons of CO2e. This quantification is formalised in Equation 1 of the SOC FM and can be broken down into two secondary objectives:

    1 Quantify SOC sequestration as $\Delta C_{SOC,\, t-0}$ .

    2 Quantify the project emissions ($PE_{t-0}$), the leakage ($LK_{t-0}$), and the buffer (BUF) emissions.

25. These guidelines are exclusively concerned with the first of the two: the quantification of the SOC sequestration as the total SOC stock change within the project's spatial and temporal boundaries and the corresponding uncertainty of this estimate. This is formalised in Equation 2 in the SOC FM, where $\Delta C_{SOC,\, t-0}$ stands for the change within the project's spatial and temporal boundaries, $SOC_0$ and $SOC_t$ stand for the SOC stock at the beginning and end of the calculation period, and UD stands for the corresponding uncertainty deduction.

**Box 2. The baseline in the SOC FM**

In SOC sequestration projects in general, the change in the project needs to be compared to a baseline. In the SOC FM, the baseline scenario is defined as the "continuation of the historical land management practices ... in the last five years before the project start date (BAU)", and the baseline SOC stocks are calculated as "the sum of stocks in [the project's spatial boundaries in the year of the project start]". As a default, the SOC FM prescribes a static baseline with the assumption that SOC stock is at equilibrium (i.e., would not change) under the BAU baseline scenario. Thus, the baseline SOC stock is set equal to the initial SOC stock within the project's spatial boundaries in the project start year: "SOC0 is equal to $SOC_{BL}$."

Since the definition of the baseline might change in newer versions of the SOC FM, the latest version shall be the definitive source to clarify the definition of the baseline scenario and the quantification of the baseline SOC stock.

26. This means the following quantities can generally be modelled within the context of the SOC FM:

    - Mean, total, and variance of SOC stock change over the project's spatial boundaries (or individual modelling units) and over calculation/crediting period(s) ($\Delta C_{SOC, t-0}$)

    - Mean, total, and variance of SOC stock over the project's spatial boundaries (or individual modelling units) for a snapshot (project start [i.e., $SOC_o$/$SOC_{BL}$], middle, or end [$SOC_t$])

    These guidelines focus on the latter (modelling snapshots) with the goal of applying Equation 2 in the SOC FM to derive $\Delta C_{SOC, t-0}$. This is in line with the SOC FM, and there are also validation and calibration data limitations to modelling the change directly, as explained here in Box 3

**Box 3. Modelling change directly versus subtracting modelled snapshots**

While the SOC stock change can be modelled directly with temporally explicit dynamic process-based models, it is quite challenging to validate (and calibrate) such model predictions because of the lack of appropriate validation studies and reference data (Garsia et al., 2023; Le Noë et al., 2023). (Also see Section 5.4 of these guidelines, "Collect the Required Data.") Overall, there is a shortage of SOC stock time series data at the spatial scale (extent and density) of typical SOC sequestration projects (Bradford et al., 2023; Lavallee et al., 2024). The projects themselves can generate this data when repeatedly measuring SOC data in situ for calculation of the sequestered SOC (see Table 3 in these guidelines for the calculation instructions), as described in Approach 1 for baseline and project scenario calculations in the SOC FM (Sections 6-7). However, creating a time series of paired soil data points requires careful planning in both sampling and measurement. The full requirements 13 are listed in Box 5 in these guidelines.

Because of these limitations, and in correspondence with the SOC FM, these guidelines focus on modelling the mean, total, and/or variance of the SOCt stock at temporal snapshots and determining the corresponding uncertainty for calculating the SOC stock change over time as the difference between two snapshot predictions.

27. To quantify the SOC stock over the spatial boundaries of the project, the SOC FM allows for two distinct quantification approaches:[1]

    "Approach 1: On-site measurements to directly document baseline and project SOC stocks.

    Approach 2: Models from peer-reviewed publications to estimate baseline and project SOC stocks. Non-obligated entities need to prove that the models are conservative and applicable to the project site and management practice. Models derived locally may be applied only if validated by direct measurements in the project area (Approach 1). Generally, non-obligated entities shall select the most specific approach possible with the data available, giving preference to local data sources and models." (SOC FM)

    In estimation statistics, these two approaches are coined design-based (Approach 1) versus model-based (Approach 2) estimation respectively (Särndal et al., 2003). From here on, we will use these terms instead of Approach 1 or 2 respectively, as they are more generally applicable beyond the scope of the SOC FM.

28. While purely model-based estimation is generally permitted under the SOC FM, it is clearly stated that the model shall be previously validated against direct measurements in the project area. When direct measurements are not available before the start of the project, such measurements can be taken and then used for dual use: for validating the model for the project and for design-based estimation. Also, prior to or during design-based estimation, a model can be used as follows:

    - To determine where and how many soil samples shall be taken in the prescribed stratified random sampling design as noted in Section 14.2 of the SOC FM. To determine how many samples (soil pits) are required, an estimate of the variance of SOC is needed as described in Section 4.4.2 of these guidelines, "Requirements for Data Collection." This estimate of the variance can be obtained through modelling.

    - In addition to the sampled ground-measured observations in a model-assisted estimation process to increase the precision of the estimate. (Brus, 2000)

29. Thus, the modelling goals can be threefold in the SOC FM (also see Box 1 in these guidelines):

    1) **Variance estimation for informing design-based estimation of SOC stocks:** Modelling the variance of SOC stock within the project's spatial boundaries to determine the number of required soil samples to take and draw the stratum boundaries (modelling units) for taking soil samples.

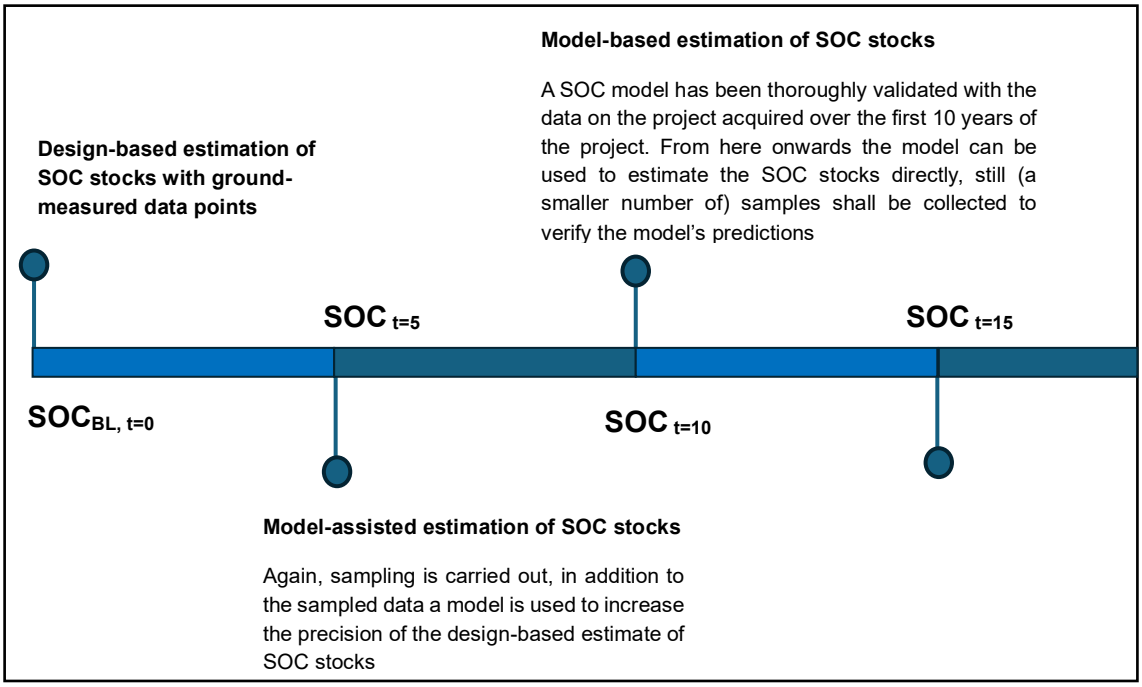
    2) **Model-assisted estimation of SOC stocks:** Modelling a regression function in addition to the ground-measured samples for design-based estimation to increase the overall precision of the estimate.

    3) **Model-based estimation of SOC stocks:** Modelling the mean or total SOC stock as a snapshot at the start of the project or prior to any following verification event to quantify the creditable SOC sequestration for the respective calculation period.

30. Throughout the life cycle of a SOC FM sequestration project, one, two, or all three of these modelling goals might be applied as shown in Box 4 below. This box presents a fictitious example and is not intended as a strict prescription. In particular, model assisted design-unbiased estimates can already be applied at t=0 without explicit model validation. Model-based estimates can also be used at t=0 if the models can be validated in accordance with Section 5.1 of these guidelines, “Validate the Model,” for the respective target variable on Data Sources B, C, or D (see Section 4.4.3 of these guidelines, ”How to Acquire Data”) with the respective safety discounts in Table 4 of these guidelines.

**Box 4. Fictitious example of modelling SOC within a SOC FM project**

When there is not sufficient validation data at the start of a project, the first quantification rounds (i.e., for the baseline quantification at t=0 and for the first calculation period, e.g., at t=5 years) can be conducted via design-based estimation, where in parallel a model is calibrated and predictions are generated solely for the purpose of validating these predictions against the ground-measured observations as described in Section 5.1 of these guidelines, “Validate the Model.” After successful validation of the model for the baseline and the first calculation period, the model might be used for quantification in later calculation/crediting periods if permitted by the ICM Administrator and/or the responsible ACVA in accordance with Section 8 of the SOC FM.




[1] Approach 3: Default factors to estimate SOC changes, relating to the general Tier 1/2 model described in the Intergovernmental Panel on Climate Change (IPCC) Guidelines for National Greenhouse Gas Inventories (IPCC 2019) is not relevant to the types of models covered in these guidelines.

31. Only the purely model-based estimation of the mean or total SOC stocks over the project's spatial boundaries requires strict model validation, as the validity and objectivity of the approach are not guaranteed without ground-measured observations to compare with the model predictions.

### 4.2.2 Defining the Modelling Context

32. In addition to determining whether the goal of the model is to assist with design-based estimation or conduct model-based estimation, the modelling context shall be specified with respect to the project's spatial and temporal extent and the intended land management changes.

33. SOC models are often developed and validated with data from specific ecosystems and land management practices. A model developed and validated with data from a specific spatial and temporal extent, climate zone and soil texture class, and/or for a specific type of agricultural land management might be valid and applicable in the same or similar context—but not necessarily in a different climate zone or soil texture class and/or for a different practice.

34. The SOC FM prescribes that the project area shall be stratified into multiple modelling units with similar soil types, climate zones, and land management practices (further differentiated in the SOC FM, Section 7). As a consequence, the modelling objectives in different modelling units of the same project may be different. Projects with multiple modelling objectives may require selection, calibration, and validation of separate models for the individual modelling units since a single model may not be suitable for all modelling objectives.

### 4.2.3 Requirements for Defining the Modelling Objectives and Context

35. To effectively define the modelling objective(s), the following requirements shall be met:

    1 Project definition: Define the spatial and temporal boundaries of the project, activities, and land management changes in line with Chapter 4 of the SOC FM.

    2 Data collection: Collect and record project data and parameters as specified in Section 14.3 of the SOC FM, which includes:

        a. Project activity/land management change

        b. Project start and end dates

        c. Details of each farm/land parcel within the project boundaries, including unique identifiers, owner information, physical address, global positioning system (GPS) coordinates/geographic shape, land area and use, agrochemical and fertiliser usage, and fossil fuel and electricity consumption for baseline activities

    3 Contextual data: Gather data to establish representativeness of validation data to the project context, including climate zones, soil texture classes, vegetation types, and management history (for every modelling unit).

### 4.2.4 How to Define the Modelling Objectives

36. Defining the project and modelling objectives involves several key steps:

    a) Defining project boundaries and activities: Clearly delineate the spatial and temporal boundaries of the project and define the specific activities undertaken.

    b) Data collection and analysis: Collect detailed data on the farms and land parcels. The exact requirements are listed in Section 6.1 of these guidelines. Most of the relevant data shall be provided by the non-obligated entity. If the data is not readily available from the non-obligated entity, then do the following:

        - Search for the data in public data records or use published maps.
        - Implement soil sampling and measurement activities to obtain the required data.

    c) Stratify the project area into modelling units in line with Chapter 6 of the SOC FM.

    d) Formulate the modelling objective(s) for the project as a whole or the individual modelling units as precisely as possible, including at least the following contextual information:

        - Target variable to be quantified (e.g., change in SOC stock)
        - Soil depth of interest
        - Project activity
        - Climate zone
        - Soil type

37. Formulating the modelling objective in this detailed way will help in selecting the most suitable model and finding representative data for model calibration and validation. If any of the contextual information listed above differs between modelling units, the modelling objectives shall be stated individually for each modelling unit. In this case, different SOC models, different calibration data, and/or different validation data might be required for the different modelling units.

### 4.2.5 Outputs of the Modelling Objective and Context Definition

38. The project context shall be defined in line with Chapter 4 of the SOC FM. The project's spatial and temporal boundaries shall be clearly delineated, and the project activity/activities shall be clearly defined. In addition to the requirements for data to be collected as listed in Sections 14.3 and 14.4 of the SOC FM, the data in this Table 1 shall be collected and submitted**.**

**Table 1.** Expected output: geospatial file(s) (geojson, geopackage, or any other open standard geospatial file format) with the following data and metadata on every farm/parcel within the project's spatial boundaries (one row per farm or parcel as appropriate

| Attribute | Data Type |
|---|---|
| Unique identifier | Numeric/Hash |
| Descriptive name (name of the farm/parcel, address, landowner/ non-obligated entity) | Text |
| Exact spatial boundaries of the total and eligible area (expected spatial accuracy within single-digit metres; if otherwise, report the spatial accuracy) | Text (i.e, well-known text representation of geometry) |
| Start date and end date of the crediting period | Text (yyyy-mm-dd) |
| Modelling objective | Text |
| Modelling unit to which the farm/parcel belongs | Text |
| Climate zone as defined in the 2019 Refinements to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (Reddy et al., 2019) | Text |
| Soil texture class and specifically the soil's clay content in % | Text/Numeric (clay %) |
| Vegetation type (e.g., grasses, legumes, non-legume broadleaf species) | Text |
| Current land use | Text |
| Management history (land use changes), historic vegetation types, tillage techniques, and fertilisation | Text |

39. In addition, the modelling goal shall be clearly communicated when the project context definition is submitted to the ICM Administrator as one of the goals noted in Section 4.2.1 of these guidelines, “The Potential SOC Modelling Goals.”

## 4.3. Select an Appropriate Tool

### 4.3.1 Model Selection Rationale

40. This section provides general guidance on how to select appropriate SOC models to achieve the modelling objectives of a SOC sequestration project. It does not provide definite recommendations to use a specific SOC model in a specific project context. However, it is important that the chosen SOC model can be independently validated. To that end, the model’s predictions and simulations shall be fully reproducible and accessible, meaning a proper versioning of the model and associated data (including input data and random seeds for stochastic models) shall be available.

### 4.3.2 Understanding SOC Models

41. Before selecting a model, it's essential to understand the different types of SOC models and what they do. Process-based SOC models simulate the accumulation and/or loss of organic carbon in the soil. Statistical models use regression or machine learning algorithms to derive a quantification formula to determine the SOC stock (change) from input data and associated weights. Both types of models vary in complexity, scale, and the factors they incorporate, such as climate variables, land use, and management practices.

42. The selection of an appropriate SOC model is critical for the success of a SOC sequestration project. The right model balances accuracy, complexity, data requirements, and resource availability. Carefully considering these factors and thoroughly evaluating available models ensure that a SOC sequestration project is based on reliable and relevant modelling outcomes.

### 4.3.3. Model Selection Requirements

43. Key Considerations for Model Selection:

    a) Project context: The SOC model shall be appropriate for the project context (i.e., project activity, climate zone, soil type). Different models are designed for various scales, such as field level, regional, or changes on a daily, monthly, yearly, or decadal basis. Whether a model is appropriate can be judged only through model validation. When a model meets the requirements noted in Section 5.2.1 of these guidelines, it is deemed appropriate to the project context.

    b) Data availability and quality: In addition to the aforementioned validation data (Section 4.4 of these guidelines), the availability of input data (e.g., predictors, features, parameter sets, independent variables) shall be assessed. Some models need extensive input data, while others are less data intensive. The choice of model may depend on the data that can be realistically obtained.

c) Model complexity and usability: Models range from being simple and statistical (e.g., linear regression, one target/predictor variable, predicting single point in time) to being complex and process-based (e.g., DayCent, multiple internal variables, predicting at daily time steps). While complex models may provide more detailed insights, they require more data and computational resources. The team's expertise and resources in model operation shall be considered.

d) Local calibration/training and validation: Check if the model can be calibrated/trained and validated with local data. A model's performance greatly depends on its applicability to the local context of the project. Reserve some local data for validation that shall not be used for calibration/training purposes.

e) Temporal and spatial resolution: Consider the temporal and spatial resolution needed for the project. Some models provide fine-scale, high-resolution outputs, while others work at coarser resolutions.

f) Integration with other tools: Determine if the model needs to be integrated with other tools or models (e.g., geographic information system software, climate models). Some SOC models are standalone, while others are designed as part of larger environmental modelling suites.

g) Model support and community: Consider the support available for the model, including documentation, user communities, and technical support. A well supported model can significantly ease the implementation process.

### 4.3.4 How to Select a Model

44. **Steps in Selecting a SOC Model**

a) Define the project context as described in Section 4.2 of these guidelines, "Define the Modelling Objectives."

b) Research available models: Conduct thorough research on existing SOC models, including their capabilities, limitations, and data requirements. (See the resources below.) c) Check prior validation: If one of the available models is already validated on the project context (e.g., same type of practice change, prior land use, soil texture class, and climate zone), it is possible to reuse the model.

c) Evaluate data availability: Match the available data with the data needs of the models, taking into account both input and output data for calibration and validation.

d) Shortlist suitable models: Based on the prior steps, create a shortlist of potential models.

e) Compare the models: If possible, test the shortlisted models with a subset of the data to compare their performance as described in Section 4.3.5 of these guidelines, "Expected Model Selection Outputs."

f) Final selection: Choose the model(s) that best fit the project needs, considering performance, data availability, and resource constraints.

References:

Garsia et al (2023) have compiled several resources on process-based SOC model selection and validation. Additional resources are available at the Model Portal of the International Soil Modelling consortium (ISMC, 2023).[2]Remote sensing–based SOC models also have become popular in recent years (Dvorakova et al., 2023; Hengl et al., 2021; Heuvelink et al., 2021; Poggio et al., 2021; Szatmári et al., 2021; Zepp et al., 2021) and might be suitable. However, take care during the validation process to test these models on the right validation data.

#### 4.3.5 Expected Model Selection Outputs

45. The output of the model selection process is one model per modelling unit of the project. The same model can be selected for multiple modelling units if the model can be validated for all of them. Any high-level model class (e.g., DayCent, RothC, Random Forest) that was considered and dismissed during the model selection process shall be recorded in a list, along with a short explanation of why it was dismissed.

### 4.4 Collect the Required Data

#### 4.4.1 Rationale of Data Collection

46. Any SOC model intended for use in a SOC sequestration project shall demonstrate reliable predictions for SOC stocks and corresponding uncertainty estimates. The only way to robustly carry out this analysis is by validating the model outputs against measured reference data. Data used for model validation is called validation data. Measured reference data can also be used during model calibration to fit a model to a specific project context or to select the best model from a set of candidate models. Data used for these purposes is called calibration data. This section provides the key requirements to consider when gathering validation and calibration data for SOC stock modelling.

47. The SOC stock change data from the project site is required to ultimately validate the whole quantification procedure. However, such data rarely exists within the project spatial boundaries before or at the project start as it would require at least two SOC stock measurements over time. Ideally, the SOC stock change data should be obtained from diachronic (i.e., paired-at-same-location time series) data (Le Noë et al., 2023), i.e., it should be derived from time series of SOC stock data measured recurrently from the same location(s) at time t0 before a land management change was initiated and at times t1 (t2, t3, …, tn) some time (1-20 years) after the specific change was implemented. Since this data hardly exists in practice, the capacity to validate change predictions of the model is limited at the start of the project. Instead, what can be measured and validated is the spatial variation of SOC stocks at one point in time and the capacity of the model to predict the parameters (e.g., mean, variance) of this spatial distribution of SOC stocks at a single point in time. From the precision of this spatial estimation of a single point in time, the uncertainty for the calculation of the change as the difference between two single points in time can be calculated. (See Annex A.1 Simulated Example)

48. Over the course of the project, sufficient validation data shall be sampled and measured over time within the project's spatial boundaries to allow for independent diachronic validation of the model, as Le Noë et al. (2023) suggest (see Figure 3 in these guidelines). This collected

[2] https://www.soil-modeling.org/resources-links/model-portal

validation data can then be used to validate a model and opt for purely model-based estimation of the SOC stock in later calculation periods (as explained in Box 6 in these guidelines).

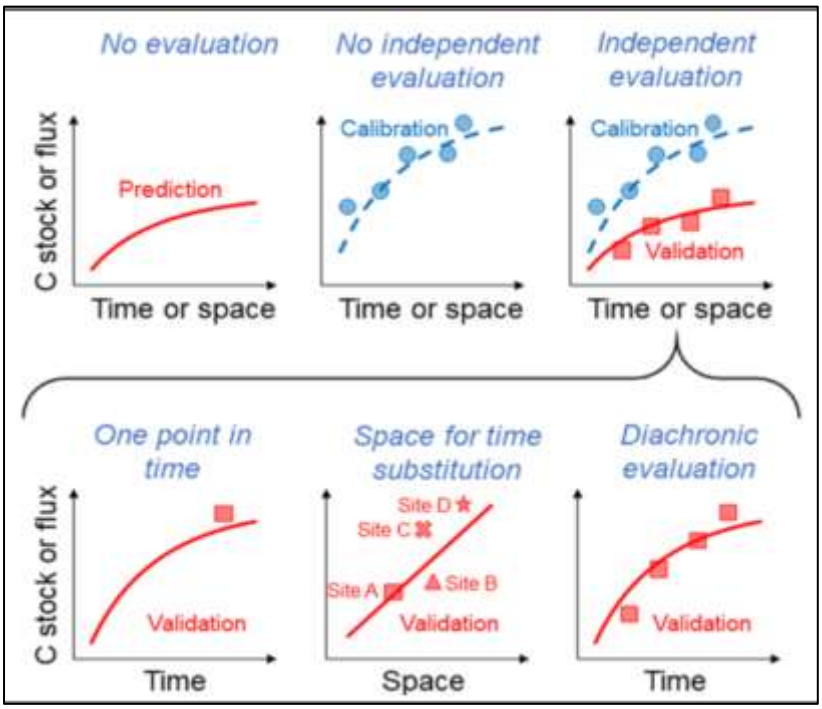



**Figure 3**. Le Noe et al. (Le Noë et al., 2023): "A schematic representation of (...) links (...) between SOC models and [the] empirical field

### 4.4.2 Requirements for Data Collection

49. The validation data shall represent the different levels of SOC stocks[3]

a) The data shall be presented in units of tonnes of carbon per hectare [tC/ha].

b) The SOC stock data shall be calculated as the product of the SOC concentration of the fine soil (sieved soil particles at <2mm in [%] with the fine soil stock (FSS), i.e, the fine soil mass per unit area in [t/ha].

1. If FSS is measured directly, Equations 3a and 3b in Table 2 in these guidelines shall be applied (Poeplau et al., 2017).

2. If FSS is not measured directly, it shall be computed as the product of the (gravimetric) bulk density (g $cm^{-3}$ ) of the fine soil and the soil depth (cm), deducted by the coarse fragment ratio (sieved soil particles at <2mm, as shown in Equation 4 in Table 2 in these guidelines.

[3] The data shall be representative of the project in terms of spatial extent (total area and radius), temporal extent (length of project and length of calculation periods), and the management practices taking place in the project scenario.

3. When comparing two different measurements of a time series, the concept of equivalent soil masses (Wendt & Hauser, 2013) shall always be considered. See Equations 5a and 5b in Table 2 in these guidelines.

c) To ensure comparability across datasets, the following data harmonization steps shall be taken prior to analysis:

1. All SOC and bulk density (BD) values shall be converted to common depth intervals aligned with the modelling objective (e.g., 0–30 cm), using pedotransfer functions or depth-weighted averaging when necessary.

2. Units shall be harmonized across all datasets (e.g., converting g/cm³ to t/ha or %SOC to g/kg where applicable).

3. Differences in analytical methods (e.g., Walkley-Black versus dry combustion) shall be documented. When possible, correction factors shall be applied based on peer-reviewed cross-lab comparison studies.

4. Metadata (e.g., sampling date, method, equipment, and lab technique) shall accompany all datasets to support transparent harmonization.

50. The sampling units of the validation data shall be consistent with the prediction units of the model that is validated. For example, a model that produces farm-level predictions shall be validated using data that was sampled on a farm level.

51. If there is a mismatch between sampling units and prediction units, the unit with the finer spatial granularity shall be aggregated (e.g., by calculating the mean SOC stock of point locations to represent the mean SOC stock at a field level) to match the unit with the coarser granularity. For example, if the SOC model produces farm-level predictions but SOC reference data is available at individual point locations sampled via probability sampling within farms, then all points within the boundaries of a farm shall be aggregated to provide a single farm-level mean reference value.

52. The validation data from the project site shall capture the variance of the SOC stock within the spatial boundaries of the project with sufficient precision to quantify the changes within a 90% confidence interval (in correspondence to the uncertainty quantification of the SOC FM, Section 9). In cases where the sampled SOC reference data is aggregated (i.e., a field- or farm-level mean from point-level measurements) to be consistent with the prediction unit of the model, the thresholds apply after aggregation. The equations to calculate the required number of samples are shown in Box 5.

53. The validation data shall be representative for the project context (modelling units):

a) The validation data shall be sampled over the soil depth interval stated in the modelling objective (e.g., 0-30 cm or 0-50 cm).

b) The validation data from within the project boundaries shall be collected in accordance with Section 14.2 of the SOC FM.

c) Directed stratified sampling (with stratification based on additional data and composite samples aggregated within each stratum) shall be used in accordance with the protocols noted in the SOC FM or Annex 3.2 of the Food and Agricultural Organization (FAO) Global Soil Organic Carbon - Monitoring, Reporting, and Verification (GSOC

MRV) Protocol (FAO, 2020, pp. 89-91, referred to as “directed stratified sampling” in this protocol.) for any validation data.

d) When previously collected data is used for model validation that was collected without following the directed stratified sampling design referenced above, justification shall be provided for how the data is unbiasedly representative of the project context (detailed further in Section 4.4.3 of these guidelines, “How to Acquire Data”). In the best case, the data was still sampled via a probability sampling design to satisfy the criterion of unbiasedness.

54. For the calibration data, there are no hard requirements, although the data shall be similarly representative of the project context as the validation data to make model calibration most effective. However, data that does not satisfy some of the requirements for validation data may still be used as calibration data.[4] This also means that calibration data does not need to come from within the project site itself.

55. The validation data shall be statistically independent of the calibration data:

a) To ensure statistical independence, the following criteria shall be fulfilled:

1. The validation data and the calibration data shall be mutually exclusive, i.e., no single data point that is part of the validation data may be part of the calibration data and vice versa.

2. The minimum geographical distance between any validation data point and any calibration data point shall be 30 metres. This entails that all data points taken at the same location at different points in time and different soil depths shall either be completely assigned to the validation data or completely assigned to the calibration data.

b) Ideally, calibration data and validation data shall be drawn randomly without replacement from the same superpopulation of data (in the best case, from the same sampling campaign(s)). Care shall be taken to enforce the minimum distance constraint mentioned above.

c) To prove that the above requirements are fulfilled:

1. The model validation report shall clearly state the minimum geographical distance (in metres) between any validation data point and any calibration data point as well as the coordinate reference system in which the distance was computed.

2. The geographical locations, time stamps, and soil depths of all validation data points and all calibration data points shall be made available as a geospatial file (geopackage, geojson, or other open standard geospatial file format) along with the model validation report allowing for simple verification of the minimum distance constraint. The actual measurements (e.g., SOC stock values) or model predictions for these data points may be omitted for reasons of confidentiality but shall be made available to the ACVA upon request for the purpose of official validation and verification.

[4] These guidelines and the SOC FM evaluate the suitability of modelling SOC stock change quantification based on the model validation and verification. Before a model is allowed to be used for direct quantification with the SOC FM, it shall be validated as noted in Section 6.1 of these guidelines

When the goal is to collect or generate a time series of SOC stock data, special care shall be taken to guarantee the adequacy of the time series data, as explained in Box 6 in these guidelines.

**Table 2:** Calculation of SOC stocks

| Method | Equation | Equation Number |
|---|---|---|
| Poeplau FSS Method<br><br>(Poepla u et al., 2017) | $$FSS = \frac{mass_{finesoil}}{volume_{sample}} x\ depth x\ 0.01$$<br>where $mass_{finesoil}$ is given in [g], $volume_{sample}$ is given in [$cm^{-3}$] and depth is given in [cm]; the resulting unit is [t/ha].<br>$$SOCstock = SOCcon_{finesoil} x\ FSS$$<br>where $SOC_{confinesoil}$ is given in [%]; the resulting unit is [t/ha] | 3a<br><br>3b |
| BD Method | $$SOC_{stock} = SOCcon_{finesoil} x\ BD_{finesoil} x\ depth x\ (1 - CFR)$$<br>where $SOC_{confine\ soil}$ is given in [%], $BD_{fine\ soil}$ is given in [g $cm^{-3}$], depth is given in [cm] and CFR is given in [%]; the resulting unit is [t/ha]. | 4 |
| Equivalent soil masses (Wendt & Hauser, 2013) | $$M_{SOIL(DL)} = \frac{mass}{area} = \frac{M_{SAMPLE(DL)}}{\pi(\frac{D}{2})^2 x\ N} X\ 1000$$<br>where<br>• $M_{SOIL(DL)}$ is the soil mass represented by a soil sample depth layer in Mg/ha.<br>• $M_{SAMPLE(DL)}$ is the dry sample mass represented by a soil sample depth layer by the area sampled by the probe or auger in g.<br>• π(D/2)2 is the cross-sectional area of the probe's or auger's inside diameter in mm.<br>• N is the number of cores sampled.<br>Furthermore:<br>$$M_{OC(DL)} = M_{SOIL(DL)} X C_{OC(DL)}$$<br>where<br>• $M_{OC(DL)}$ is the mass of organic carbon represented by a soil sample depth layer in kg/ha.<br>• $C_{OC(DL)}$ is the concentration of organic carbon represented by a soil sample depth layer in g/kg. | 5a<br><br>5b |

Box 5. Confidence interval-based estimation of sample size versus minimum detectable difference

While many SOC sampling protocols and methodologies recommend the use of a minimum detectable difference (MDD) formula for determining the required sample size, we do not recommend this. The MDD is mainly intended for use in a statistical t-test, which is not required as part of the SOC FM. Instead, a 90% confidence interval is required (SOC FM Section 9), which shall be sufficiently narrow to allow for robust quantification of SOC stock changes and to avoid excess uncertainty deductions. To determine the required sample size (for design-based estimation) for this 90% confidence interval, two things are needed:

1. The desired degree of precision (or margin of error) in terms of SOC stock change

2. An estimate of either the variance of SOC stock change (paired sampling scenario) or an estimate of the spatial variance of the SOC stock at the two temporal snapshots

Correspondingly, there are two potential formulas for determining the required sample size:

1. For paired sampling, with a change (temporal) variance estimate:

$$n = \frac{z_{\alpha/2}^2 s^2}{MoE^2}$$ equation 6

2. For independent sampling, with a spatial (snapshot) variance estimate:

$$n = \frac{z_{\alpha/2}^2 s^2}{MoE^2}$$ equation 7

Where:

= required sample size

Z $\alpha$ /2 = z-value of the normal distribution at confidence level 100(1-$\alpha$)% (two-sided)

$\alpha$ = significance level (for the SOC FM 90%CI $\alpha$= 0.1)

$S^2$ = estimate of the variance of 1. SOC stock change 2. SOC stock snapshot

n MoE = Desired margin of error (degree of precision) We strongly recommend that such derived sample sizes are the basis for any quantification approach (even model-based), as this allows for switching to a design-based approach if a model falls short during verification.

Box 6. Requirements for collecting SOC stock time series data

While high-quality time series data can greatly simplify the model validation process for SOC sequestration projects, it is quite difficult to obtain such data. Creating a time series of paired soil data points requires careful planning in both sampling and measurement. Between the multiple measurements, consistency is key, i.e., the same sampling and measurement protocols shall be applied and preferably also the same devices and personnel to carry out the work.

Sites shall be representative and accessible and shall be consistently sampled at fixed locations and regular intervals (at least in five-year intervals). Because of the well-documented micro-scale variability of SOC stocks, composite samples within a spatial radius are required to establish SOC stock time series data. The spatial radius to cover the composite shall be recorded and shall be at least as large as the spatial imprecision of the GPS/global navigation satellite system (GNSS) device used to measure the (central) spatial location of the composite sample. Standardised protocols shall be used for collecting soil samples at consistent depths. Repeated sampling campaigns shall consider the equivalent soil mass principle (see Table 2 in these guidelines). Proper sample handling, including preservation and accurate labelling, is essential to prevent degradation. Detailed records of sampling conditions and environmental factors shall be maintained, and a robust data management system shall be used to organise data. Quality control measures, such as duplicates and calibration of equipment, are necessary to ensure accuracy. Finally, appropriate statistical methods shall be applied to analyse temporal trends in the data. This approach ensures reliable, consistent, and interpretable soil data over time.

### 4.4.3 How to Acquire Data

56. Validation data that fulfils the requirements noted above can be obtained from different sources. Depending on which data source is used, the data will be more or less representative of the project area. The following shall be considered for SOC model validation:

    - Data Source A: Direct SOC measurements[5] from within the project area
    - Data Source B: Direct SOC measurements from similar reference sites adjacent to the project area covering the same management changes as in the project scenario over the same spatial and temporal extent.
    - Data Source C: Direct SOC measurements from similar reference sites not adjacent to the project area but covering the same management changes as in the project scenario over the same spatial and temporal extent.

[5] ”Direct” means field-based, on-the-ground measurements using widely accepted soil sampling protocols (Aynekulu et al., 2011; FAO, 2020). Soil spectrometry may be accepted if the statistical relationship with traditionally accepted field measurements is characterised as strong and verified for the project area.

- Data Source D: Direct SOC measurements from similar reference sites not adjacent to the project area, covering the same management changes as in the project scenario but over a different spatial or temporal extent.

57. It is strongly recommended to use validation data from Data Sources A or B to ensure maximal representativeness for the project area. Depending on the data source, additional uncertainty deductions might be applied. The inclusion of Data Sources B, C, and D is a pragmatic approach for allowing non-obligated entities to validate and use models for estimation in the absence of validation data from the project site itself (i.e., in the first years of a project before resampling is conducted). As the project progresses and data is sampled from within the project's spatial boundary over time for verification, the model shall be revalidated on this data from within the project's spatial boundary to continue using the model for model-based estimates.

58. For Data Sources B, C, and D, the similarity of reference sites to the project area shall be justified on the basis of the respective modelling unit defined in accordance with Sections 6 and 7 of the SOC FM, based on a common climate zone, soil texture class, vegetation type, and management history. This justification shall be included in the model validation report to allow independent assessment of the representativeness of the validation data for the project area.

### 4.4.4 Outputs of Data Collection

59. The main output of the data acquisition step is electronic files with all collected SOC reference data points used for model calibration and validation, i.e.:

    a) The measured SOC stock data points for all sampling units

    b) The exact dates when the individual samples were collected (and, optionally, the dates when they were analysed in the laboratory)

    c) The exact locations of the sampling units (point coordinates or exact spatial boundaries)

    d) The exact soil depth intervals at which the samples were collected.

    Furthermore, the following metadata shall be recorded in the model validation report:

    e) The full description of the sample preparation before the lab analysis (e.g., compositing of samples, homogenisation, sieving, drying)

    f) The full description of the lab methods and measurements carried out (e.g., International Organization for Standardization [ISO] 10694 dry combustion, volumetric/gravimetric quantification of BD/coarse fragment ratio)

    g) The uncertainty of the lab measurements (if possible, the standard error of the mean of the lab measurements)

    Lastly, the model validation report shall contain:

    h) Proof that the validation data and the calibration data are statistically independent as described in Section 4.4.2 of these guidelines.

i) If Data Sources B, C, or D are used, justification that the sampled reference sites are similar to the project area, as described in Section 4.4.3 of these guidelines.

## 4.5 Calibrate the Model

### 4.5.1 Rationale of Model Calibration

60. Calibration/training is a critical step in ensuring the accuracy and reliability of a SOC model for a sequestration project. It involves adjusting the model parameters so that its outputs closely match observed data. Data representative of the project context and algorithms can be used to test and adjust the model parameters, such that the model predictions match the observed validation data. This section is a guide for the process of calibrating a SOC model for a specific project.

61. Model calibration adjusts model parameters within plausible ranges to minimise the discrepancy between the model outputs and observed field data. This process finetunes the model to local conditions, enhancing its predictive accuracy.

62. Calibration is a crucial process for adapting a SOC model to a specific sequestration project, thereby enhancing its predictive accuracy and reliability. Carefully gathering data, understanding model parameters, and iteratively refining the model through calibration and validation ensures that a SOC model provides valuable insights for effective carbon sequestration management. Calibration is not just a technical exercise; it's a critical step in aligning the model with the real-world dynamics of a project site.

**Calibration Challenges**

- Data limitations: Limited or low-quality data can constrain calibration accuracy.
- Model structure: Inherent limitations in the model structure may prevent perfect calibration. Understanding these limitations is essential for interpreting model results.
- Computational resources: Some calibration methods

### 4.5.2 Calibration Requirements

63. **Considerations for Effective Calibration**

- Data quality: Ensure the quality and representativeness of the data used for calibration. High-quality data leads to more reliable calibration.
- Temporal and spatial scale: Calibrate the model at a temporal and spatial scale that matches the project requirements.
- Model complexity: The complexity of the model can impact the ease and effectiveness of calibration. More complex models may offer more detailed insights but may require more extensive calibration.
- Uncertainty analysis: Acknowledge and analyse uncertainties that are inherent in the model and input data. Understanding these uncertainties is crucial for interpreting model predictions.

### 4.5.3 How to Calibrate a Model

**A. Steps in SOC Model Calibration**

a) Gather local data: Collect field data relevant to SOC dynamics in the project area, including soil carbon measurements, soil properties (texture, pH, moisture), climate data, land use history, and management practices. This data forms the basis for calibration.

b) Understand the model parameters: Get familiar with the parameters of the chosen SOC model. Understand which parameters are most influential and how they relate to the physical processes in the project area.

c) Make initial parameter setting: Begin with default or literature-based parameter values. These values provide a starting point for calibration.

d) Choose a calibration method: Common methods include manual calibration (trial and error), automated algorithms (e.g., genetic algorithms, Monte Carlo simulations, machine learning models), or a combination of both.

e) Calibrate against measured reference data: Adjust the model parameters to minimise the difference between model predictions and measured reference data. Section 4.4 of these guidelines, “Collect the Required Data,” provides details on how to obtain calibration data. Use the statistical metrics from Section 4.3.3 of these guidelines, “Model Selection Requirements,” to quantify the model performance on the calibration data.

f) Iterative Refinement: Calibration is an iterative process. Continuously refine the parameters and rerun the model until a satisfactory level of agreement is reached between the model outputs and observed data.

g) Validation: After calibration, validate the model using a held-out validation dataset as described in Section 4.4 of these guidelines, “Collect the Required Data.” Successful validation ensures that the model is reliable and not just overfitted to the calibration data set.

### B. Outputs of Model Calibration

64. The foremost output is the final calibrated model. In addition, the calibration process shall be documented, and the documentation shall be submitted in the context of the project. The model’s name and version shall be documented together with a description of the data and the method that was used to calibrate/train the model.

65. With the calibrated model, SOC stock snapshots or changes can be modelled in the project area and context. However, to officially use the model for quantification of SOC within the context of a SOC FM project, the model shall be appropriately validated, such that the uncertainty of model outputs and predictions can be quantified appropriately. Therefore, model validation, predictions, uncertainty quantification, and model verification are all covered in depth in the following section on uncertainty quantification.

# 5. Uncertainty Quantification

## 5.1 **Validate the Model**

66. This and the following sections cover the official validation, prediction, and verification process when a model shall be used for purely model-based estimation of the SOC stocks within a project's spatial boundaries. For simple design-based or model-assisted estimation, no official model validation or verification is required, as the SOC stock quantification is ultimately based on the sample of ground-measured observations. This holds true even when a model is used to quantify the variance to determine strata and the required sample size or for model-assisted estimation after the sampling process.

### 5.1.1 Rationale of Model Validation

67. The goal of model validation is to demonstrate that a SOC model is suitable for achieving the stated modelling objective for a given modelling unit within the project area. The SOC model selected for a modelling unit of the project area shall be able to accurately predict SOC stocks within that modelling unit and provide reliable uncertainty estimates. In the model validation process, the predictive accuracy of the model shall be assessed using independent observations (validation data) that are 32 representatives for the specific modelling unit and were not used for model selection or calibration. (See Section 4.4.2 of these guidelines, "Requirements for Data Collection.") The model validation metrics and thresholds are chosen to ensure that (1) the model is sufficiently calibrated to allow for the quantification of SOC stocks in the modelling units and (2) the model provides reliable uncertainty estimates.

### 5.1.2 Validation Metrics

68. The three metrics that shall be computed and assessed for model validation are:

    1. Bias
    2. Root mean squared error (RMSE)
    3. Coefficient of determination ($R^2$ score/R-squared)

    Bias describes the average difference between the model predictions and the measured data. RMSE represents the approximate average error of the model on the level of the prediction unit. The R 2 score describes the fraction of the variation in the measured data that is accounted for by the linear regression of predicted to ground-measured data. Definitions and short descriptions of every metric are given below. All required equations are listed in Table 3 in these guidelines.

69. When computing these validation metrics, it is important to maintain consistency between the prediction unit of the model and the sampling unit of the validation data. Ideally, the prediction unit of the model is the same as the sampling unit of the validation data. For example, a model that provides farm-level SOC predictions should be validated using a dataset of farm-level SOC reference data, where each data point is a composite sample across a farm.

70. If the prediction unit of the model differs from the sampling unit of the validation data, the finer unit shall be aggregated to match the coarser unit before the validation metrics are computed. For example, if the model produces predictions for individual points (in geographical space) and the validation data contains farm-level SOC reference data, all point predictions within the boundaries of a farm shall be aggregated to produce a farm-level prediction. The validation metrics are then computed on a farm level.

71. A SOC model can be validated only up to the spatial level of granularity (e.g., point, field, farm) of the sampling units of the validation data (or coarser), regardless of the prediction units. For example, a SOC model that produces predictions for individual points but was validated on farm-level SOC reference data is considered as validated only for farm-level predictions. The point predictions cannot be viewed as reliable unless they are validated on point-level SOC reference data.

72. For simplicity, in the remainder of this section, it is assumed that the prediction units and the sampling units of the validation data are the same.

**Bias**

The Bias quantifies the structural error that leads to systematic overestimation or underestimation of the measured SOC reference data. The Bias is calculated by averaging the signed model errors/residuals (differences between model predictions and measured reference data). See Equation 17 in Table 3 in these guidelines. A positive Bias means the model tends to overestimate the measured values, whereas a negative Bias means the model tends to underestimate them. The absolute value of the Bias indicates how strongly the model overestimates or underestimates the mean of the measured reference data in units of the data (i.e., SOC stock [t/ha]). The Bias shall be close to zero to show that the model does not systematically overestimate or underestimate the measured values.

**RMSE**

The RMSE quantifies how close, on average, the predicted SOC values are to the observed SOC values. It is calculated by taking the square root of the mean squared error (MSE). See Equation 18 in Table 3 in these guidelines. Since the individual errors are squared before averaging, larger errors are more strongly penalised than smaller errors. The MSE can be decomposed into the squared Bias and the variance of the model errors; for this reason, the RMSE jointly captures the Bias and the precision of a model. The RMSE has the same unit as the data (i.e., SOC stock [t/ha]) and ranges from zero to infinity. Lower RMSE values indicate a better model fit; the closer to zero, the better.

**$R^2$ score**

The R2 score quantifies the proportion of the variation of the measured reference data that is explained by the model. It can be used to assess how well a SOC model explains the variability of SOC on the ground. The $R^2$ is score is dimensionless and can take values between minus infinity and one. The value shall be positive for the model to be useful in practice. Higher $R^2$ values indicate a better fit of the model to the data; a value close to one means that the SOC model effectively explains all of the variability of SOC on the ground. When the mean of the ground measured observations is used as a prediction for all values in the dataset, the $R^2$ value will be zero. The $R^2$ score can be interpreted as a rescaled variant of RMSE that relates the model errors to the variance of the measured data. See Equation 19 in Table 3 in these guidelines.

**Table 3: Accuracy and uncertainty metrics**

| Symbol/Name | Equation | Equation Number |
|---|---|---|

| | | |
|---|---|---|
| Single measured data point | $X$, where $X \in \mathbb{R}$ | 8 |
| Sequence of measured data points and indexed data points | $(X_n)_{n\in\mathbb{N}} = (X_1, X_2, X_3, , X_n)$, where<br>$n$ = total number of data points or sample size<br>and<br>$X_i \in (X_n)_{n\in\mathbb{N}}$, where:<br>$i$ = unique index of an individual data point | 9<br>9b<br>9c |
| Significance level | $\alpha$, typically chosen as 0.05 to correspond to 95% confidence | 10 |
| Single prediction at the prediction unit | $\hat{X}$ where $\hat{X} \in \mathbb{R}$ | 11 |
| Mean model prediction | $\bar{\hat{X}} = \frac{1}{n}\sum_{i=1}^{n} \hat{X}_i$ | 12 |
| Validation data mean | $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ | 13 |
| Variance of the validation data | $s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \bar{X})^2$ | 14 |
| Standard deviation of the validation data | $s = \sqrt{s^2}$ | 15 |
| Single model error/residual | $r_i = \hat{X}_i - X_i$ | 16 |
| Bias | $Bias = \frac{1}{n}\sum_{i=1}^{n} r_i = \bar{\hat{X}} - \bar{X}$ | 17 |
| RMSE | $RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{X}_i - X_i)^2} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} r_i^2}$ | 18 |
| $R^2$ | $R^2 = 1 - \frac{\sum_{i=1}^{n}(\hat{X}_i - X_i)^2}{\sum_{i=1}^{n}(\bar{X} - X_i)^2}$ | 19 |
| Standard error of the validation data mean | $SE = \sqrt{\frac{1}{n}s^2}$ | 20 |
| Margin of error (MoE) of the two-sided 90% confidence interval around the validation data mean | $MoE = SE \times t_{(1-0.9)/2,n-1}$ | 21 |

| | | |
|---|---|---|
| Prediction interval with coverage probability 1-α | $I_{1-\alpha}(X_i) = [a_i, b_i]$<br>where $a_i$ is the upper bound of the interval (how these are calculated depends on the model) | 22 |
| Prediction interval coverage probability (mean of the interval coverage indicator function) | $PICP = \frac{1}{n}\sum_{i}^{n} \delta(a_i \leq X_i \leq b_i)$<br>with the indicator function<br>$\delta(t) = \begin{cases} 1, \text{if t is TRUE} \\ 0, \text{else} \end{cases}$ | 23a<br>23b |
| Standard deviation of the interval coverage indicator function | $s_{PICP} = \sqrt{\frac{1}{n-1}\sum_{i}(\delta(a_i \leq X_i \leq b_i) - PIC}$ | 24 |
| Test statistic for the t-test to verify that the prediction interval coverage probability (PICP) is no less than 90% | $t = \frac{PICP - 0.9}{s_{PICP}/\sqrt{n}}$ | 25 |

### 5.1.3. Requirements for Model Validation

73. For successful model validation, all of the metrics described above shall be computed and assessed on independent validation data, i.e., data that was not used for model selection or calibration. Section 4.4 of this document, “Collect the Required Data,” details the exact requirements for validation data and how it should be collected. If the modelling objectives of the project differ across modelling units, model validation shall be carried out separately for every modelling unit. All results of the model validation process shall be collected in a model validation report and made available for independent review.

74. In a best-case scenario, there is an abundance of data that can be split into multiple validation datasets to (1) allow validation of candidate models during model selection and calibration and (2) for final model validation. If there is data scarcity, cross validation can be used (Arlot and Celisse, 2010). See “Annex B: Requirements for Cross-Validation” in these guidelines.

75. Once the final validation data has been determined, the metrics from Section 5.1.2 of these guidelines, “Validation Metrics,” shall be computed on that data. The following sections provide details on how the metrics shall be reported and which thresholds shall be satisfied.

76. **Requirements for Bias**

    The model validation report shall demonstrate that the Bias of the model is close enough to zero. More specifically, the absolute value of the Bias on the validation data shall not be greater than the half-width of the two-sided 90% confidence interval around the validation data mean. This requirement ensures that the mean model prediction over the validation data lies within the confidence interval around the validation data mean and that the overall model error is smaller or equal to the accepted margin of error, as noted in Section 9 of the SOC FM.

    The model validation report shall contain a table showing the following:

a) Number of validation data points

b) Mean model prediction over the validation data points (see Equation 12 in these guidelines)

c) Validation data mean (see Equation 13 in these guidelines)

d) Standard error of the validation data mean (see Equation 20 in these guidelines)

e) Half-width of the two-sided 90% confidence interval around the validation data mean (see Equation 21 in these guidelines)

f) Bias value (see Equation 17 in these guidelines)

"Example 1: Validation of Bias" in these guidelines demonstrates how the validation results can be presented in practice, including a visualisation of the above quantities. It is recommended but not required to add such a visualisation to the model validation report to facilitate interpretation of the results.

77. **Requirements for RMSE and $R^2$ score**

The model validation report shall demonstrate that the model captures variability of the validation data so that the individual model predictions are better than a mean prediction.

More specifically, the RMSE computed on the validation data shall be less than the standard deviation of the validation data (Chai and Draxler, 2014). Equivalently, the $R^2$ score shall be greater than zero.

78. The model validation report shall clearly state the relevant quantities and visualise the relationship between model predictions and measured data. In particular, the report shall contain a table showing:

a) Number of validation data points

b) Standard deviation of the validation data (see Equation 15 in these guidelines)

c) RMSE computed on the validation data (see Equation 18 in these guidelines)

d) $R^2$ score computed on the validation data (see Equation 19 in these guidelines)

Furthermore, the validation report shall contain two scatter plots, each with corresponding linear regression lines (optionally including their confidence intervals) and 1:1 line for comparison:

1. A regression plot shows the measured reference values of the validation data on the vertical axis (y-axis) and the model predictions on the horizontal axis (x-axis); the 1:1 line is the straight line that satisfies $y = x$.

2. A residual plot shows the residuals on the vertical axis (y-axis) and the model predictions on the horizontal axis (x-axis); the 1:1 line is the straight line that satisfies $y = 0$.

Systematic differences between the above-referenced plots and the corresponding 1:1 line may indicate model misspecification and shall be explained in the report. If there is evidence for heteroscedastic residuals in the plots (e.g., larger residual variance for

larger predictions), the model validation report shall state how heteroscedasticity is addressed (e.g., by log-transforming the data), particularly in the light of providing conservative estimates for **Δ**CSOC, t-0[6].

Example 2 demonstrates how the validation results can be presented in practice and is based on the same data as Example 1. Both examples follow.

**Table E1.1. Bias validation results**

| | |
|---|---|
| Number of validation data points | 93 |
| Mean model prediction | 10.29 t/ha |
| Measured validation data mean | 11.81 t/ha |
| Standard error of the validation data mean | 1.19 t/ha |
| Half-width of the 90% confidence interval around the validation data mean | ±2.36 t/ha |
| Bias | -1.52 t/ha |

[6] The presence of systematic patterns or heteroscedasticity in the regression plot or residual plot indicates that model residuals are not statistically independent of the model predictions. A lack of this statistical independence would invalidate the statistical analyses commonly performed with empirical SOC models to quantify uncertainties of SOC stock or SOC stock change predictions, including model based geostatistical approaches (Szatmári et al., 2021) and design-based approaches as described in Annex A.3

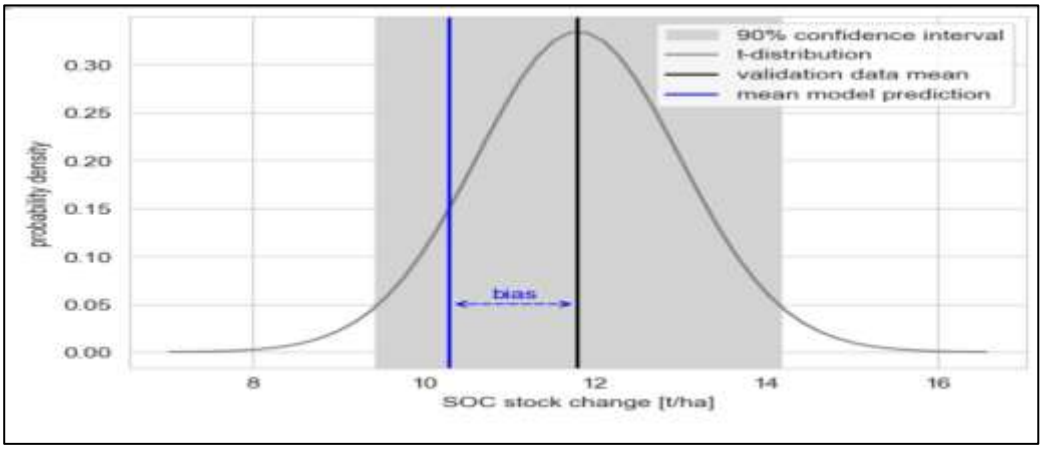



**Figure E1.2.** Visualisation of the Bias validation results.

The absolute value of the Bias (1.52 t/ha) is not greater than the half-width of the 90% confidence interval around the estimated validation data mean (2.36 t/ha). The mean model prediction lies within the confidence interval. The location and scale of the t-distribution are given by the validation data mean and its standard error; the degrees of freedom are given by n - 1, where n is the number of validation data points.

**Example 2**: Validation of RMSE and $R^2$ score

**Table E2.1.** RMSE and $R^2$ score validation results

| | |
|---|---|
| Number of validation data points | 93 |
| Standard deviation of the validation data | 11.42 t/ha |
| RMSE | 5.31 t/ha |
| $R^2$ score | 0.78 |

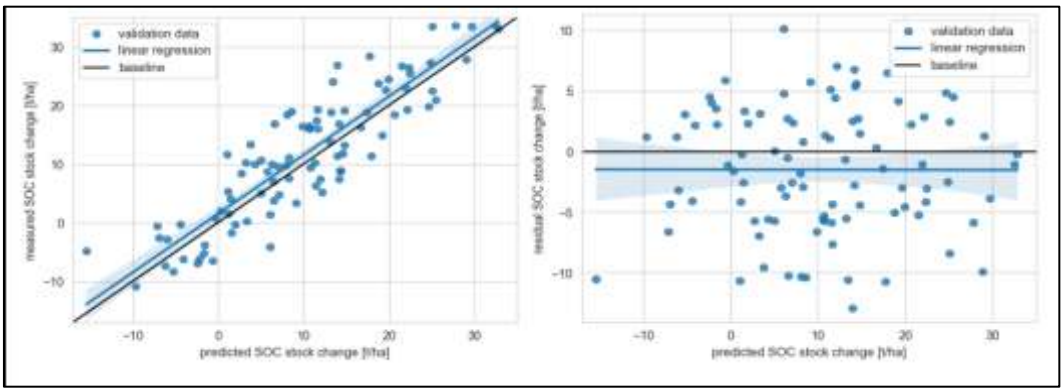


**Figure E2.2.** Scatter plots for RMSE and $R^2$ validation.

The plot on the left demonstrates that the model well approximates the measured values; the variability around the regression line is small compared to the total variability. In both plots, the regression lines are parallel to the respective 1:1 lines but do not match the 1:1 line exactly due to the non-zero Bias. Apart from the Bias, there is no apparent structure in the plots, meaning that the model is well specified. In particular, there is no evidence for heteroscedasticity.

### 5.1.4 How to Validate a Model

79. **Steps in the Final Model Validation**

For every modelling unit of the project, a suitable model shall be validated by performing the following steps:

1. Use a hold-out validation dataset (a dataset that is independent of the data used at any stage during model calibration), as outlined in Section 4.4 of these guidelines, “Collect the Required Data.”

2. Apply the final calibrated model on the validation data to obtain model predictions for every validation data point. The exact details of this step are model-dependent.

3. Calculate all validation metrics noted in Section 4.3.3 of these guidelines, “Model Selection Requirements,” as well as the threshold values and visualisations in Section 4.3.4 these guidelines, “How to Select a Model.”

4. Make sure that all requirements from “How to Select a Model” are satisfied.

Note that a single model may be applied in multiple modelling units but that the model shall be validated separately for each modelling unit in which it is applied.

In cases where model validation does not succeed, other quantification approaches shall be used for the project, e.g., the **design-based estimation approach** (Approach 1 in the SOC FM) or the **model-assisted estimation approach** (as noted in Box 1 in these guidelines). Potentially, collecting more in situ data from the project area will enable better calibration and a successful validation of the model for later calculation/crediting periods. There is also the possibility of retrying validation with a different model selection, as described in Section 4.3 of these guidelines, “Select an Appropriate Model.”

### 5.1.5 Outputs of Model Validation

80. A model validation report shall include for every modelling unit of the project:

a) A table summarising the Bias validation result as described in Section 4.3.4 of these guidelines, “How to Select a Model,” with an optional visualisation of the quantities (see Example 1 in these guidelines).

b) A table and two scatter plots summarising the RMSE and $R^2$ score validation results as described in “How to Select a Model” (see Example 2 in these guidelines).

c) If cross-validation is used, a table summarising the distances between cross validation folds as described in “How to Select a Model” (see Example 3 in these guidelines).

d) All data used to calculate the validation metrics (i.e., all model predictions with the corresponding measured reference values) shall be made available, along with the model validation report in a format that allows simple reproduction of the model validation results by an independent reviewer. Geographical locations, time stamps, soil depths, and any other information that would uniquely identify the data points in space and time may be omitted for reasons of confidentiality.

## 5.2 Make Predictions and Estimate the Uncertainty of the Predictions

### 5.2.1 Rationale of Predictions and Uncertainty Estimation

81. Upon successful model selection, calibration, and validation, the SOC models shall be applied in the project context to issue carbon credits for the quantification period. For that purpose, model predictions shall be made for every prediction unit in every modelling unit, and all quantities required to issue carbon credits for the project shall be derived from these model predictions. In particular, the uncertainties of the predictions and all derived quantities shall be estimated to apply the required uncertainty deductions described in Section 9 of the SOC FM. An additional safety discount based on the source of the validation data shall be applied to account for the uncertainty about the applicability of a model in the project context.

82. Uncertainty in SOC model outputs arises from various sources, including measurement errors in input data, variability in environmental conditions, model structural inadequacies, and parameter uncertainties. All of these uncertainties can propagate through the model and affect the model predictions and derived quantities. Therefore, statistically sound quantification of the uncertainties of the SOC model outputs is critical for any model-based SOC sequestration project.

83. Details on how uncertainties shall be expressed and which uncertainties shall be quantified are given below, along with examples for different uncertainty quantification approaches. Whether an uncertainty quantification approach is suitable for a specific model highly depends on the model and must be theoretically well justified. The chosen quantification approach shall be tested for its reliability within the project context as part of the model validation process. (See Section 4.3 of these guidelines, “Select an Appropriate Model.”)

84. A type of higher-level uncertainty that is particularly hard to quantify is the uncertainty about whether the model is applicable in the project context (and provides reliable uncertainty estimates in the first place). Model applicability crucially depends on the representativeness of the validation data for the project context. Therefore, these guidelines define an additional safety discount depending on the data source of the validation data. (See Section 4.4.3 of these guidelines, “How to Acquire Data.”) The safety discount must be added to the uncertainty deductions noted in Section 9 of the SOC FM.

### 5.2.2 Requirements for Predictions and Uncertainty Estimation

85. The following SOC model outputs must be computed:

a) SOC stock predictions for every prediction unit of every modelling unit of the project

b) The mean of these SOC stock predictions over the entire project area

Uncertainty estimates shall be provided for all of these SOC model outputs. Uncertainties shall be expressed using two-sided (bounded) intervals that contain the quantity of interest with a nominal coverage probability of 90%. The exact way the intervals are constructed and interpreted depends on the model and the underlying probability interpretation (frequentist or Bayesian). The following types of intervals are allowed for uncertainty quantification:

- **Confidence intervals,** whenever the model regards a quantity as a fixed but unknown population parameter (frequentist perspective)
- **Prediction intervals**, whenever the model regards a quantity as an unobserved random variable (frequentist perspective)
- **Credible intervals (prior intervals, posterior intervals),** whenever the model regards a quantity as an unobserved random variable (Bayesian perspective)

Which type of interval is suitable for a specific SOC model output depends on the model:

- In a classical regression approach based on probability samples, the individual predictions would be considered random variables (with associated prediction intervals), while the mean values and mean difference values would be considered fixed but unknown population parameters (with associated confidence intervals).
- When following a geostatistical model-based approach, the individual predictions at each modelling unit and their means would be random variables, and prediction intervals for all of these quantities would be provided.
- In a process-based model, uncertainties would be computed in a Bayesian way by assuming prior distributions on the input parameters and constructing credible intervals for all quantities using Monte Carlo simulations (effectively treating all model outputs as unobserved random variables).

The process of model validation should ensure that the model is suitable for the declared modelling objectives within the project's spatial boundaries. For model-based estimation approaches, spatial, temporal, and spatial-temporal autocorrelation shall be properly evaluated and taken into account during prediction and uncertainty estimation. For an example of how to do this, see "Annex A: Quantification of SOC Stock Change Using Digital Soil Mapping" in these guidelines.

### 5.2.3 How to Predict and Estimate Uncertainties

86. The exact steps and equations to obtain the SOC model outputs and associated uncertainties described above are highly model-dependent. Therefore, only general guidance is given here:

1. Run the SOC model(s) on every prediction unit of every modelling unit to obtain SOC stocks for the start and end of the respective calculation period.[7]

2. Calculate the total SOC stocks for the project scenario across all modelling units with Equation 6 of the SOC FM for the start and end of the respective calculation period.

3. Estimate the uncertainties for the mean SOC stock change using well-established statistical methods from textbooks or peer-reviewed literature that are suitable for the specific SOC model. The methods may involve one or more of the following:

    - Monte Carlo simulations to obtain credible intervals for process-based model outputs: Run the model multiple times with randomised inputs and internal parameter sets that follow strong prior distributions. Use the empirical quantiles of the simulated SOC model outputs to construct interval bounds.

    - Bootstrapping to obtain confidence intervals for population parameters in frequentist models: Create multiple bootstrap samples from the calibration data by random selection with replacement and estimate the parameter independently from every bootstrap sample. Use the empirical quantiles of the bootstrap replicates of the parameter estimate to construct interval bounds.

    - Quantile Regression (Koenker, 2005) to obtain prediction intervals in a regression model: Algorithms such as the Quantile Regression Forest (Meinshausen, 2006) or Conformalised Quantile Regression (Romano et al., 2019) directly estimate prediction intervals from covariates.

4. Calculate the model output uncertainty (UNC) defined in the SOC FM, Chapter 9, Step 3, Equation 11, by dividing the half-width of the interval around the SOC stock change prediction by the total SOC stock change estimate.

5. Obtain the total uncertainty (TU) by adding a safety discount to UNC based on the data source of the validation data (Table 4 below) and proceed with Step 4 in the SOC FM, Chapter 9, using the TU.

**Table 4** Safety discounts based on data.

| Data Source | Safety discount |
|---|---|
| A | 0% |
| B | 10% |
| C | 30% |
| D | 50% |

Table 4 provides a heuristic orientation for the safety discount, and the values of the safety discounts by data source for validation may be adjusted in coordination with the

[7] Model-based estimation SOC stock predictions must be sent to the ICM Administrator and the ACVA before the validation/verification data is sampled from the project area. This means that there can be a long time period between running the model for the start and end of the calculation period. On request by the ACVA, the exact reproducible script or sequence of steps for running the model for any prediction used for quantification purposes in the project must be made available by the non-obligated entity

ICM Administrator and the respective project's ACVA. As the project continues, model validation shall be conducted on Data Source A as soon as it becomes available within the project.

#### 5.2.4 Outputs of Predictions and Uncertainty Estimation

a) All SOC model outputs listed in Section 5.2.3 of these guidelines, "How to Predict and Estimate Uncertainties," along with intervals that capture their uncertainties.

b) UNC as defined in the SOC FM

c) TU that combines UNC and the safety discount

## 5.3 Verify the Model Predictions

### 5.3.1 Rationale for Model Verification

87. Verifying the predictive performance of a SOC model within the context of a specific soil carbon sequestration project is the final critical step in model application. This process involves comparing the model predictions with actual outcomes to assess accuracy and reliability in real-world conditions. This is sometimes referred to as "model true-up" (Lavallee et al., 2024). This section provides a comprehensive approach to verifying the prediction performance of a SOC model in a SOC FM project.

88. **Understanding Model Verification**

Verification differs from calibration and validation as it specifically focuses on how well the model performs in predicting the quantities to calculate SOC stock change outcomes in the project. Verifying the prediction performance of a locally calibrated and validated SOC model is essential to ensure its effectiveness in a specific soil carbon sequestration project. This process not only tests the model accuracy in real-world scenarios but also builds confidence among stakeholders in its predictions. Successful verification involves a thorough comparison of model outputs with actual SOC data, continuous monitoring, and adaptability to refine the model as more data becomes available. Rigorously verifying the SOC model enhances its value as a reliable tool for guiding soil carbon sequestration efforts and contributing to broader climate change mitigation strategies.

### 5.3.2 Requirements for Model Verification

89. **Considerations for Effective Verification**

a) Data quality and representativeness: The accuracy of model verification is heavily dependent on the quality and representativeness of the independent data used. Only data from Data Source A in Section 4.4.3 of these guidelines, "How to Acquire Data," is allowed for model verification.

b) Realistic expectations: Set realistic expectations for model performance. All models have inherent limitations and uncertainties.

c) Adaptability: Be prepared to adapt the model if verification reveals significant discrepancies between predictions and observations.

d) Stakeholder engagement: Involve stakeholders in the verification process to ensure transparency and build trust in the model's predictions.

90. **Challenges in Model Verification**

a) Data limitations: Limited availability of relevant, high-quality independent data can be a significant challenge.

b) Dynamic environmental conditions: Changes in environmental conditions, such as unusual weather events or management practices, can impact model performance.

c) Model complexity: Highly complex models might perform well in calibration and validation but might struggle in real-world applications.

### 5.3.3 How to Verify a Model

91. Steps in Model Verification

a) Re-establish the verification criteria:

1. The model must be unbiased (Bias < half-width of 90% confidence interval).
2. The model error must be small (RMSE < Standard deviation of the verification data).
3. The model must capture variability in the data (R2 > 0).

b) Collect and prepare independent data: Gather independent SOC data from the project site (Data Source A) that was not used in the model calibration or validation. This data shall represent the conditions under which the model is expected to operate. The data shall be collected in line with well-established soil sampling protocols (Aynekulu et al., 2011; FAO, 2020).

c) Run the model for prediction: Use the model to make predictions for the period and conditions corresponding to the independent data collected.

d) Compare predicted and observed data: Compare the model's predictions with the observed SOC data. Quantify the model's predictive performance using the metrics from Section 5.1.2 of these guidelines, “Validation Metrics”: Bias, RMSE, $R^2$ score.

e) Analyse discrepancies: If discrepancies exist between predicted and observed values, analyse these differences to understand their causes. This may involve examining model assumptions, data quality, or unexpected environmental changes. If the model violates any of the verification criteria, the non-obligated entity must submit a plan to address potential over- or under-issuance of CCCs due to the model’s predictions.

f) Continuously monitor and update: Model verification is an ongoing process. The model's performance shall be verified at least every five years, and the model shall be updated as more data becomes available or as project conditions change.

### 5.3.4 Outputs of Model Verification

92. The required output of verification must be the verification data with all its metadata as described in Section 4.4.4 of these guidelines, “Outputs of Data Collection.” Additionally, the calculated verification metrics and documentation of whether the criteria were met shall be submitted.

93. In case the model verification criteria were not met, a plan shall be submitted to describe how over- or under-crediting (issuance of CCCs) due to the model predictions is addressed by the non-obligated entity. The latest version of the performance shortfall guidelines shall provide the specific requirements for such potential over- or under-crediting. If model predictions have failed verification, the non-obligated entity must notify the ICM administrator and ACVA immediately. Within a 90-day period, the non-obligated entity can conduct a root-cause analysis of the model's shortcomings, improve the model via recalibration, and attempt to revalidate. Upon successful revalidation, the model can be applied for model-based estimations in the project context and same verification cycle/stage again. The revalidation shall follow the same procedure as noted in Section 4.1 of these guidelines and shall include the data from the verification event where the model predictions have failed. If the model cannot be revalidated, the non-obligated entity shall switch the quantification approach to direct measurements until a new version of the model can successfully be validated and applied again in the project context.

# 6. Monitoring Methodology

## 6.1 Data and Parameters Monitored

| | |
|---|---|
| **Data/Parameter** | $\Delta C_{SOC, t-0}$ |
| **Data unit:** | Mass of SOC in tonnes (Mg) |
| **Description** | SOC mass change between time t=t and t=0 |
| **Source of data:** | Directly modelled or derived as difference of SOCt - SOC0 (See Equation 2) |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $SOC_t$ |
| **Data unit:** | Total mass of SOC in tonnes (Mg) |
| **Description** | SOC mass snapshot at time t=t |
| **Source of data:** | Design-based (sampling), model-assisted, or model-based estimate |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period |
| **QA/QC procedures:** | - |

| | |
|---|---|
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | *UD* |
| **Data unit:** | Unitless ratio ([0,1]) |
| **Description** | Uncertainty deduction: proportion of $\Delta C_{SOC,\ t-0}$ that cannot be issued as credits and that shall be deducted because of the quantification uncertainty |
| **Source of data:** | Uncertainty estimation in line with a) design-based (sampling), b) model-assisted, or c) model-based estimates as noted in Section 5.2 of these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $M_{OC(DL)}$/*SOC stock* |
| **Data unit:** | Mass of SOC in tonnes per hectare (ha) (Mg/ha) |
| **Description** | SOC stock: mass of SOC C per unit area over a fixed depth column |
| **Source of data:** | Modelled directly or calculated from Equation 3b, 4, or 5b in these guidelines |
| **Measurement procedures (if any):** | Measurements shall be in line with the FAO GSOC MRV protocol. |
| **Monitoring frequency:** | Every calculation period |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $M_{SOIL(DL)}$ |
| **Data unit:** | Equivalent mass of fine soil in tonnes per ha (Mg/ha) |
| **Description** | Mass of the fine soil (dried and sieved) |
| **Source of data:** | Measured from whole soil core used for organic carbon analysis, dried, and sieved |
| **Measurement procedures (if any):** | |

| | |
|---|---|
| **Monitoring frequency:** | Every calculation period, if a variant of design-based estimation is used |
| **QA/QC procedures:** | - |
| | |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $C_{OC(DL)}$/*SOCcon* |
| **Data unit:** | SOC concentration in % (gC/gSoil * 100%) |
| **Description** | SOC concentration as measured by dry combustion and inorganic carbon removed |
| **Source of data:** | Measured fine soil mass (g, whole sample, dried and sieved) |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if a variant of design-based estimation is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | *FSS* |
| **Data unit:** | FSS in tonnes per hectare (Mg/ha) |
| **Description** | Fine soil stock, calculated as the ratio of the mass of a dried and sieved soil core per the known volume of the untreated soil core |
| **Source of data:** | See Equation 3a in these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if the FSS method is used to determine the SOC stock |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $BD_{\ fine\ soil}$ |
| **Data unit:** | BD of the fine soil in grams per $cm^3$ ($g/cm^3$) |
| **Description** | Bulk Density, the mass of soil per unit volume, measured on a sub-sampled aliquot |

| | |
|---|---|
| **Source of data:** | Measured with the intact core method on field sampled data |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if the BD method is used to determine the SOC stock (see Equation 4 in these guidelines) |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $n$ |
| **Data unit:** | Count (unitless) |
| **Description** | (Estimated) number of (composite) samples in a design- 50 based estimation (sample size) |
| **Source of data:** | Estimated with Equations 6, 7 in these guidelines or determined by constraints of feasibility |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if a variant of design-based estimation is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $\alpha$ |
| **Data unit:** | Significance level (%) |
| **Description** | The probability of incorrectly rejecting the null hypothesis (calling an effect significant although it is not) |
| **Source of data:** | Determined by non-obligated entity |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Once determined, but reported every calculation period, if $\alpha$ changes for any reason |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $X_i$ |

| | |
|---|---|
| **Data unit:** | Instance of a random variable (e.g., SOC stock or SOC stock change, units of the variable of interest apply, e.g., Mg/ha) |
| **Description** | Random variable, e.g., SOC stock or SOC stock change |
| **Source of data:** | Ground measured data (+ derived through Equations 3b, 4, 5a in these guidelines) |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if some form of design-based estimation is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | *$s^2(X)$; $s^2$ for short* |
| **Data unit:** | (Estimated) variance of a random variable (squared unit of measurement, e.g., $(Mg/ha)^2$ for SOC stock variance) |
| **Description** | (Estimated) variance of a random variable—measure of variability of the random variable around the arithmetic mean (sum of squared differences of measured data from the mean) |
| **Source of data:** | Estimated (e.g., through model predictions or based on design-based estimation) via Equation 14 in these guidelines |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if some form of design-based estimation is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $\bar{X}$ |
| **Data unit:** | Arithmetic mean (e.g., SOC stock or SOC stock change; units of the variable of interest apply, e.g., Mg/ha) |
| **Description** | Arithmetic mean of measured instances of a random variable |
| **Source of data:** | See Equation 13 in these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if some form of design-based estimation is used |

| | |
|---|---|
| **QA/QC procedures:** | - |
| **Any comment:** | - |
| **Data/Parameter** | $\overline{SE(X)}$ |
| **Data unit:** | Standard error of the mean (e.g., SOC stock or SOC stock change; units of the variable of interest apply, e.g., Mg/ha) |
| **Description** | Uncertainty measure of ground-measured (validation) data: standard error as standard deviation of the measured data divided by the square root of the number of samples n |
| **Source of data:** | See Equation 20 in these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, if some form of design-based estimation is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $\overline{\hat{X}_i}$ |
| **Data unit:** | Instance of a model prediction of a random variable (e.g., SOC stock or SOC stock change; units of the variable of interest apply, e.g., Mg/ha) |
| **Description** | Instance of a model prediction of SOC stock or SOC stock change |
| **Source of data:** | Model prediction |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, when modelling is used for deriving the variance or mean estimate of SOC stock or SOC stock change |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $r_i$ |
| **Data unit:** | Instance of a residual (e.g., SOC stock or SOC stock change; units of the variable of interest apply, e.g., Mg/ha) |
| **Description** | Instance of a residual of SOC stock or SOC stock change: difference between a measured instance of a random variable and a prediction of that instance of the random variable |

| | |
|---|---|
| **Source of data:** | See Equation 16 in these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, when modelling is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | *Bias* |
| **Data unit:** | Units of the random variable in question, e.g., SOC stock or SOC stock change: Mg/ha |
| **Description** | Model structural error: arithmetic mean of the residuals (without squaring residuals or taking the absolute value) |
| **Source of data:** | See Equation 17 in these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, when modelling is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | *RMSE* |
| **Data unit:** | Units of the random variable in question, e.g., SOC stock or SOC stock change: Mg/ha |
| **Description** | Square root of the arithmetic mean of the squared residuals, showing the average error of the model and penalising greater residuals over lower residuals |
| **Source of data:** | See Equation 18 in these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, when modelling is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

| | |
|---|---|
| **Data/Parameter** | $R^2$ *score* |

| | |
|---|---|
| **Data unit:** | Unitless ratio |
| **Description** | Shows relationship of the mean squared error to the variance in the measured data |
| **Source of data:** | See Equation 19 in these guidelines. |
| **Measurement procedures (if any):** | - |
| **Monitoring frequency:** | Every calculation period, when modelling is used |
| **QA/QC procedures:** | - |
| **Any comment:** | - |

# 7. Annexes

## Annex A: Quantification of SOC Stock Change Using Digital Soil Mapping

94. Recent advances in digital soil mapping (DSM) have made it possible to quantify SOC stocks on a farm scale, regional scale and national scale with decent accuracies. These 56 DSMs are often produced with remote sensing-based statistical models and can be recomputed annually as soon as the required remote sensing data becomes available. For this reason, DSM is an appealing approach to quantifying SOC stock changes over time.

95. This section exemplifies (1) how to validate a statistical model for DSM in the project context and (2) how to obtain statistically sound SOC stock change estimates from SOC stock maps produced by that model, including uncertainties. The statistical analysis follows a model-based approach and can be applied only when probability samples from the project area are available.

### A.1 Simulated Example

96. To illustrate the approach, the spatial distribution of SOC stocks in a fictitious project area ("ground-truth") and DSMs produced by a SOC model were simulated for two points in time, 2015 and 2020, before a hypothetical project start. In the simulation, it is assumed that SOC is in an equilibrium state and has roughly the same mean (~50 t/ha) and the same standard deviation (~12 t/ha) in both years. Figure A.1 visualises the simulated ground-truth, model predictions, and model residuals, i.e., the differences between ground-truth and predictions. In practice, only the model predictions (DSMs) are available for the complete project area. Information about the ground-truth and residuals must be obtained by collecting soil samples.

### A.2 Model Validation with SOC Stock Data

97. Validation of the model is demonstrated using SOC stock data only (not SOC stock change data). It is assumed that 100 soil samples have been taken with simple random sampling[8] in 2015 and another 100 samples in 2020 for the sole purpose of model validation. These samples have not been used to calibrate the SOC stock model that is used to produce the DSMs.

98. Figure A.2 depicts the locations of the soil samples taken in both years. It also shows the regression plots and residual plots mentioned in Section 4.3.4 of this document, "How to Select a Model," in this case, separately for the SOC stock predictions in the two years. Figure A.3 shows the regression plot and residual plot jointly for all 200 SOC stock validation data points. Table A.4 shows the validation metrics and other quantities listed in "How to Select a Model," computed separately for the SOC stock predictions in both years and as well as jointly for the two years.

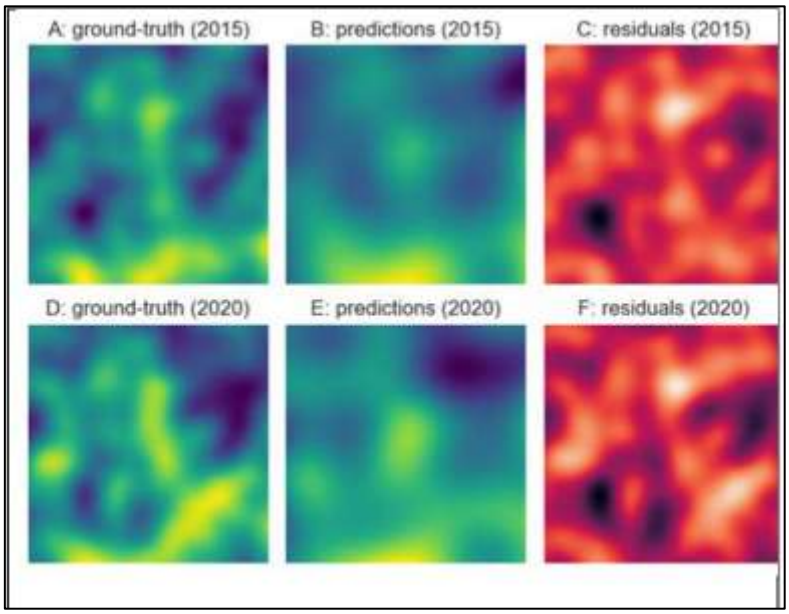



**Figure A.1.** Simulated ground-truth that shows the spatial distribution of SOC stocks in a fictitious project area in 2015 and 2020, before project start (A, D). Simulated model predictions (DSM) for the project area at the same points in time (B, E). Model residuals (C, F)

99. The regression plots and scatter plots closely follow the expected patterns (diagonal lines and horizontal lines, respectively; there is no heteroscedasticity). They provide visual evidence that the model properly explains the variability of SOC stocks, individually at each time point and jointly across all time points. The validation metrics confirm this observation quantitatively in all cases:

- The (absolute) Bias values are well below the margins of error of the validation data means (half-widths of the 90% confidence intervals).
- The RMSE values are well below the standard deviations of the validation data.

[8] Using simple random sampling or stratified simple random sampling with proportional allocation. Neyman allocation is not compatible with the statistical analysis used in this example. Random locations shall be selected independently in every time step. A sampling design in which the same locations are revisited in subsequent sampling rounds would also invalidate the analysis.

- the $R^2$ scores are well above zero.

With these observations, the SOC stock model is successfully validated for the project context.

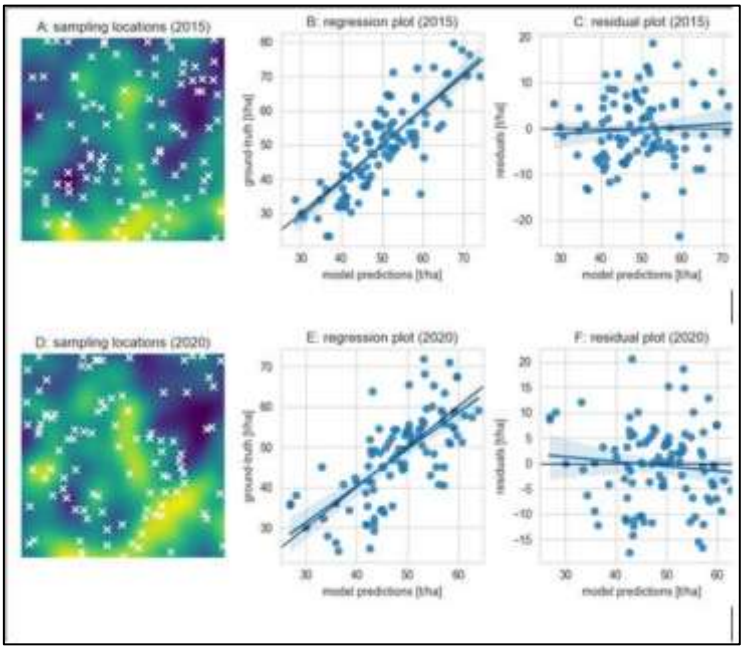



**Figure A.2**. Sampling locations of the validation data overlaid on the SOC stock ground truth for 2015 and 2020 (A, D), along with regression plots (B, E) and residual plots (C, F) created separately for the two years. The regression plots and residual plots contain linear trend lines fitted to the respective data (blue lines), along with the corresponding 95% confidence intervals (shaded blue areas) and 1:1 line for comparison (thin black lines)

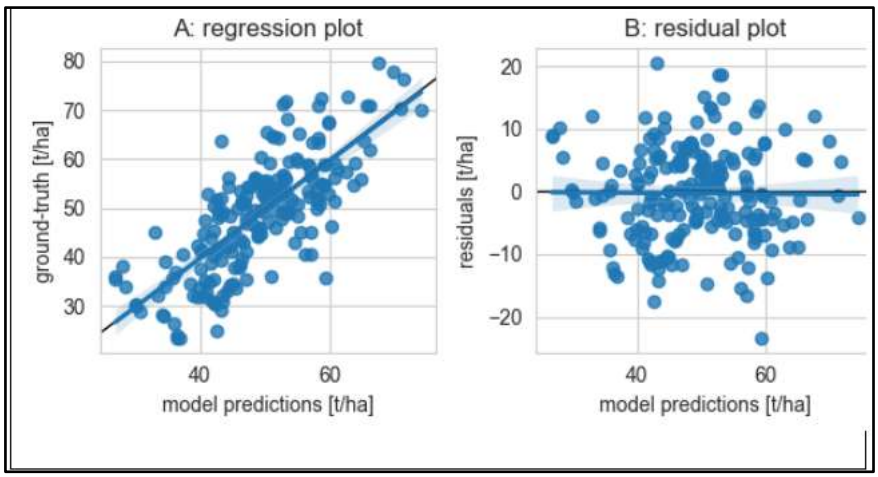


**Figure A.3**. Regression plot (A) and residual plot (B) across all 200 SOC stock data points taken in 2015 and 2020, along with linear trend lines (blue lines), 95% confidence intervals (shaded blue areas), and 1:1 line for comparison (thin black lines).

**Table A.4. Model validation metrics on SOC stock data.**

| Year | 2015 | 2020 | 2015 + 2020 |
|---|---|---|---|
| Number of validation data points | 100 | 100 | 200 |
| Mean model prediction | 49.82 t/ha | 48.42 t/ha | 49.12 t/ha |
| Validation data mean | 49.69 t/ha | 48.01 t/ha | 48.85 t/ha |
| Standard error of the validation data mean | 1.26 t/ha | 1.08 t/ha | 0.83 t/ha |
| Half-width of the 90% confidence interval around the validation data mean | ±2.09 t/ha | ±1.80 t/ha | ±1.37 t/ha |
| Validation data standard deviation | 12.57 t/ha | 10.85 t/ha | 11.74 t/ha |
| Bias | -0.13 t/ha | -0.40 t/ha | -0.27 t/ha |
| RMSE | 7.40 t/ha | 7.76 t/ha | 7.59 t/ha |
| $R^2$ score | 0.65 | 0.48 | 0.58 |

## A.3 Change Prediction and Uncertainty Estimation Exemplified

100. The goal of this section is to showcase exemplarily how to obtain model-based estimates for the mean SOC stock change and its uncertainty in compliance with Section 4.2 of these guidelines, using the following:

- **DSMs** produced by the SOC stock model during the project period.
- **Soil samples** taken with simple random sampling before the project period.

101. In this scenario, the main challenge is to temporally extrapolate the residuals computed from the soil samples into the project period and take them into account for statistically sound change prediction and uncertainty estimation. A geostatistical approach is not required due to the use of probability samples. Instead, two assumptions are made:

1. The residuals are statistically independent of the model predictions.

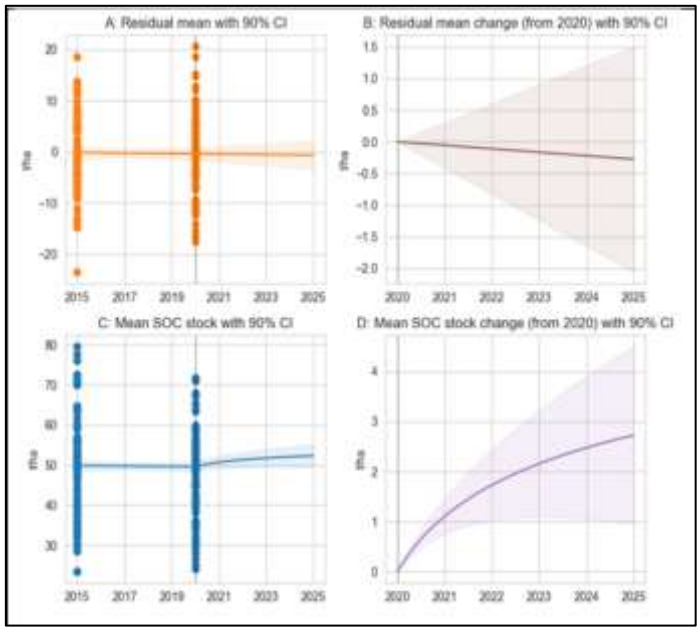
2. The residuals follow a temporal pattern that can be forecast into the project period.

102. The first assumption is supported by the residual plots: There is no apparent relationship between the residuals and the model predictions and no indication of heteroscedasticity. (See “Annex B: Requirements for Cross-Validation” and Section 4.3.4, “How to Select a Model,” in these guidelines.) The second assumption is much harder to verify without taking additional samples during the project period and will always be speculative until

true-up. By assuming a model for the temporal behaviour of the residuals, the approach becomes model based. There is a plethora of time series models (e.g., AR, ARIMA, etc.) that can be employed for this purpose (Hamilton 1994). The approach yields accurate estimates and uncertainties only if the model adequately represents the true temporal dynamics of the residuals.

103. For simplicity, this example assumes that the residuals follow a linear trend over time.[9] For this purpose, a linear regression model can be fitted to the validation data residuals (as a function of time).[10] The linear regression model is then used to forecast the residual mean into the project period. During the project period, the mean SOC stock in any given year is estimated as the sum of the mean SOC model prediction (mean of the DSM for that year) and the residual mean forecast from the linear regression model for that year. The mean SOC stock change from project start to any given year within the project period is estimated by subtracting the two SOC stock means obtained in the previous step. Uncertainties of the mean SOC stock predictions and the mean SOC stock change predictions are determined from the uncertainty of the residual mean forecast.

104. The approach outlined above is visualised in this Figure A.4 for the simulated example from the previous section. All required equations to obtain SOC stock change estimates with uncertainties (as in Figure A.4.D) are listed in Table A.6.




[9] A linear trend in the residuals captures models that steadily improve over time (e.g., because more data becomes available to recalibrate a model) as well as models that steadily deteriorate (e.g., because a model was calibrated on historic data and is applied in conditions that are increasingly different from the historic calibration data). It also captures models with a constant performance over time (a zero trend), without assuming that the model performance is constant.

[10] A linear regression model can be fitted only if validation data from at least two points in time is available. If validation data from only one point in time is available, e.g., soil samples taken at project start, there is no information about the temporal behaviour of the residuals. As a workaround, if there is reason to believe that SOC was in an equilibrium state for, e.g., five years before the project start, it is permissible to assume that the soil samples taken at project start are also representative for the SOC distribution five years before. In this case, the validation data residuals should be split randomly (50:50), and half of the residuals should be treated as if the samples were taken five years before when fitting the linear regression model.

**Figure A.4.** Visualization of a model-based SOC stock change quantification and uncertainty estimation approach based on SOC stock predictions, assuming a linear temporal trend in the residuals

The dotted vertical lines in the plots indicate the project start, solid lines indicate mean predictions, and shaded areas indicate 90% confidence intervals (CIs) for the respective means. **A:** SOC stock validation data residuals shown as a function of time, along with a linear regression fit and a forecast of the residual mean into the 62-project period. **B:** The linear regression model allows estimating the change of the residual mean from project start to any year within the project period. **C**: The mean SOC stocks in any given year (before or after the project start) are estimated by adding the residual mean prediction of the linear regression model to the mean SOC stock prediction obtained from the SOC stock model. **D**: Resulting estimates for the mean SOC stock change from project start to any year within the project period.

**Table A.6: Equations for SOC stock change quantification with a linear residual**

| **Inputs** | | **Equation Number** |
|---|---|---|
| Validation data residuals of the SOC stock model | $e_n$ | 26 |
| Validation data sampling times | $t_n$ | 27 |
| Validation data residual mean | $\bar{e} := \frac{1}{N}\sum_n e_n$ | 28 |
| Validation data mean sampling time | $\bar{t} := \frac{1}{N}\sum_n t_n$ | 29 |
| SOC stock model mean prediction in the project area at time t (obtained from the DSM) | $\mu_{\text{pred}}(t)$ | 30 |
| **Linear regression parameters** | | |
| Slope | $\widehat{\beta_1} := \frac{\sum_n (t_n - \bar{t})(e_n - \bar{e})}{\sum_n (t_n - \bar{t})^2}$ | 31 |
| Intercept | $\widehat{\beta_0} := \bar{e} - \widehat{\beta_1} \cdot \bar{t}$ | 32 |
| Regression error variance | $\widehat{\sigma^2}_{\text{resid}} := \frac{1}{N-2}\sum_n \left(e_n - \left(\widehat{\beta_0} + \widehat{\beta_1} \cdot t_n\right)\right)^2$ | 33 |
| **SOC stock change estimator** | | |
| SOC stock mean change estimate from time t to time t' | $\widehat{\mu}_\Delta(t,t') := \mu_{\text{pred}}(t') - \mu_{\text{pred}}(t) + (t'-t)\widehat{\beta_1}$ | 34 |
| **Uncertainty** | | |
| Margin of error of the SOC stock mean change estimate | $T_{\alpha/2,N-2}\sqrt{\frac{(t'-\bar{t})^2}{\sum_n (t_n-\bar{t})^2}\widehat{\sigma^2}_{\text{resid}}}$ | 35 |

## Annex B: Requirements for Cross-Validation

105. Cross-validation (Arlot and Celisse, 2010) may be used during model calibration and for validation of the final model. Cross-validation is a statistical technique to deal with the problem of data scarcity. During cross-validation, a dataset is randomly split into multiple non-overlapping data subsets ("folds"). The folds are used in an iterative process of model calibration and validation; in iteration, a single fold is used for model validation, while the other folds are used for model calibration. The overall model performance is then estimated by averaging the performance metrics obtained in each iteration. The following general requirements apply for cross-validation:

- A minimum of five cross-validation folds shall be used (10 recommended) to reduce the cross-validation uncertainty.
- The minimum number of data points per fold must be 20. In the special case of leave-one-out cross-validation, each fold may contain a single data point only.
- Overall, a minimum of 100 data points shall be used for model calibration when opting for a cross-validation approach.

106. The procedure that splits the data into different folds shall ensure that there is no data leakage from the training folds to the validation fold. During cross-validation, data leakage can happen in two ways:

1. The exact same data point occurs in multiple cross-validation folds.
2. A cross-validation fold contains a data point that is geographically very close to a data point that lies within a different fold.

The first type of data leakage ca be avoided by following the standard cross validation protocol. The second type of data leakage shall be addressed explicitly by enforcing a minimum geographical distance between the data points across different folds. The model validation report must demonstrate that the smallest geographical distance between any two data points from two different folds is greater than 30 meters.[11]

107. For this purpose, the report shall contain a table with the smallest distances (in metres) between any two data points across every pair of folds, as in Example 3 below. In the special case of leave-one-out cross-validation, where each fold contains a single data point only, the report shall clearly state the smallest distance between any two validation points; there is no need to include the table with all pairwise distances. In any case, the report shall indicate in which coordinate reference system the distances have been computed.

**Example 3:** Validation of distances for five-fold cross-validation

[11] This threshold was chosen to avoid data leakage due to imprecision of the GPS devices used to record the geographical locations. For example, two points may be taken at different depth intervals at the same location and stored along with two separate GPS devices. readings. Due to GPS device imprecision, the stored locations may be several metres apart. The threshold was explicitly not chosen to avoid *spatial autocorrelation* and achieve *spatial independence* under a *specific geostatistical model*.

**Table E3.1.** Smallest pairwise point distances between cross-validation.

| Fold | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **1** | 0 m | - | - | - | |
| **2** | 242 m | 0 m | - | - | - |
| **3** | 301 m | 187 m | 0 m | - | - |
| **4** | 193 m | 115 m | 221 m | 0 m | - |
| **5** | 267 m | 152 m | 199 m | 239 m | 0 m |

## Annex C: Table With Critical Values From the T-Distribution

**Figure A.7**. Table 6 from the SOC FM: "t-values ($t_{np}$) applicable in Equation 8. Select appropriate tnp value depending on the number of samples (np) measured for parameter p.

| $n_p$ | $t_{np}$ | $n_p$ | $t_{np}$ | $n_p$ | $t_{np}$ | $n_p$ | $t_{np}$ | $n_p$ | $t_{np}$ | $n_p$ | $t_{np}$ | $n_p$ | $t_{np}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 31 | 1.6973 | 61 | 1.6706 | 91 | 1.6620 | 121 | 1.6577 | 151 | 1.6551 | 181 | 1.6534 |
| | | 32 | 1.6955 | 62 | 1.6702 | 92 | 1.6618 | 122 | 1.6575 | 152 | 1.6550 | 182 | 1.6533 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 2.9200 | 33 | 1.6939 | 63 | 1.6698 | 93 | 1.6616 | 123 | 1.6574 | 153 | 1.6549 | 183 | 1.6533 |
| 4 | 2.3534 | 34 | 1.6924 | 64 | 1.6694 | 94 | 1.6614 | 124 | 1.6573 | 154 | 1.6549 | 184 | 1.6532 |
| 5 | 2.1319 | 35 | 1.6909 | 65 | 1.6690 | 95 | 1.6612 | 125 | 1.6572 | 155 | 1.6548 | 185 | 1.6532 |
| 6 | 2.0150 | 36 | 1.6896 | 66 | 1.6686 | 96 | 1.6610 | 126 | 1.6571 | 156 | 1.6547 | 186 | 1.6531 |
| 7 | 1.9432 | 37 | 1.6883 | 67 | 1.6683 | 97 | 1.6609 | 127 | 1.6570 | 157 | 1.6547 | 187 | 1.6531 |
| 8 | 1.8946 | 38 | 1.6871 | 68 | 1.6679 | 98 | 1.6607 | 128 | 1.6570 | 158 | 1.6546 | 188 | 1.6531 |
| 9 | 1.8595 | 39 | 1.6859 | 69 | 1.6676 | 99 | 1.6606 | 129 | 1.6568 | 159 | 1.6546 | 189 | 1.6530 |
| 10 | 1.8331 | 40 | 1.6849 | 70 | 1.6673 | 100 | 1.6604 | 130 | 1.6568 | 160 | 1.6545 | 190 | 1.6529 |
| 11 | 1.8124 | 41 | 1.6839 | 71 | 1.6669 | 101 | 1.6602 | 131 | 1.6567 | 161 | 1.6544 | 191 | 1.6529 |
| 12 | 1.7959 | 42 | 1.6829 | 72 | 1.6666 | 102 | 1.6601 | 132 | 1.6566 | 162 | 1.6544 | 192 | 1.6529 |
| 13 | 1.7823 | 43 | 1.6820 | 73 | 1.6663 | 103 | 1.6599 | 133 | 1.6565 | 163 | 1.6543 | 193 | 1.6528 |
| 14 | 1.7709 | 44 | 1.6811 | 74 | 1.6660 | 104 | 1.6598 | 134 | 1.6564 | 164 | 1.6543 | 194 | 1.6528 |
| 15 | 1.7613 | 45 | 1.6802 | 75 | 1.6657 | 105 | 1.6596 | 135 | 1.6563 | 165 | 1.6542 | 195 | 1.6528 |
| 16 | 1.7530 | 46 | 1.6794 | 76 | 1.6654 | 106 | 1.6595 | 136 | 1.6562 | 166 | 1.6542 | 196 | 1.6527 |
| 17 | 1.7459 | 47 | 1.6787 | 77 | 1.6652 | 107 | 1.6593 | 137 | 1.6561 | 167 | 1.6541 | 197 | 1.6527 |
| 18 | 1.7396 | 48 | 1.6779 | 78 | 1.6649 | 108 | 1.6592 | 138 | 1.6561 | 168 | 1.6540 | 198 | 1.6526 |
| 19 | 1.7341 | 49 | 1.6772 | 79 | 1.6646 | 109 | 1.6591 | 139 | 1.6560 | 169 | 1.6540 | 199 | 1.6526 |
| 20 | 1.7291 | 50 | 1.6766 | 80 | 1.6644 | 110 | 1.6589 | 140 | 1.6559 | 170 | 1.6539 | ≥200 | 1.6525 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 21 | 1.7247 | 51 | 1.6759 | 81 | 1.6641 | 111 | 1.6588 | 141 | 1.6558 | 171 | 1.6539 | | |
| 22 | 1.7207 | 52 | 1.6753 | 82 | 1.6639 | 112 | 1.6587 | 142 | 1.6557 | 172 | 1.6538 | | |
| 23 | 1.7172 | 53 | 1.6747 | 83 | 1.6636 | 113 | 1.6586 | 143 | 1.6557 | 173 | 1.6537 | | |
| 24 | 1.7139 | 54 | 1.6741 | 84 | 1.6634 | 114 | 1.6585 | 144 | 1.6556 | 174 | 1.6537 | | |
| 25 | 1.7109 | 55 | 1.6736 | 85 | 1.6632 | 115 | 1.6583 | 145 | 1.6555 | 175 | 1.6537 | | |
| 26 | 1.7081 | 56 | 1.6730 | 86 | 1.6630 | 116 | 1.6582 | 146 | 1.6554 | 176 | 1.6536 | | |
| 27 | 1.7056 | 57 | 1.6725 | 87 | 1.6628 | 117 | 1.6581 | 147 | 1.6554 | 177 | 1.6536 | | |
| 28 | 1.7033 | 58 | 1.6720 | 88 | 1.6626 | 118 | 1.6580 | 148 | 1.6553 | 178 | 1.6535 | | |
| 29 | 1.7011 | 59 | 1.6715 | 89 | 1.6623 | 119 | 1.6579 | 149 | 1.6552 | 179 | 1.6535 | | |
| 30 | 1.6991 | 60 | 1.6711 | 90 | 1.6622 | 120 | 1.6578 | 150 | 1.6551 | 180 | 1.6534 | | |

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**METHODOLOGICAL TOOL**

BM-T-0XXX

# Soil Organic Carbon Activity Tool for Cover crops

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INDIAN Carb•n MARKET

Publication Date: XXXX

Version 1.0

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# 1. Introduction

1. This tool focuses on cover crops planted to cover the bare soil during fallow periods to increase soil organic carbon (SOC).

2. In a typical baseline situation, crop fields are left bare (i.e., without plant growth) from the time of harvest until the next crop is planted (fallow period). The project introduces cover crops to be planted during the fallow period to cover the soil and keep roots in the soil, which improves soil condition and increases SOC.

3. This activity tool addresses different categories of cover crops (legumes, nonlegumes, grasses, and brassicas) and options for cover crop application depending on and aligning with the associated practices (reduced till, no-till) that are implemented. Cover cropping can comprise a single species or a mixture of species and can use annual, biennial, or perennial vegetation. At the end of the fallow period, cover crops are usually terminated to prevent competition with the primary crop and to allow the timely planting of the next crop. In no-till approaches, cover crops are left on the soil surface to support weed control and nitrogen inputs before a primary crop is planted. Grazing may also be used to terminate cover crops.

4. In line with the Soil Organic Carbon Framework Methodology, this tool provides three approaches for quantification of SOC increase. This accommodates that not all relevant measurements and parameters may be available to projects. Approach 1 requires on-site measurements to directly document pre-project and project SOC stock levels; Approach 2 uses datasets, parameters, and/or models from peer-reviewed publications to quantify baseline and project SOC stock levels; and Approach 3 applies default factors to quantify SOC changes relating to the general methodology described in the Intergovernmental Panel on Climate Change (IPCC) Guidelines for National Greenhouse Gas Inventories (hereafter referred to as “IPCC 2019”), using a Tier 2–level approach whenever possible. For Approaches 2 and 3, non-obligated entities shall document that the coefficients applied are conservative and applicable to the project site and management practice.

# 2. Definitions

5. This tool uses the following specific definition:

   i. **Cover crop -** A cover crop is a close-growing, dense canopy crop that provides soil protection, seedling protection, and soil improvement between periods of normal crop production (Soil Science Society of America, 2008). A cover crop is not harvested, and biomass is not removed from a field (this is particularly relevant to SOC).

   ii. **Farm** - A farm is an area of land that is devoted primarily to agricultural processes with the primary objective of rearing livestock or producing food and other crops. Under this tool, ‘farm’ encompasses all such lands legally registered as part of the farm’s holdings as well as associated leased lands with a valid contract.

   iii. **Full tillage (full-till) -** Full tillage refers to field practices leading to substantial soil disturbance with full inversion and/or frequent, within-year tillage operations, while leaving <30% of the surface covered by residues at the time of planting (IPCC 2019, Volume 4, Chapter 5, Section 5.2.3.3).

iv. **Reduced tillage (reduced-till) -** Reduced tillage refers to field practices with primary and/or secondary tillage but with reduced soil disturbance that is usually shallow and without full soil inversion; this normally leaves surface with >30% coverage by residues at planting (IPCC 2019, Volume 4, Chapter 4, Section 5.2.3.3).

v. **Zero Tillage pasture (no-Till) -** Zero tillage/no-till is an agricultural technique for growing crops or pasture without mechanically disturbing the soil through tillage (including disturbance from non-turning tillage such as rippers and disc harrows). Crop is sown directly into soil that has not been tilled since the harvest of the previous crop. Zero tillage has more than 30% coverage of the soil surface with residues.

# 3. Applicability

## 3.1. Applicability

6. A project applying this tool shall comply with the applicability conditions specified below and with those in the Soil Organic Carbon Framework Methodology. In addition, the project shall comply with applicable requirements specified in the detailed procedure for offset mechanism under CCTS.

7. Applicability conditions

   i. Projects must be implemented in India.

   ii. Generally, entire farms or selected fields/areas of a farm are eligible (see Section 4.1: Spatial Boundary).

   iii. The tool shall not be applicable to farms which have been using cover crops for more than one season within five years prior to project start, even if such application was on fields not included in the project.

   iv. In line with the soil organic carbon framework methodology, this tool accounts for only those benefits in the SOC pool. No other pools or emission reductions (e.g., in nitrous oxide [$N_2O$] due to reduced fertiliser need) can be accounted for as benefits.

   v. Cover cropping can comprise a single species or a mixture of species and can use annual, biennial, or perennial vegetation.

8. Other than cover crops, organic carbon inputs to the field shall not be changed as part of the project by more than 5% from baseline period average. If this tool is stacked with an additional ICM methodology or tool which uses the same quantification approach in the project (measurement or compatible SOC model allowing distinction of impacts between cover crops and organic inputs from other practice changes) and considers respective leakage effects, this applicability condition may be omitted.

   i. If the project includes change of tillage practices, the same quantification approach (measurements, models, or defaults) shall be used for both activities and shall be fully aligned to ensure that overall benefits are quantified correctly and not double-counted. Changes in tillage practices shall be documented and shall follow quantification approaches according to a respective ICM approved methodology or tool.

9. Application of herbicides to terminate cover crops shall be minimised, and negative environmental impacts shall be prevented. Use of herbicides related to the project,

specifically for crop termination and weed control, shall not be increased by more than 5% from baseline period average.

### 3.2. Methodology Tool Approval Date

10. The date of adoption of this document shall be effective from XX/XX 2026.

## 4. Project Boundaries

### 4.1. Spatial boundary

11. For spatial boundaries, the rules and requirements defined in the SOC Framework Methodology apply.

12. Spatial boundary description specific to this tool:

    i. Generally, entire farms or selected fields/areas of a farm are eligible. If the project is implemented only on selected fields, all applicability conditions, and requirements in the Soil Organic Carbon Framework Methodology (including maintaining yield) and from this tool shall be applied to these specific fields. These fields shall remain in the project during the entire crediting period (i.e., no change of participating fields within a farm during the crediting period is allowed).

    ii. This tool requires greenhouse gas (GHG) sources, sinks, and reservoirs (SSR) as identified in Figure 04 to be considered, with a focus on their impact on SOC and increase of emissions from the project. The boundary thus includes all impacted field activities (machine use, tillage, application of fertiliser and pesticides), including main crop practices. As SOC increase from cover crops is driven primarily through inputs of biomass (root and residue) as well as rhizodeposits, growth and residue management for cover crops is essential to assess SOC impact. For the same reason, other organic inputs (e.g., manure, compost) shall be monitored. As the project newly introduces cover crops to the field, emissions from the production and transport of seeds shall be considered. As no changes are expected to be made to farm infrastructure or handling changes are expected to be made to farm infrastructure or handling of main crops, these activities are not included in the project boundary for quantification.

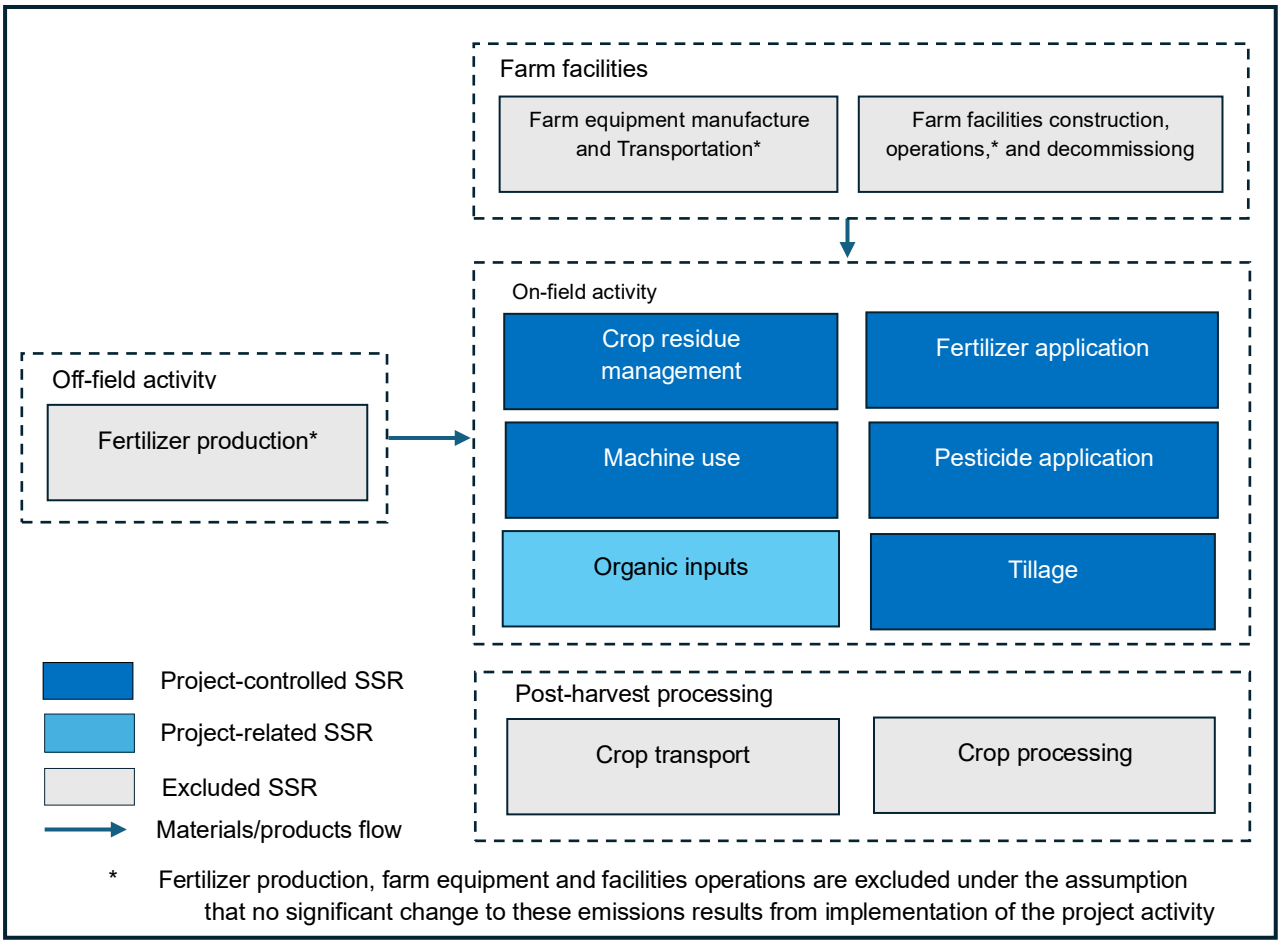



**Figure 04: Spatial project boundary**

## 4.2. Temporal boundary

13. This tool requires a crediting period of 10 years to account for the slow SOC buildup and to ensure long-term SOC impacts. The renewable crediting period of five years for each period and maximum renewal of two additional periods, that is total crediting period length of fifteen years.

## 4.3. Carbon pools

14. Carbon pool description:

i. In line with the Soil Organic Carbon Framework Methodology, this tool accounts for only those benefits in the SOC pool. Additional pools (e.g., biomass and litter) may be used for calculation of SOC change, but no GHG sequestration in pools other than SOC shall be accounted for as benefits.

ii. No emission reduction from other GHG sources (e.g., fossil fuel use or fertiliser application) shall be reported for benefits under this tool.

### 4.4. Greenhouse gases

15. Following the SOC Framework Methodology, all GHG sinks and sources affected by its activities shall be monitored.

16. For this tool, this specifically includes any significant increases in carbon dioxide ($CO_2$), methane ($CH_4$), and $N_2O$ emissions from the project.

## 5. Emissions Reduction Quantification Approach

17. Calculations for overall benefits follow the equations in the Emissions Reduction Quantification Approaches section of the Soil Organic Carbon Framework Methodology. The following sections describe the approaches and calculations that are specific to this tool.

### 5.1. Approaches for Baseline and Project Scenario Quantification

18. This tool allows application of all three quantification approaches as described in the Soil Organic Carbon Framework Methodology. The non-obligated entities shall select the most specific approach possible with the data available, giving preference to local data sources and models as outlined in the decision tree in Figure 2 of the Soil Organic Carbon Framework Methodology.

### 5.2. Approach 1 (On-Site Measurements of Soil Organic Carbon)

19. Additional requirements:

    i. SOC measurements shall be taken at a suitable time of year, e.g., at the beginning of the season for main crops, with repeated measurements performed at the same time of the year to avoid bias due to short-term variation in soil carbon pools (e.g., after harvest of main crops and tilling of crop residue into the soil). The project documentation shall specify the timing of soil sampling, i.e., the season and crop rotation phase, as applicable, to be implemented by all sampled farms in the project or program.

    ii. To avoid overestimation of SOC stocks, measurements shall always be performed before any significant carbon inputs such as organic or partially organic fertilisers, including compost and manure, as well as other carbon sources, e.g., in lime or other soil inputs and soil disturbance events (tillage).

    iii. Bulk density shall be considered in all quantification approaches. Quantification shall be done on an “equivalent mass basis” to properly quantify carbon impacts, i.e., if increased bulk density is measured in the project scenario, respective layer depth shall be reduced proportionally when calculating SOC change.

20. Special care shall be taken to ensure that litter on the soil surface (crop residue) is not included in the SOC quantification.

### 5.3. Approach 2 (Application of Models, Research Data)

21. Additional requirements:

i. Non-obligated entities shall provide evidence that calculation approaches, datasets, parameters, and models from peer-reviewed publications to estimate baseline and project SOC stocks in the project are conservative and applicable to the project site conditions and management practices.[12] In accordance with the detailed procedure for offset mechanism under CCTS requirements, "conservativeness" shall mean that from a range of available data, the value resulting in lower GHG benefits shall be applied.

ii. Models and datasets applied shall be calibrated locally, i.e., using project specific data. Global models without such local parametrisation and validation shall not be applied. Application of global defaults for soil respiration, e.g.,"$Q_{10}$" values to quantify temperature effects on respiration, is not allowed.

iii. Project documentation shall include evidence that the sources used are applicable to main crops and levels of yield (production intensity), management practices (e.g., tillage[13]), and cover crop species and practices (e.g., duration of cover crop growth) used in the project area for baseline and project scenarios. Evidence shall also be provided that datasets and models have sufficient temporal resolution (i.e., multiple data points per year) to account for impacts of seasonal variations in practices and other factors such as changing weather conditions. If the model used does not allow seasonal crop and practice variations, data shall be aggregated to a representative annual value with and without cover crops.

iv. Parameters for process-based models shall be calibrated and verified following best practice, using at least five years of data for calibration and three years of independent data for verification. Both calibration and verification datasets shall match project conditions as outlined above. Any gaps in datasets and procedures and how they were addressed in the model calibration and verification shall be reported at validation.

v. SOC measurements for a representative sample in each stratum of the project area shall be made at project start and at least once every five years, applying proven methodology (see Section 5.2 above) and using statistically assessed model results or research data for quantification. If significant differences from modelled results or research data are found, models shall be refined.

## 5.4. Approach 3 (Default Factors and IPCC Models)

22. Additional requirements: For Approach 3, this tool provides two alternative calculations in addition to the IPCC Tier 1/2 approach as listed in the Soil Organic Carbon Framework Methodology:

i. Approach 3.1 follows the method described in the Soil Organic Carbon Framework Methodology but requires Tier 2 data to allow calculation for cover crops.

ii. Approach 3.2 applies the "Steady State Method" that was also introduced in IPCC 2019. This approach is somewhat more complex in calculation but uses a different set of data that is more readily available for cropping systems and is also applicable to cover crops.

[12] Refer to stratification criteria (Section 7.1.4) for factors determining applicability to project conditions. All factors listed for stratification shall be assessed to confirm applicability.

[13] Changes in tillage due to cover crops shall be represented in the model parameters at the level required by the specific model applied.

23. **Approach 3.1** applies default equations to estimate SOC changes as described for Approach 3 in the Soil Organic Carbon Framework Methodology, based on a Tier 2 approach according to IPCC 2019. Cover crop residue shall be considered as "organic inputs" only. However, as no specific stock change factors for cover crops are available in the IPCC guidelines, national or regional Tier 2 parameters for organic input stock change factors specific to cover crops must be used, in line with the Soil Organic Carbon Framework Methodology. This tool thus requires that national or regional Tier 2 SOC reference values ($SOC_{REF}$) and stock change factors shall be applied in Equations 4 and 6 of the Soil Organic Carbon Framework Methodology. If tillage practices are changed due to the introduction of cover crops, stock change factors for tillage practices (see Section 2 for definitions) shall also be applied in accordance with the Tier 2 data source.

24. **Approach 3.2** applies the IPCC Tier 2 "Steady State Approach" according to IPCC 2019, Volume 4, Chapter 5 (see Table 01 for references). When applying the Steady State Method at the project level, the following interpretation and additional guidance shall be applied to the IPCC 2019 descriptions and procedures:

    i. The IPCC 2019 term "grid cell or region" shall be interpreted as field strata or individual fields, as appropriate to achieve conservative results at the field level within uncertainty requirements of SOC framework methodology.

    ii. Quantification shall consider all inputs to the fields for baseline and project scenarios at the field or field stratum level, including inputs from all crops separately (harvested crops for baseline; harvested crops plus cover crops for project scenario), as well as potential other organic inputs such as green manure, compost, or animal manure.

    iii. Change in tillage practices (see Section 2 for definitions) due to cover crop management shall be considered per the guidance provided in IPCC 2019 and in accordance with the Tier 2 data source.

    iv. Project-specific data and parameters shall be applied if available. Evidence for applicability of such data shall be provided with the project documentation.

    v. If no project-specific data is available, Tier 2 parameters listed in IPCC 2019 (see Table 01 below) may be applied, considering parameter uncertainties as listed. It is recommended to perform a sensitivity analysis with minimum and maximum input parameters to ensure conservative benefits estimations in case of high uncertainties. Note: Crop volumes, soil, and weather data used in the quantification shall always be based on local information.

    vi. All input parameters used in this approach shall be monitored and reported, including source and rationale for selection (see Section 11).

    vii. If the project includes tillage changes, Approach 3.2 shall be applied only if no separate calculation of tillage impacts on SOC is performed (e.g., not applying any separate SOC modules under the SOC Framework Methodology or other tillage-focused methodology to account for SOC benefits from tillage change).

    viii. For reference, the calculation spreadsheet and example data provided in IPCC 2019 may be consulted (see reference in Table 01). Application shall require the use of appropriate datasets as described above.

**Table 01: Key References to IPCC 2019 Steady State Method**

| TOPIC | References IPCC 2019, VOL. 4, CHAPTER 5 |
|---|---|
| Description of Steady State Method | Pages 5.15 to 5.18 |
| Step-by-step procedure | Pages 5.23 to 5.26 |
| Equations for Steady State Method | Pages 5.18 to 5.23 |
| Default value table for model parameters | Table 5.5A, page 5.31 |
| Default value table for nitrogen and lignin contents in crops | Table 5.5B, page 5.32 |
| Default value table for carbon to nitrogen ratios and lignin contents in livestock manure | Table 5.5C, page 5.33 |
| Tier 2 Steady State Method spreadsheet (MS Excel) | Separate tool[14] |

# 6. Baseline Scenario

## 6.1. Baseline Calculations

25. Quantification for SOC in the baseline ($SOC_{BL,y}$) shall follow the rules, approaches, calculations, and parameters in the Baseline Scenario section of the SOC Framework Methodology.

26. Baseline data is required for five years per the SOC Framework Methodology. If five-year baseline data is not fully available for a field within a stratum (e.g., it was leased less than five years prior to project start), evidence shall be provided that average baseline conditions for the stratum apply to the field lacking data, referencing the stratification criteria described below.

27. Baseline period may be extended to cover full rotation periods; avoid including fragments of rotations in the baseline quantification, e.g., in a three year crop rotation system, a six-year baseline period will deliver more consistent results, including all relevant practices, inputs, and crop yields for two full rotations.

28. To determine the baseline of the eligible project area, the land shall be stratified into modelling units (MUs) according to criteria in Section 6 of the Soil Organic Carbon Framework Methodology. In addition, the following criteria shall be considered in stratification:

[14] https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/Vol4_Ch5_Tier2_Steady_State_MethodSpreadsheet.xlsx

i. Mineral soil type[15]

ii. Tillage practices

iii. Specific crops (and crop rotations) and production period

iv. Productivity (e.g., net primary productivity or crop yield)

v. Fallow (bare) period

For each stratum (MU), SOC measurements shall be performed (Approach 1) and/or model parameters shall be identified and verified (Approach 2 or 3).

29. If a model is used to quantify baseline SOC (Approach 2), full transparency shall be provided on input data and parametrisation. Application of "black box" models is not allowed.

# 7. Project Scenario

## 7.1. Project Calculations

30. Quantification for SOC in the project scenario ($SOC_{t,y}$) shall follow the rules, approaches, calculations, and parameters in the Project Scenario section of the SOC Framework Methodology.

31. This tool addresses different categories of cover crops (legumes, non-legumes, grasses, and brassicas) and options for cover crop application depending on and aligning with the associated practices (reduced till, no-till) that are implemented. Cover cropping can comprise a single species or a mixture of species and can use annual, biennial, or perennial vegetation. Quantifications for all approaches shall consider specific cover crops and practices (including cover crop termination and residue cover) applied. When measuring (Approach 1), respective criteria shall be considered in the project area stratification for sampling. For model application (Approach 2 or 3), calculation and parameters shall be adapted to the specific project situation.

32. If other agricultural activities besides cover crops are introduced in a project (e.g., tillage change), interactions shall be considered by applying the same SOC quantification approaches (Approach 1, 2, or 3). If models are applied (Approach 2 or 3), they shall specifically cover all activities in the project scenario.

33. For the project scenario, the eligible project area shall be stratified into MUs according to the criteria in Section 7 of the SOC Framework Methodology. In addition, the following criteria shall be considered in stratification. Baseline strata with bare land shall be subdivided based on cover crop type and practices.

i. Mineral soil type

ii. Tillage practices

iii. Specific crops (or crop rotations) and production period

iv. Productivity (e.g., net primary productivity or crop yield)

[15] e.g. World Reference Base for Soil Resources (WRB) Reference Soil Groups (RSG)

v. Fallow (bare) period (relevant only if part of the project area lands remain uncovered for all or part of the year, e.g., due to environmental conditions preventing cover crops at the time)

vi. Cover crop period

vii. Cover crop type

For each stratum (MU), SOC measurements shall be performed (Approach 1) and/or model parameters shall be identified and verified (Approach 2 or 3).

34. If a model is used to quantify project SOC (Approach 2), full transparency shall be provided on input data and parametrisation. Application of "black box" models is not allowed.

# 8. Uncertainty

35. Calculation of uncertainty shall follow the rules and equations in the SOC Framework Methodology.

36. If no information on Standard Deviation (SD) or Standard Error (SE) is known for a parameter, SE of 50% of the parameter value shall be assumed. For the calculations of the upper and lower confidence intervals, a t-value of 3 shall be applied. Exceptions to this rule are accepted default values that are considered constant (e.g., physical conversion rates, global warming potentials).

# 9. Other Emissions

## 9.1. Project Emissions

37. Significant additional GHG emissions (>5% total) due to the project shall be accounted for. For this tool, that explicitly includes emissions from cover crop seed production and transport, increased fertiliser input, fuel and electricity use, and other agrochemical emissions.

38. As such emissions may differ between strata, calculation for this tool shall follow the rules and equations described in Equations 1 through 8, which replace the area-wide calculation in Equations 12 through 18 of the SOC Framework Methodology. They follow the same approach but expand calculation to include cover crop seed production and transport, and they allow application of different emission factors and quantities for different strata.

$$PE_{t-0} = \Delta FE_{t-0} + \Delta FU_{t-0} + \Delta SPT_{t-0} + \Delta AE_{t-0} \quad (1)$$

Where:

$PE_{t-0}$ = emissions from project activities in the calculation period [tCO2e]

$\Delta FE_{t-0}$ = emissions from increased fertilizer use in the calculation period [$tCO_2e$]

$\Delta FU_{t-0}$ = emissions from increased fuel and electricity use in the calculation period [$tCO_2e$]

$\Delta SPT_{t-0}$ = emissions from cover crop seed production and transport in the calculation period [tCO2e]

$\Delta AE_{t-0}$ = other agrochemical emissions in the calculation period [$tCO_2e$]

## 9.2. Increased Nitrogen Fertilizer Input

39. Emissions from increased nitrogen (N) fertiliser input in the project scenario as compared to the baseline scenario are calculated as follows. No differentiation is made between synthetic and organic N fertiliser. This equation shall not be applied for decreases in N fertiliser input, in which case $\Delta FEt_{-0}$ is considered 0. To account for reductions in fertiliser input (and the respective GHG emissions reductions), a separate ICM methodology shall be applied.

$$\Delta FE_{t-0} = \sum y\left[EF_{FE,y} \times \sum Ta = 1\left(FE_{PR,y,a} - FE_{BL,y}\right)\right] \quad (2)$$

Where:

$\Delta FE_{t-0}$ = emissions from increased fertiliser use in the calculation period [tCO2e]. Must be ≥ 0 in this tool (i.e. no accounting of reductions).

$FE_{PR,y,a}$ = N fertiliser input in stratum y under the project scenario in year a of the calculation period [kgN]

$FE_{BL,y}$ = mean annual N fertiliser input in stratum y under the baseline scenario [kgN]

$EF_{FE,y}$ = Conversion factor for emissions from N fertiliser in stratum y [tCO2e kgN-1]. IPCC 2019 aggregated default value[16] for EFFE is 0.01. Disaggregated default values in IPCC 2019 Table 11.1 may be used if fertiliser inputs are known per fertiliser type.

$T$ = number of years in the calculation period [yr]

*FEPR* and *FEBL* shall be documented by the non-obligated entity. For *FEBL*, mean annual input is calculated based on respective management records for five years prior to project start. If no adequate documentation can be provided, *FEBL* shall be no more than 50% of *FEPR*.

## 9.3. Increased Combustion of Fossil Fuels and Electricity Use

40. Additional CO2 emissions from use of fossil fuel and electricity in the project (e.g., fuel used by farm machines due to a need for stronger tractors or additional passes to close/treat the surface; fuel/electricity for irrigation pumps) shall be accounted for, unless the non-obligated entity can demonstrate that fossil fuel/electricity used in the project scenario is less than or does not differ significantly from fossil fuel/electricity used in the baseline, in which case $\Delta FUt_{-0}$ is considered 0.

[16] IPCC 2019, Vol. 4 AFOLU, Table 11.1 (aggregated default value)

$$\Delta FU_{t-0} = \sum y\left[\sum Ta = 1\left(FU_{PR,y,a} - FU_{BL,y}\right) + \left(EU_{PR,y,a} - EU_{BL,y}\right)\right] \quad (3)$$

Where:

$\Delta FU_{t-0}$ = emissions from increased fossil fuel and electricity use in the calculation period [tCO2e]

$FU_{PR,y,a}$ = emissions from use of fossil fuels in stratum y under the project scenario in year a of the calculation period [tCO2e]

$FU_{BL,y}$ = mean annual emissions from use of fossil fuels in stratum y under the baseline scenario [tCO2e]

$EU_{PR,y,a}$ = emissions from use of electricity in stratum y under the project scenario in year a of the calculation period [tCO2e]

$EU_{BL,y}$ = mean annual emissions from use of electricity in stratum y under the baseline scenario [tCO2e]

$T$ = number of years in the calculation period [yr]

41. *FUPR* and *FUBL* shall be documented by the non-obligated entity and generally calculated with the equation below, based on fuel consumption by machine type and fuel emission factor.

$$FU_{i,y,a} = \sum MT\, FUL_{i,y,MT,a} \times FEF_{i,y,MT} \quad (4)$$

Where:

$FU_{i,y,a}$ = emissions from use of fossil fuels in stratum y in year a [tCO2e ha-$^{1}$ ]

$FUL_{i,y,MT,a}$ = fuel consumption in stratum y by the machinery type MT used in year a [litres]

$FEF_{i,y,MT}$ = emissions factor for the fuel used in stratum y in machinery MT [tCO2e litres$^{-1}$]

$MT$ = machinery type (gasoline two-stroke, gasoline four-stroke, diesel)

*i* = formula used for baseline (i=BL) as well as project scenario (i=PR)

For *FUBL*, mean annual emissions are calculated based on respective management records for five years prior to project start. If this is not available, the amount of fuel combusted can be estimated using fuel efficiency (for example l/100 kilometres (km), l/tonne-km, l/hour) of the vehicle and the appropriate unit of use for the selected fuel efficiency (e.g. km driven if efficiency is given in l/100 km). If no adequate documentation can be provided, *FUBL* shall be no more than 50% of *FUPR*.

Non-CO2 GHG emissions caused by the use of fossil fuel from project activities (management operations, machinery, etc.) are insignificant and may thus be neglected.

42. *EU$_{PR}$* and E*U$_{BL}$* shall be documented by the non-obligated entity and generally calculated with the equation below, based on electricity consumption by appliance and respective emission factor. If electricity is generated on-site using fossil fuels (e.g. in diesel generators for irrigation pumps), emissions from fuel combustion shall be calculated instead, following the approach described above.

$$EU_{i,y,a} = \sum SE\ EUW_{i,y,SE,a} \times EEF_{i,y,SE} \quad (5)$$

Where:

$EU_{i,y,a}$ = emissions from electricity consumption in stratum y in year a [tCO2e ha$^{-1}$]

$EUW_{i,y,SE,a}$ = electricity consumption in stratum y from source SE in year a [kWh]

$EEF_{i,y,SE}$ = emissions factor for the electricity used in stratum y in source SE [tCO2e kWh$^{-1}$]

$SE$ = electricity source type (grid, fossil fuel generator, etc)

*i* = formula used for baseline (i=BL) as well as project scenario (i=PR)

For *EUBL*, mean annual emissions are calculated based on respective management records for five years prior to project start. If no adequate documentation can be provided, *EUBL* shall be no more than 50% of *EUPR*.

## 9.4. Cover Crop Seed Production and Transport

43. As applicability conditions for this tool do not allow cover crops in the baseline scenario, emissions from seed production and transport (SPT) in the baseline scenario are considered 0. Change in emissions $SPT_{t-0}$ thus equals emissions from seed production and transport in the project scenario $SBT_{PR}$, calculated as follows:

$$\Delta SPT_{t-0} = \sum y\left[\sum Ta = 1\ \left(SPT_{PR,y,a}\right)\right] \quad (6)$$

44. Annual project emissions from seed production and transport $SPT_{PR,y,a}$ are calculated based on amount of seed per crop for each stratum:

$$SPT_{PR,y,a} = \sum c\left[\sum\left(SPT_{PR,y,a,c} \times A_{PR,y}\right) \times \left(EF_{SP,y,c} + EF_{ST,y,c} \times DistST_{PR,y,a,c}\right)\right] \quad (7)$$

Where:

$SPT_{PR,y,a}$ = Emissions from seed production and transport in stratum y in year a of the project scenario [tCO2e]

$SPT_{PR,y,a,c}$ = Amount of seed for crop c applied in stratum y in year a of the project scenario [kg ha$^{-1}$]

$A_{PR,y}$ = Area in stratum y in the project scenario [ha]

$EF_{SP,y,c}$ = Emission factor for production of seed for crop c applied in stratum y in year a of the project scenario [tCO2e kg$^{-1}$]

$EF_{ST,y,c}$ = Emission factor for transport of seed for crop c applied in stratum y in year a of the project scenario [tCO2e $km^{-1}$ $kg^{-1}$]

$DistST_{PR,y,a,c}$ = Transport distance for seed for crop c applied in stratum y in year a of the project scenario [km]

45. **Amount of seed per crop $S_{PR,y,a,c}$** per stratum y shall be monitored and documented for the project. If no detailed information is available for seed crops, specifically for pre-mixed seeds, conservative and documented expert opinion may be used to estimate average amounts for each cover crop species applied in the mix.

46. **Emission factors for seed production per crop $EF_{SP,y,c}$** shall be documented for the project. If no specific emission factors are available for a crop type, conservative default emission factors may be applied (e.g., based on emission factors quantifications for related crops. For details, see the monitoring requirements for $EF_{SP,y,c}$ in Section 11).

47. **Emission factors for seed transport per seed type $EF_{ST,y,c}$** shall be documented for the project. National values specific to the transport means (vehicle type, capacity, etc.) shall be applied. If no national emission factors are available, default emission factors can be applied conservatively (e.g., using tools or emission factors published by the Greenhouse Gas Protocol[17]).

48. **Seed transport distance from production to field ($DistST_{PR,y,a,c}$)** shall be documented based on field locations for each stratum y using verifiable distance calculation, e.g., geographic information system (GIS)/map distances. Average distance in each stratum weighted by field size shall be used for calculation. If emission factor for seed production ($EF_{SP,y,c}$) already includes transportation (e.g., to distribution centers), the remaining transport distance to the fields shall be calculated for $Dist_{PR,y,a,c}$. If different transport types (air, sea, ground) are used, respective distances shall be calculated separately, and respective emission factors ($EF_{ST,t,0}$) shall be applied.

## 9.5. Other Agrochemical Emissions

49. Additional agrochemical emissions (AE) related to the project from increased use of agrochemicals, especially pesticides or non-N fertilizers, shall be accounted for unless the non-obligated entity can demonstrate that agrochemicals used in the project scenario are less than or do not differ significantly from agrochemicals used in the baseline, in which case $\Delta AE_{t-0}$ is considered 0.

50. If use of agrochemicals (herbicides, pesticides) or non-N fertilizer is significantly higher in the project than in the baseline scenario, the non-obligated entity shall calculate respective emissions by using specific amounts and emission factors. Emission factors applied shall be based on manufacturer information or scientific sources.

$$\Delta AE_{t-0} = \sum y\left[\sum_{a=1}^{T} \left(AE_{PR,y,a} - AE_{BL,y}\right)\right] \quad (8)$$

Where:

[17] "Emission Factors from Cross-Sector Tools" (EF list) or "GHG Emissions from Transport or Mobile Sources" (calculation tool, results provided correspond to $EF_{FT,t}$*$Dist_{FT,t}$) published at https://ghgprotocol.org/calculationtools#cross_sector_tools_id

$\Delta AE_{t-0}$ = additional emissions from project activity in the calculation period [tCO2e]

$AE_{PR,y,a}$ = other emissions in stratum y under the project scenario in year a of the calculation period [tCO2e]

$AE_{BL,y}$ = other emissions (annual mean) in stratum y under the baseline scenario [tCO2e]

$T$ = number of years in the calculation period [yr]

51. $AE_{PR}$ and $AE_{BL}$ shall be documented for each emitter type (agrochemical) by the non-obligated entity and calculated with the equation below, based on emission type, underlying quantity, and respective emission factor.

$$AE_{i,y,a} = \sum ET\ AQ_{i,y,ET,a} \times AEF_{i,y,ET} \quad (9)$$

Where:

$AE_{i,y,a}$ = emissions from use of other agrochemicals in stratum y in year a [tCO2e ha$^{-1}$]

$AQ_{i,y,ET,a}$ = quantity of agrochemicals in stratum y for emitter type ET applied in year a [kg]

$AEF_{i,y,ET}$ = emissions factor of the agrochemical used in stratum y (for emitter type ET) [tCO2e kg$^{-1}$]

$ET$ = emitter type (specific pesticide, fertilizer, or other agrochemical)

$i$ = formula used for baseline (i=BL) as well as project scenario (i=PR)

52. For $AE_{BL}$, mean annual emissions are calculated based on respective management records for five years prior to project start. If no adequate documentation can be provided, $AE_{BL}$ shall be no more than 50% of $AE_{PR}$.

# 10. Leakage

53. Calculation of leakage shall follow the rules and equations set out in the SOC Framework Methodology. If a reduction in yield is detected in a performance certification, it is assumed that the lost production capacity has to be made up for on land outside the project area. Emissions caused by such a shift shall be accounted for as leakage according to the equation listed in the SOC Framework Methodology.

54. Additional requirements: Under this tool, leakage shall be assessed at stratum level. Instead of Equation 19 in the SOC Framework Methodology, leakage shall thus be calculated for each stratum separately, using Equation 10:

$$LK_{t-0} = \sum y \left[\frac{CY_{min,y} - CY_{t,y}}{CY_{BL,y}} \times \bar{A}_{y,t}\right] \times \left(\Delta BC_{LA} + \Delta SOC_{LA,t-0} + \Delta FE_{LA,t-0} + \Delta FU_{LA,t-0}\right) \quad (10)$$

Where:

$LK_{t-0}$ = emissions due to shift of production to non-project lands (leakage area) [tCO2e]

$CY_{t,y}$ = crop yield in stratum y at time t (five-year average*) [kg ha$^{-1}$]

$CY_{min,y}$ = lowest crop yield in stratum y in any calculation period since project start (5-year average*) [kg ha$^{-1}$]

$CY_{BL,y}$ = crop yield in stratum y under the baseline scenario (five-year average*) [kg ha$^{-1}$]

$\bar{A}_{y,t}$ = eligible project area in stratum y (five-year average*) [ha]

$\Delta BC_{LA}$ = change in biomass carbon stocks in leakage area [tCO2e ha$^{-1}$]

$\Delta SOC_{LA,t-0}$ = change in soil organic carbon stocks in leakage area [tCO2e ha$^{-1}$]

$\Delta FE_{LA,t-0}$ = change in emissions from use of fertilizer in leakage area [tCO2e ha$^{-1}$]

$\Delta FU_{LA,t-0}$ = change in emissions from fuel use in leakage area [tCO2e ha$^{-1}$]

* If the baseline period is extended to cover full rotation cycles, crop yield and stratum area shall be calculated as the average across an equal length period.

# 11. Monitoring

55. Monitoring approach and parameters in the Soil Organic Carbon Framework Methodology shall be followed. Also, the non-obligated entity shall collect and document evidence that this tool's applicability conditions are met at all times.

56. Additional requirements: The following parameters shall be monitored in this tool, depending on the quantification approach selected. Where parameters listed below overlap with the Soil Organic Carbon Framework Methodology monitoring section, the following descriptions shall be applied under this tool to ensure stratum-level quantification and appropriate data sources.

**Data / Parameter table 1.**

| **Data / Parameter:** | $A_{PR,y}$ |
|---|---|
| Unit: | Ha |
| Description: | Area in stratum y in the project scenario |
| Source of data: | Farm records (GPS data, GIS files) |
| Value(s) applied | |
| Measurement procedures | Documentation allowing unambiguous identification and documenting area size at field level |
| Monitoring frequency | At project start; reviewed annually |
| QA/QC procedures | |
| Additional comments | Used in Equation 7 |

**Data / Parameter table 2.**

| **Data / Parameter:** | $\bar{A}_{y,t}$ |
|---|---|
| Unit: | Ha |
| Description: | Area in stratum y at time t (five-year average*) |
| Source of data: | Farm records (GPS data, GIS files) |
| Value(s) applied | |
| Measurement procedures | Documentation allowing unambiguous identification and documenting area size at field level |
| Monitoring frequency | Annually (for calculation of five-year average*) |
| QA/QC procedures | |
| Additional comments | Used in Equation 10 |

* If the baseline period is extended to cover full rotation cycles, area shall be calculated as the average across an equal length period.

**Data / Parameter table 3.**

| **Data / Parameter:** | $\boldsymbol{AQ_{i,y,ET,a}}$ |
|---|---|
| Unit: | Km |
| Description: | Quantity of agrochemicals in stratum y for emitter type ET applied in year a |
| Source of data: | Farm records |
| Value(s) applied | |
| Measurement procedures | Documentation of quantity and type of non-N–fertiliser agrochemicals used for cover crops (e.g., for termination) |
| Monitoring frequency | Annually |
| QA/QC procedures | |
| Additional comments | Used in Equation 9 |

**Data / Parameter table 4.**

| **Data / Parameter:** | $\boldsymbol{AEF_{i,y,ET}}$ |
|---|---|
| Unit: | $tCO_2e\ kg^{-1}$ |
| Description: | Emissions factor of the agrochemical used in stratum y (for emitter type ET) |
| Source of data: | Supplier information |
| Value(s) applied: | |
| Measurement procedures: | Supplier information for production and transport<br>If no supplier information is available, national or regional default values or third-party emission factors (e.g., from applicable studies on respective crop emissions) may be applied. |
| Monitoring frequency: | Annually |

| | |
|---|---|
| QA/QC procedures: | |
| Additional comments: | Used in Equation 9 |

**Data / Parameter table 5.**

| **Data / Parameter:** | $CY_{BL,c,y}$ |
|---|---|
| Unit: | kg $ha^{-1}$ |
| Description: | Average annual crop yield for crop type c per ha in stratum y in the project area during the baseline period (five-year average) |
| Source of data: | Farm records, e.g., field records, sales receipts |
| Value(s) applied: | |
| Measurement procedures: | Yield is recorded for each crop season and cumulated annually (as sum of each crop type). For the baseline, an average annual yield calculated across the five-year baseline period. |
| Monitoring frequency: | Project start |
| QA/QC procedures: | |
| Additional comments: | This parameter replaces parameter $CY_{BL}$ in the Soil Organic Carbon Framework Methodology (Equation 19). |

**Data / Parameter table 6.**

| **Data / Parameter:** | $CY_{t,y}$ |
|---|---|
| Unit: | kg $ha^{-1}$ |
| Description: | Average annual crop yield per ha in stratum y in the project area (five-year average*) |
| Source of data: | Farm records, e.g., field records, sales receipts |
| Value(s) applied: | |
| Measurement procedures: | Yield is recorded for each crop season and cumulated annually (as sum of each crop type). Average annual yield is calculated for each stratum across the five-year reporting period.<br>If cover crops are partially grazed or harvested (e.g., as feed or biomass crop), this yield may be included in $CY_{t,y}$ and, as relevant in quantification Approach 2 or 3, may be deducted from organic inputs to the soil (i.e., the respective model parameter).<br><br>* If the baseline period is extended to cover full rotation cycles, crop yield shall be calculated as the average across an equal length period. |
| Monitoring frequency: | Annually |
| QA/QC procedures: | |
| Additional comments: | Used in equation 10. This parameter replaces parameter $CY_t$ in the Soil Organic Carbon Framework Methodology (Equation 19). |

**Data / Parameter table 7.**

| **Data / Parameter:** | $CY_{min,y}$ |
|---|---|
| Unit: | kg $ha^{-1}$ |
| Description: | Minimum annual crop yield per ha in stratum y since project start |
| Source of data: | Farm records, e.g., field records, sales receipts |
| Value(s) applied: | |
| Measurement procedures: | Yield is recorded for each crop season and cumulated annually<br>(as sum of each crop type). $CY_{min,y}$ represents lowest average annual yield per stratum since project start and serves as threshold to assess yield reduction. |
| Monitoring frequency: | Annually |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 8. This parameter replaces parameter $CY_{min}$ in the Soil Organic Carbon Framework Methodology (Equation 19). |

**Data / Parameter table 8.**

| **Data / Parameter:** | $DistST_{PR,y,a,c}$ |
|---|---|
| Unit: | km |
| Description: | Transport distance for seed for crop c applied in stratum y in year a of the project scenario |
| Source of data: | |
| Value(s) applied: | |
| Measurement procedures: | Seed transport distance from production to field shall be documented based on field locations for each stratum y using verifiable distance calculation, e.g., GIS/map distances. Average distance in each stratum weighted by field size shall be used for calculation. If emission factor for seed production ($EF_{SP,y,c}$) already includes transportation (e.g., to distribution centers), the remaining transport distance to the fields shall be calculated for $Dist_{PR,y,a,c}$. If different transport types (air, sea, ground) are used,<br>respective distances shall be calculated separately, and respective emission factors ($EF_{ST,t,0}$) shall be applied. |
| Monitoring frequency: | Annually |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 7 |

**Data / Parameter table 9.**

| **Data / Parameter:** | $EF_{FE,y}$ |
|---|---|
| Unit: | tCO2e kgN-1 |
| Description: | Conversion factor for emissions from N fertiliser in stratum y |
| Source of data: | IPCC 2019 |

| | |
|---|---|
| Value(s) applied: | IPCC 2019 aggregated default value for $EF_{FE}$ is 0.01.<br><br>Disaggregated default values in IPCC 2019 Table 11.1 may be used if fertiliser inputs are known per fertiliser type. |
| Measurement procedures: | |
| Monitoring frequency: | Project start |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 2 |

**Data / Parameter table 10.**

| **Data / Parameter:** | $EF_{SP,y,c}$ |
|---|---|
| Unit: | $tCO2e\ kg^{-1}$ |
| Description: | Emission factor for production of seed for crop c applied in stratum y in year a of the project scenario |
| Source of data: | Seed producer information/product information |
| Value(s) applied: | |
| Measurement procedures: | If no specific emission factors are available for a crop, conservative default emission factors may be applied, e.g., emission factors from national or regional sources, peer-reviewed scientific publications for related (commercial) crops with comparable management practices (i.e., fertiliser use and field management/tillage). |
| Monitoring frequency: | Project start; updated annually |
| QA/QC procedures: | The seed production emission factors shall be assessed by an expert who shall be part of the ACVA team at project validation. |
| Additional comments: | Used in Equation 7 |

**Data / Parameter table 11.**

| **Data / Parameter:** | $EF_{ST,y,c}$ |
|---|---|
| Unit: | $tCO2e\ km^{-1}\ kg^{-1}$ |
| Description: | Emission factor for transport of seed for crop c applied in stratum y in year a of the project scenario |
| Source of data: | National transport emission factors |
| Value(s) applied: | |

| | |
|---|---|
| Measurement procedures: | Documented for the project<br>National values specific to the transport means (vehicle type, capacity, etc.) shall be applied.<br>If no national emission factors are available, default emission factors can be applied conservatively, e.g., by using tools or emission factors published by the Greenhouse Gas Protocol: Emission Factors from Cross-Sector Tools (EF list) or GHG Emissions from Transport or Mobile Sources (calculation tool; results provided correspond to $EF_{FT,t}$*$Dist_{FT,t}$), published at https://ghgprotocol.org/calculation-tools#cross_sector_tools_id.<br>Uncertainty of factor applied shall be determined based on parameter source (e.g., scientific source for emission factor used in tools listed above) or, if not available, conservatively assumed to be 50%. |
| Monitoring frequency: | Project start; reviewed annually |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 7 |

**Data / Parameter table 12.**

| **Data / Parameter:** | $\boldsymbol{EEF_{i,y,SE}}$ |
|---|---|
| Unit: | $tCO_2e\ kWh^{-1}$ |
| Description: | Emissions factor for the electricity used in stratum y in source SE |
| Source of data: | a) National GHG inventory sources<br>b) If no national data is available, default emission factors in IPCC 2006, Volume 2, Chapter 3, Table 3.3.1 may be applied (taking into account listed uncertainties). |
| Value(s) applied: | |
| Measurement procedures: | |
| Monitoring frequency: | Project start; reviewed annually for national updates |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 5 |

**Data / Parameter table 13.**

| **Data / Parameter:** | $\boldsymbol{EUW_{i,y,SE,a}}$ |
|---|---|
| Unit: | kWh |
| Description: | Electricity consumption in stratum y from source SE in year a |
| Source of data: | Farm records |
| Value(s) applied: | |

| | |
|---|---|
| Measurement procedures: | 1) Document electric equipment and power source used for cover cropping in each stratum.<br>2) Quantify electricity used for cover crop activity based on:<br>a. Direct records (electricity documentation), or<br>b. Duration of equipment use (hours) and power consumption per equipment type (kW). |
| Monitoring frequency: | Use-based documentation, aggregated (sum) per year |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 5 |

**Data / Parameter table 14.**

| **Data / Parameter:** | $FE_{BL,y}$ |
|---|---|
| Unit: | kgN |
| Description: | Mean annual N fertiliser input in stratum y under the baseline scenario |
| Source of data: | Farm records (field level) |
| Value(s) applied: | Average annual N input in stratum y, calculated across baseline period (five years) |
| Measurement procedures: | |
| Monitoring frequency: | Project start |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 2 |

**Data / Parameter table 15.**

| **Data / Parameter:** | $FE_{PR,y,a}$ |
|---|---|
| Unit: | kgN |
| Description: | N fertiliser input in stratum y under the project scenario in year a |
| Source of data: | Farm records (field level) |
| Value(s) applied: | Sum of N inputs in stratum y in year a |
| Measurement procedures: | |
| Monitoring frequency: | Use-based; aggregated annually |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 2 |

**Data / Parameter table 16.**

| **Data / Parameter:** | $FEF_{i,y,MT}$ |
|---|---|
| Unit: | $tCO_2e$ $liters^{-1}$ |
| Description: | Emissions factor for the fuel used in stratum y in machinery MT |
| Source of data: | a) National GHG inventory sources<br>b) If no national data is available, default emission factors in IPCC 2006, Volume 2, Chapter 3, Table 3.3.1 may be applied (taking into account listed uncertainties). |
| Value(s) applied: | |
| Measurement procedures: | |
| Monitoring frequency: | Project start; reviewed annually for national updates |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 4 |

**Data / Parameter table 17.**

| **Data / Parameter:** | $FUL_{i,y,MT,a}$ |
|---|---|
| Unit: | Liter |
| Description: | Fuel consumption in stratum y by the machinery type MT used in year a |
| Source of data: | Farm records |
| Value(s) applied: | |
| Measurement procedures: | 1) Document machines and fuel type for equipment used for cover cropping in each stratum.<br>2) Document amount of equipment use for cover crops from:<br>a. Direct records (fuel use), or<br>b. Equipment usage (hours or distance) and fuel consumption (liters/hour or liters/km). |
| Monitoring frequency: | Baseline: Project start<br>Project: Use-based documentation, aggregated (sum) per year |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 4 |

**Data / Parameter table 18.**

| **Data / Parameter:** | $S_{PR,y,a,c}$ |
|---|---|
| Unit: | kg $ha^{-1}$ |

| | |
|---|---|
| Description: | Amount of seed for crop c applied in stratum y in year a of the project scenario |
| Source of data: | Farm records |
| Value(s) applied: | |
| Measurement procedures: | Document for each cover crop type separately.<br>If no detailed information is available for seed crops, specifically for pre-mixed seeds, conservative and documented expert opinion may be used to estimate average amounts for each cover crop species applied in the mix. |
| Monitoring frequency: | Use-based documentation, aggregated (sum) per year |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 7 |

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**METHODOLOGICAL TOOL**

BM-T-0XXX

# Soil Organic Carbon Activity Tool for Increasing Soil Carbon Through Improved Tillage Practices

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INDIAN Carbon MARKET

Publication Date: XXXX

Version 1.0

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# 1. Introduction

1. This tool provides an approach to quantify greenhouse gas (GHG) emissions from agriculture by changing soil tillage practices within agricultural systems. This tool shall be applied in conjunction with the Soil Organic Carbon Framework Methodology.

# 2. Definitions

2. This tool uses the following specific definition:

i. **Conservation Tillage -** Includes any form of minimum or reduced tillage, where residue, mulch, or sod is left on the soil surface to protect soil and conserve moisture. After planting, at least 30 percent of the soil surface remains covered by residue to reduce soil erosion by water.

ii. **Conventional Tillage** - Seedbed preparation using cultivation instruments such as harrows, mouldboard ploughs, offset harrows, subsoilers, and rippers. Conventional tillage methods, involving extensive seedbed preparation, cause the greatest soil disturbance and leave little plant residues on the surface.

iii. **No Tillage -** A way of growing crops or pasture without tillage (no turning of topsoil), minimizing soil disturbance. Also called no-till farming or zero tillage.

# 3. Applicability

3. A project applying this tool shall comply with the applicability conditions specified below and within the SOC Framework Methodology. In addition, the project shall comply with applicable requirements specified in the detailed procedure for offset mechanism under CCTS.

i. Soil type:

- Proposed projects on sites with organic soils (Histosols), as defined by the World Reference Base for Soil Resources (FAO 2015)[18], are ineligible. Only mineral soil types are eligible.

ii. Cropping system:

- Managed cropping systems (e.g. single crop or crop rotation) must have been in place for at least 5 years prior to project implementation, i.e. the project does not lead to land use change.

iii. Tillage practice:

- Under this tool, conservation tillage methods are applied, meaning forms of minimum or reduced tillage, where residue, mulch, or sod is left on the soil surface

[18] World reference base for soil resources 2014 (Update 2015):

http://www.fao.org/3/i3794en/I3794en.pdf

to protect soil and conserve moisture. After planting, at least 30 percent of the soil surface remains covered by residue to reduce soil erosion by water.

- Due to the uncertainties on the actual carbon benefits of no-till techniques, this tool is not applicable to no-till techniques, including strip tillage and direct drill practices.

### 3.1. Methodological Tool Approval Date

4. The date of adoption of this document shall be effective from XX/XX 2026.

## 4. Project Boundaries

### 4.1. Spatial boundary

5. For spatial boundaries, rules and requirements defined in the SOC Framework Methodology apply

### 4.2. Temporal boundary

6. The project crediting period for SOC projects shall be as per the Ecosystem-based approaches *(currently in draft stage)* section in the "Detailed Procedure for Offset Mechanism". The draft currently mentions fixed period of 20 years and renewable crediting period of 10 years for each period and maximum renewal of two additional periods, which is total crediting period length of 30 years.

### 4.3. Carbon Pools

7. This SOC tool focuses entirely on the soil carbon pool. The project shall account for carbon pools for assessment in line with the SOC Framework Methodology.

## 5. Emissions Reduction Quantification Approaches

8. Calculations for overall benefits follow the equations set out in Section: Emissions Reduction Quantification Approaches of the SOC Framework Methodology. Sections below specify approaches and calculations specific to this tool.

### 5.1. Approaches for baseline and project scenario quantification

9. To accommodate that soil measurements are not always available to projects, especially for small community-based activities, this tool allows selection of any of the three approaches to baseline and project activity quantification as described in the SOC Framework Methodology. If a different approach is used for baseline and project scenarios in a stratum, conservativeness and comparability have to be ensured.

   i. **Approach 1:** Requires on-site measurements to directly document pre-project and project SOC stocks.

   ii. **Approach 2:** Uses peer-reviewed publications to quantify pre-project SOC stocks and project impact. The non-obligated entity shall prove that the research results are conservative and applicable to the project site and management practice.

iii. **Approach 3:** Applies default factors to quantify SOC changes from improved tillage, relating to the general methodology described in the IPCC 2019 Guidelines for National Greenhouse Gas Inventories (IPCC 2006a). However, instead of IPCC default SOC reference values ($SOC_{REF}$), a project oriented $SOC_{REF}$ value shall be used in connection with IPCC impact factors.

Generally, non-obligated entity shall apply the most specific approach possible with the data available, giving preference to local data sources and models. A decision tree to determine an eligible approach is provided in the SOC Framework Methodology.

# 6. Baseline Scenario

10. Under the additionality and applicability conditions set in the SOC Framework Methodology and this tool, the relevant baseline scenario is the continuation of the historical cropping practices where, in the absence of the project activity, conventional tillage is done in a business as usual (BAU) manner.

11. To determine the baseline of the eligible project area the land shall be stratified into modelling units (MU) according to:

    i. Mineral soil type

    ii. Climate zone

    iii. Land management / cropping system

    iv. Input levels (e.g. fertilization)

    v. Tillage practices

    For each stratum (MU), SOC measurements have to be performed (Approach 1) and/or model parameters identified and verified (Approach 2 or 3).

## 6.1. Baseline Calculations

12. Quantification for soil organic carbon in the baseline ($SOC_{BL,y}$) shall follow the rules, approaches, calculations, and parameters set out in Section: Baseline Scenario of the SOC Framework Methodology.

# 7. Project Scenario

13. Under the project scenario, SOC relevant practices are applied in the project area.

14. As with the baseline, the eligible project area shall be stratified into modelling units (MU) according to:

    i. Mineral soil type

    ii. Climate zone

    iii. Cropping systems

    iv. Input levels (e.g. fertilization)

v. Tillage practices

For each stratum (MU), SOC measurements have to be performed (Approach 1) and/or modelling parameters identified (Approach 2 or 3).

### 7.1. Project Scenario Calculations

15. Quantification for soil organic carbon in the project scenario ($SOC_{t,y}$) shall follow the rules, approaches, calculations, and parameters set out in Section: Project scenario of the SOC Framework Methodology.

## 8. Uncertainty

16. Calculation of uncertainty shall follow the rules and equations set out in the SOC Framework Methodology.

## 9. Other Emissions

17. Significant additional greenhouse gas emissions (>5% total) due to the project activity need to be accounted for. For this tool, this explicitly includes emissions from increased fertilizer input, fossil fuel combustion, and other agrochemical emissions. Calculation thereof shall follow the rules and equations set out in the SOC Framework Methodology.

## 10. Leakage

18. Calculation of leakage shall follow the rules and equations set out in the SOC Framework Methodology.

19. For this SOC tool, leakage from C runoff is considered 0 as projects are not allowed on wetlands. Also, as the project site is being actively maintained for commodity production during the project-crediting period, yield related leakage risks are relatively small. Crop producers are commonly risk averse and are unlikely to intentionally suffer reduced crop yields. Moreover, according to SOC Framework Methodology, projects shall not lead to a decrease in agricultural productivity, thus all projects shall be set up to maintain or increase yield. Thus, for initial project calculations, LKt-0 is considered equal 0.

20. If a reduction in yield is detected in a performance certification, it is assumed that the lost production capacity will have to be made up for on land outside the project area. Emissions caused by such a shift shall be accounted for as leakage according to the equation listed in the SOC Framework Methodology.

## 11. Monitoring

**21.** Monitoring approach and parameters shall be followed as set out in the SOC Framework Methodology. In addition to the data and parameters listed in the SOC Framework Methodology, the following parameters need to be monitored and recorded:

## 11.1. Data and Parameters collected for baseline calculation and when project areas (farms/land parcels) are being added.

**Data / Parameter table 4.**

| **Data / Parameter:** | $F_{MG,BL,y}$ |
|---|---|
| Unit: | [dimensionless] |
| Description: | tillage factor before project start in stratum y |
| Source of data: | IPCC defaults or national / local studies (preferred) |
| Value(s) applied | |
| Measurement procedures | |
| Monitoring frequency | Project start |
| QA/QC procedures | - |
| Additional comments | |

## 11.2. Data and parameters monitored

**Data / Parameter table 2.**

| **Data / Parameter:** | $F_{MG,PR,y}$ |
|---|---|
| Unit: | [dimensionless] |
| Description: | tillage factor after project start in stratum y |
| Source of data: | IPCC defaults or national / local studies (preferred) |
| Value(s) applied: | |
| Measurement procedures: | |
| Monitoring frequency: | Annually |
| QA/QC procedures: | |
| Additional comments: | |

Also, the non-obligated entity shall collect and document evidence that this SOC tool's applicability conditions are met at all times, especially that:

- Measures are taken to prevent soil erosion
- Adequate input of organic crop residue, mulch, sod or other organic C source is applied to the project area fields

# Annex A: Soil Carbon Accumulation Examples

Table 02 indicates the variation across climate and soil types as well as the rather broad ranges within these categories. This corroborates that in order to apply such parameters, projects must match the sources conditions, i.e. climate, soil, cropping system and any other parameters in the reference (e.g. soil clay content). It is the non-obligated entities' responsibility to provide evidence that this requirement is met.

Please note that the results in table 01 are focused on India, with the intention to provide startup information for smallholder projects. Yet these values are rough indications only, with more specific research needed for individual projects.

**Table 02: Soil carbon (SOC) accumulation for improved tillage approaches for climate and soil combinations**

| IPCC Climate Zone | IPCC Soil Type | Country | SOC accumulation range ($tCO_2e\ ha^{-1}\ yr^{-1}$) | Crop - Biomass productivity y | Reference |
|---|---|---|---|---|---|
| Tropical Dry | HAC | India | 0.83 | No change | Kurothe et al., 2014 |
| Tropical Dry | LAC | India | 1.94 to 2.24 | No change | Bhattacharyya et al., 2009 |
| Tropical Dry | LAC | India | 0.89 to 1.38 | No change | Das et al., 2013 |
| Tropical Dry | LAC | India | 1.36 | Higher | Bhattacharyya et al., 2012 |

# Annex B: Additional References

***NOTE: This list is provided for information only and has not been updated since its original release. As soil and tillage research is progressing rapidly, this list does not replace a thorough literature review for up-to-date studies and results.***


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**METHODOLOGICAL TOOL**

BM-T-0XXX

# Soil Organic Carbon Activity Tool - Biostimulants for Soil Revitalisation

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INDIAN Carbon MARKET

Publication Date: XXXX

Version 1.0

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# 12. Introduction

1. This tool prescribes requirements and guidance to quantify and monitor greenhouse gas (GHG) emissions and soil organic carbon (SOC) changes resulting from change in soil management practices within agricultural systems through application of biostimulants for soil revitalisation. The eligible activities are intended to achieve net carbon sequestration in the soil carbon pool. This tool shall be applied in conjunction with the Soil Organic Carbon Framework Methodology.

# 13. Definitions

2. For the purpose of this tool, the following definitions apply:

(a) **Biostimulants:** A biostimulant refers to a product (substances and/or microorganisms) independent of the product's nutrient content that used to stimulate natural processes and metabolism in soil to improve tolerance of abiotic stresses and optimise crop quality, facilitate the availability of nutrients, conditionate the rhizosphere and enhance soil microflora activity**.**

(b) **Life Cycle Assessment (LCA):** A life cycle assessment (LCA) is a method to evaluate the environmental impact of a product over its entire life cycle. It is a holistic approach that takes into account all the activities involved in the creation of a product, such as raw material extraction, manufacturing, transportation and distribution, use and disposal. For the purpose of this activity module, LCA is based only on greenhouse gases.

# 14. Scope & Applicability

## 14.1. Scope

3. This activity module is applicable to project activities that use biostimulants for soil revitalisation

## 14.2. Applicability

4. A project applying this Activity Module shall comply with the applicability conditions specified below and within the SOC Framework Methodology. In addition, the project shall comply with applicable requirements specified in the detailed procedure for offset mechanism under CCTS.

5. The Activity module is applicable under the following conditions:

(a) Geographic location – The activity module is applicable in India.

(b) Project area - The Project Area shall NOT include;

i. riparian or other buffer zones located within such eligible areas, and excluding areas set aside for conservation.

ii. wetlands.

(c) Soil type – the proposed activity involving sites with

i. Only mineral soil types are eligible.

ii. organic soils (Histosols), as defined by the World Reference Base for Soil Resources (FAO 2015) are NOT eligible.

(d) Land use – the proposed activity shall

i. have managed cropping systems (e.g. single crop or crop rotation) in practice for at least 5 years prior to project implementation.

ii. NOT lead to land use change E.g., conversion of existing natural grasslands to croplands

(e) Soil management practices - Under this SOC tool, biostimulants for soil revitalisation may be introduced in project areas in conjunction with sustainable use and/or management of the soil that leads to an improvement in the soil characteristics indicator(s) throughout the project, as demonstrated using Annex 1 of this activity module. If it is not possible, non-obligated entity shall explain in the PDD why improvement in soil characteristic indicators was not envisaged/ intended.

(f) The biostimulants

i. shall have approval or be registered with the competent authority as a biostimulant or fertilising product with a biostimulant effect or soil improver function and not as a pesticide.

ii. shall have proven efficacy for soil revitalisation supported by peer reviewed research literature[19] and/or published studies that are based on activity in the region or comparable region.

iii. shall NOT have an adverse effect on human, animal or plant health, on safety, or on the environment, under reasonably foreseeable conditions of storage or use. This may be demonstrated via compliance to any regulations in the approval /registration process of the biostimulant or compliance to similar national/regional regulations.

iv. may include in its composition material from natural or mineral origin.

v. shall not include any Genetically Modified Organism (GMO) as a part of its composition.

[19] Peer-review publications in reputable journals that are indexed in the Web of Science: Science Citation Index (SCI; available at https://mjl.clarivate.com). Peer-review shall cite literature that is relevant to the climate zone.

### 14.3. Methodological Tool Approval Date

6. The date of adoption of this document shall be effective from XXXXX.

# 15. Project Boundaries

### 4.1 Spatial boundary

7. For spatial boundaries, rules and requirements defined in the SOC Framework Methodology apply.

### 4.2 Spatial boundary

8. The maximum crediting period, depending on the efficacy of the biostimulant, shall be as follows:
   i. 20 years – fixed crediting period is required, or
   ii. 10 years – crediting period renewal is required.

9. The crediting period starts either with the Project Start Date or three years prior to the date of Project Design Certification, whichever occurs later.

## 4.3 Carbon Pools

10. The project shall account for carbon pools for assessment in line with the SOC Framework Methodology. The soil organic carbon pool applicable to this module are as shown in Table 03

**Table 03:Emissions sources included in or excluded from the project boundary**

| Scenario | Carbon pools | $CO_2$ | $CH_4$ | $N_2O$ |
|---|---|---|---|---|
| Baseline | SOC | Yes | No | No |
| | SOC | Yes | No | No |
| Project | Emissions from production and application of biostimulants | Yes | Yes | Yes |
| | Emissions from transportation of biostimulant from production facility to farm | Yes | Yes | Yes |

## 4.4 GHGs emissions

11. The carbon pools other than SOC, applicable to the activity module are as follows:

(a) Production and application of biostimulants

i. GHGs emission from the production and application of biostimulants shall be included in the project scenario. The emissions during production and application of biostimulants are attributed to electricity and/or fossil fuel use.

ii. GHGs emission from application of biostimulants can be neglected if the application of biostimulants is through one of the baseline practices and there is

no additional use of energy (fossil fuel/electricity). E.g., Biostimulants are applied at the same time as seeding.

(b) Transportation of biostimulants

i. GHGs emission from transportation of biostimulants are considered insignificant (if less than 5% of the net emission reductions and sequestration) and may be omitted from monitoring.

# 16. Measurement-based methods for quantification of emission reduction

12. Calculations for overall GHGs sequestration follow the equations set out in Section: Emissions Reduction Quantification Approaches of the SOC Framework Methodology. Sections below specify approaches and calculations specific to this SOC tool.

## 5.1 Approaches for baseline and project scenario quantification

**13.** The baseline and project scenario SOC stocks measurements shall be done using either approach 1 or approach 2 as described in the SOC Framework Methodology**.**

**14.** **Approach 1** - Requires on-site measurements to directly document pre project and project SOC stocks.

**15.** The projects using approach 1 shall demonstrate compliance to requirements established in SOC Framework Methodology for both baseline and project quantification. For baseline quantification, SOC stock measurements shall be done prior to application of biostimulants or any other practice being introduced in the project scenario alongside application of biostimulants.

**16.** On-site SOC measurements will be carried out using:

(a) an approved protocols for on-site measurements as listed in Annex 1 of SOC Framework Methodology, or

(b) Annex 3 Soil sampling sub-protocol- GSOC-MRV Protocol[20] : It provides a standardised tool to support SDG 15.3.1., as well as any project related to SOC sequestration.

17. For soil sampling the projects activity shall apply best practices, such as those described in Smith & al., 2020[21]or FAO, 2019[22]. The FAO protocol allows for stratified simple random sampling and directed stratified sampling designs. Composite samples are required. As a minimum, samples for SOC concentration determinations shall be obtained from 0-10 cm and 10-30 cm with the FAO protocol. However, samples from deeper layers up to 1 m can

[20] FAO. 2020. A protocol for measurement, monitoring, reporting and verification of soil organic carbon in agricultural landscapes – GSOC-MRV Protocol. Rome. https://openknowledge.fao.org/handle/20.500.14283/cb0509en

[21] How to measure, report and verify soil carbon change to realize the potential of soil carbon sequestration for atmospheric greenhouse gas removal. Smith, P. & al., 2020. Global Change Biology. Available at: https://onlinelibrary.wiley.com/doi/full/10.1111/gcb.14815

[22] Measuring and modelling soil carbon stocks and stock changes in livestock production systems: Guidelines for assessment (Version 1). FAO, 2019. Available at: https://www.fao.org/3/ca2934en/CA2934EN.pdf

be collected. Sampling frequency must be as a minimum every 4 years and samples must be taken at the same time of year. For estimating bulk density, direct measurement methods shall be used, specifically the undisturbed (intact) core method and the excavation method.

18. Approach 2: The projects using approach 2 shall demonstrate compliance to requirements established in SOC Framework Methodology for both baseline and project quantification

19. Appropriateness of models, datasets or calculation approaches and management practices changes proposed by the non-obligated entities to conservatively quantify both the baseline and project SOC stocks in the region(s) where the project is located shall be demonstrated through scientific peer reviewed publications.

20. The selected models, datasets or calculation approaches shall be able to replicate project model simulations, maintain version control and fully report all parameter datasets. Where multiple parameters are utilised, selection and evidence for values shall be demonstrated via the use of peer reviewed literature. Also, appropriate description of statistical approaches used to set parameter values shall be provided. All models/calculation approaches must be validated against project data, and the same parameters must be used in both the baseline and project scenario.

21. Non-obligated entities shall prove that the research results are conservative and applicable to the project area and management practices using at least two peer reviewed papers from indexed journals.

## 5.2 Calculation of Baseline Scenario

**22.** The soil organic carbon in the baseline ($SOC_{BL,y}$) shall be measured following the requirements described in Section:6 Baseline Scenario of the SOC Framework Methodology.

**23.** The applicable baseline scenario is the continuation of the historical land management practices that are being followed in last 5 years before the project start date. To determine the baseline of the eligible project area the land shall be stratified into modelling units (MU) according to:

(a) Mineral soil type

(b) Climate zone

(c) Land management / cropping system

(d) Input levels (e.g. fertilisation)

(e) Tillage practices

(f) Any other land management practices (e.g., irrigation)

24. For each stratum (MU), SOC stock measurements shall be performed (Approach 1).

## 5.3 Calculation of Project Scenario

25. The soil organic carbon in the project scenario ($SOC_{t,y}$) shall be measured following the requirements described in Section 7: Project scenario of the SOC Framework Methodology.

26. Under the project scenario, as with the baseline, the eligible project area shall be stratified into modelling units (MU) according to:

(a) Mineral soil type

(b) Climate zone

(c) Land management / cropping system

(d) Input levels (e.g. fertilisation)

(e) Tillage practices

(f) Any other land management practices (e.g., irrigation)

27. For each stratum (MU), SOC stock measurements shall be performed (Approach 1).

## 5.4 Other emissions

28. Project activity emissions consist of GHGs emission from production, transportation from the industrial facility to the farm, and application of the biostimulant. Significant greenhouse gas emissions (>5% of the total) due to the project activity need to be accounted for. GHG emissions considered insignificant (less than 5% of total emission reduction and sequestration) may be omitted from monitoring. Non-obligated entities shall calculate GHG emissions using the following approach:

(a) Life Cycle Assessment

i. A life cycle assessment (LCA) shall be conducted

(a) considering all the activities involved in the product life cycle, including at minimum production, transport, and application of biostimulants. AND

(b) following the latest version of the recognised international standards, such as PAS 2050 issued by British Standards Institute (BSI)[23], GHG Protocol Product

[23] Specification for the assessment of the life cycle greenhouse gas emissions of goods and services - PAS 2050. British Standards Institute (BSI), 2011.

Life Cycle Standard[24], ISO standards (ISO 14040:2006)[25] and ISO 14044:2006[26],ISO 14067:2018[27].

ii. The project emissions are determined as follows:

$$PE_t = \sum^{n}(A_{(y)} x\ Q_{PR,i,t,y}\ xLCA_i) \quad \text{Eqn (1)}$$

Where

| | |
|---|---|
| $PE_t$ | total project emissions in the calculation period (tCO2e) |
| $A_{(y)}$ | Area of Stratum y (ha) |
| $Q_{PR,i,t,y}$ | Quantity of biostimulant for soil revitalisation of type i used, before dilution, in each stratum y in the calculation period (t of biostimulant/ha) |
| $LCA_i$ | Life Cycle Analysis emissions for the biostimulant for soil revitalisation of type i [tCO2e/t of biostimulant] |

iii. The following requirements apply for conducting LCA:

(a) LCA can be performed either by the producer of the product directly, non-obligated entity or a third party hired by the producer of the product. Where, the LCA is performed by the producer of the product or the non-obligated entity, the LCA shall be independently reviewed.

(b) LCA used by the non-obligated entity shall not be more than one year old from the start date of the project.

(c) LCA shall be repeated at least every 3 years or when there is a significant change (more than 5% of the original process) in the production, transportation, or application process of the biostimulant, whichever is earlier.

29. Any approach change between baseline and project scenario calculations shall be in compliance with requirements of section 8: Procedure for approach change of SOC Framework Methodology.

30. The uncertainty estimation shall be conducted following requirements described in Section 9: Uncertainty of SOC Framework Methodology.

## 5.5 Calculation of Other Emissions

31. Significant additional greenhouse gas emissions (>5% total) due to the project activity need to be accounted for. For this SOC tool, this explicitly includes emissions from increased fertiliser input, fossil fuel combustion, and other agrochemical emissions.

[24] Product Life Cycle Accounting and Reporting Standard. Greenhouse Gas Protocol, 2011. Available at: https://ghgprotocol.org/product-standard

[25] ISO 14040:2006 - Environmental management — Life cycle assessment — Principles and framework. ISO, 2016. Available at: https://www.iso.org/standard/37456.html

[26] ISO 14044:2006 - Environmental management — Life cycle assessment — Requirements and guidelines. ISO, 2016. Available at: https://www.iso.org/standard/38498.html

[27] ISO 14067:2018 - Greenhouse gases — Carbon footprint of products — Requirements and guidelines for quantification. Available at: https://www.iso.org/standard/71206.html

Calculation thereof shall follow requirement described in Section 10: Other emissions of the SOC Framework Methodology.

32. LCA for biostimulants includes emissions from increased fertiliser input and other agrochemical emissions at the application stage. Emissions from increased fossil fuel combustion as compared to the baseline scenario are also included in the Life Cycle Assessment.

## 17. Estimation of leakage emission

33. The leakage emission estimation shall be conducted following the requirements described in Section 11. Leakage of SOC Framework Methodology.

34. The project site is being actively maintained for crop production during the project-crediting period, yield related leakage risks should be negligible, if any. Crop producers are commonly risk averse and are unlikely to intentionally suffer reduced crop yields. Moreover, projects shall not lead to a decrease in agricultural productivity, thus all projects shall be set up to maintain or increase yield. Thus, for initial project calculations, $LK_{t-0}$ is considered equal to 0.

35. If a reduction in yield is detected in a performance certification, it is assumed that the lost production capacity will have to be made up for on land outside the project area. Emissions caused by such a shift shall be accounted for as leakage according to the equation listed in the SOC Framework Methodology.

## 18. Data and Parameters Used in this Tool

36. Monitoring parameters and approach shall be identified and designed following the SOC Framework Methodology.

37. In addition to the data and parameters listed in the SOC Framework Methodology, the following parameters shall be monitored and recorded. Data and Parameters collected for baseline calculation and when project areas (farms/land parcels) are being added and at renewable of crediting period if required.

## 6.1 Data and Parameters Fixed

| | |
|---|---|
| **Data/parameter:** | Cropland Management Practices $_{BL,y}$ |
| **Unit** | [dimensionless] |
| **Description** | Baseline management practices - Cropland management practices (such as tillage, manure inputs, cover crops) before project start in stratum y |
| **Source of data** | IPCC defaults or national / local studies / Farmer own records (preferred) |
| **Value(s) applied** | - |
| **Measurement procedures** | Farmer records. Where farmer records are not available, interviews can be conducted |
| **Monitoring frequency** | Project start |
| **QA/QC procedures** | - |
| **Additional comments** | - |

## 6.2 Data and Parameters monitored

| | |
|---|---|
| **Data/parameter:** | $LCA_i$ |
| **Unit** | tCO2e/kg or tCO2e/L |
| **Description** | Life Cycle Analysis emissions for the biostimulant of type i |
| **Source of data** | Life Cycle Analysis performed by producer or product, non-obligated entity or third party |
| **Value(s) applied** | As per the conducted LCA |
| **Measurement procedures** | As per the latest version of the recognised international standards, such as PAS 2050 issued by British Standards Institute (BSI)[28], GHG Protocol |

[28] Specification for the assessment of the life cycle greenhouse gas emissions of goods and services - PAS 2050. British Standards Institute (BSI), 2011.

| | |
|---|---|
| | Product Life Cycle Standard[29], ISO standards (ISO 14040[30] ISO 14044 [31] and ISO 14067 [32]) |
| **Monitoring frequency** | Project start, repeated at least every 3 years or when there is material change in production, transportation or application procedure of the biostimulant. |
| **QA/QC procedures** | LCA shall not be more than 1 year old at the start of the project. |
| **Additional comments** | Non-obligated entity should provide the latest version of the LCA applicable at the time of start of project and any subsequent performance review and renewal of the crediting period. |

| | |
|---|---|
| **Data/parameter:** | $Q_{PR,i,t,y}$ |
| **Unit** | t/ha |
| **Description** | Quantity of biostimulant for soil revitalisation of type i used, before dilution, in each stratum y in the calculation period |
| **Source of data** | Proof of product use (such as farm accounting, farmer records, product invoice or farmers' declarations) |
| **Value(s) applied** | As per source of data |
| **Measurement procedures** | - |
| **Monitoring frequency** | Annually |
| **QA/QC procedures** | Cross check that biostimulant use is not more than the prescribed limit. Verifying data entry (such as cross-checking farmer records with invoices, where applicable), and analysis techniques, ensuring data maintenance and archiving project time |
| **Additional comments** | - |

[29] Product Life Cycle Accounting and Reporting Standard. Greenhouse Gas Protocol, 2011. Available at: https://ghgprotocol.org/product-standard

[30] ISO 14040:2006 - Environmental management — Life cycle assessment — Principles and framework. ISO, 2016. Available at: https://www.iso.org/standard/37456.html

[31] ISO 14044:2006 - Environmental management — Life cycle assessment — Requirements and guidelines. ISO, 2016. Available at: https://www.iso.org/standard/38498.html

[32] ISO 14067:2018 - Greenhouse gases — Carbon footprint of products — Requirements and guidelines for quantification. Available at: https://www.iso.org/standard/71206.html

| | |
|---|---|
| **Data/parameter:** | Cropland Management Practices $_{PR,y}$ |
| **Unit** | [dimensionless] |
| **Description** | Project scenario Cropland management practices (such as tillage, manure inputs, cover crops) after project start in stratum y |
| **Source of data** | Farmer own records |
| **Value(s) applied** | - |
| **Measurement procedures** | Farmer records |
| **Monitoring frequency** | Annually |
| **QA/QC procedures** | - |
| **Additional comments** | - |

# Annex 1: Guidance on Demonstrating Positive Impact of Biostimulant Application on Soil Characteristics

41 Improving soil characteristics is one of the possible positive impacts generated through application of biostimulants in combination with other regenerative agricultural practices such as cover cropping, reduced tillage/no-tillage, etc. Where a project activity is to demonstrate improved soil characteristics in the project scenario as compared to baseline scenario, the project activity may use the following as guidance.

42 The non-obligated entity can select a relevant soil characteristics indicator. Examples of indicators from Section 6 of the Status of the World's Soil Resources[33] are:

| Topic | Example of indicator |
|---|---|
| Soil erosion | Soil erosion rates (tonnes of soil/ha/year) |
| Soil contamination | Quantity of soil contaminants such as metallic trace-elements and radionuclides, and organic compounds like xenobiotic molecules (tonnes/ha) |
| Soil acidification | Surface (0-10 cm) and depth (10-30 cm or lower) pH (dimensionless) |
| Soil salinisation and sodification | Electrical conductivity of the soil (S/m) or exchangeable sodium percentage (dimensionless) |
| Soil biodiversity | Presence of macrofauna (number and traces of activity through earthworm sampling) (Number of earthworms per m3 of soil) |

[33] Status of the World's Soil Resources. FAO and ITPS, 2015. Available at: http://www.fao.org/3/i5199e/i5199e.pdf

| | |
|---|---|
| Soil nutrient | Nutrient concentration in soils (such as nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulphur (S)) (tonnes/ha) |
| Soil compaction | Porosity (mass of carbon/m $^3$ of soil) |
| Soil-water quantity and quality | Volumetric Water Content (m $^3$/m $^3$) |

Monitoring: The non-obligated entity can monitor the soil characteristics indicator at regular intervals such as annually. The following tables present requirements relating to data and parameters for soil characteristics indicators.

**a. Data and Parameters fixed:**

| | |
|---|---|
| **Data/parameter:** | Cropland Management Practices $_{PR,y}$ |
| **Unit** | [dimensionless] |
| **Description** | Baseline management practices - Cropland management practices (such as tillage, manure inputs, cover crops) before project start in stratum y |
| **Source of data** | Farmer own records |
| **Value(s) applied** | - |
| **Measurement procedures** | Farmer records. |
| **Monitoring frequency** | Annually |
| **QA/QC procedures** | - |

**b. Data and Parameters not monitored:**

| | |
|---|---|
| **Data/parameter:** | Soil characteristics $_{PR,j,y}$ |
| **Unit** | [dimensionless] |
| **Description** | Project scenario, Soil characteristics indicator of type j during project scenario in stratum y |
| **Source of data** | Soil analysis |

| | |
|---|---|
| **Value(s) applied** | As per the soil analysis |
| **Measurement procedures** | As per the international or national applicable guidelines for the selected soil characteristics indicator j |
| **Monitoring frequency** | At every performance certification cycle, preferably annually |
| **QA/QC procedures** | As per the international or national applicable guidelines for the selected soil characteristics indicator j |

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**METHODOLOGICAL TOOL**

BM-T-0XXX

# Soil Organic Carbon Activity Module- Application of Organic Soil Improvers from Pulp and Paper Mill Sludges

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INDIAN Carb•n MARKET

Publication Date:

Version 1.0

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# 1. Introduction

1. This tool presents requirements and guidance to monitor and calculate the climate impact of the application of organic soil improvers from pulp and paper mill sludges. The tool applies to activities that involve application of organic soil improvers from pulp and paper mill sludges to improve soil organic matter content in agricultural systems thus increasing soil organic carbon content (creating a soil carbon sink)

# 2. Definitions

2. For the purpose of this tool the following definitions apply:

   (a) **Pulp and paper mill sludges:** Organic waste residue streams from pulp and paper mills.

   (b) **Organic soil improvers:** Organic soil improver products that are processed, as required, from fiber residues from pulp and paper industries. These products are used to improve soil in agricultural systems, increasing soil organic carbon content (creating soil carbon sink).

# 3. Scope & Applicability

## 3.1. Scope

3. This tool is applicable to project activities that use soil improvers produced by processing and/or recycling of pulp and paper mill sludges from industrial operations in pulp and paper manufacturing to increase soil organic matter and soil organic carbon and improve soil physical properties without negatively affecting the soil health.

4. This tool is not applicable to reduced or avoided emissions from organic waste residue streams handling and management. To quantify and monitor the reduced or avoided GHG emissions from the avoided incineration or landfilling of the industrial residues, if applicable, the non-obligated entity shall apply an approved ICM methodology and

applicable requirements specified in the detailed procedure for offset mechanism for such emissions.

## 3.2. Applicability

5. A project applying this tool shall comply with the applicability conditions specified below and within the SOC Framework Methodology. In addition, the project shall comply with applicable requirements specified in the detailed procedure for offset mechanism.

6. The tool is applicable under the following conditions:

   (a) Geographic location: Projects implemented in India are eligible under the tool.

   (b) Soil type: Project Area(s) shall not be comprised of organic soils (Histosols) as defined by the World Reference Base for Soil Resources (FAO 2015)[34]. Only mineral soil types are eligible[35]

   (c) The certified Project Area shall be limited to eligible areas, excluding riparian or other buffer zones located within such eligible areas, and excluding areas set aside for

[34] World reference base for soil resources 2014 (Update 2015): http://www.fao.org/3/i3794en/I3794en.pdf

[35] https://esdac.jrc.ec.europa.eu/content/european-soil-database-v20-vector-and-attribute-data

conservation in accordance with detailed procedure for offset mechanism requirements.

7. Under this tool, organic soil improvers processed from pulp and paper mill sludges are applied to improve soil in agricultural systems, increasing soil organic carbon content (creating a soil carbon sink). The organic soil improvers must

   (a) be processed from pulp and paper mill sludges

   (b) be eligible for agricultural use according to local legislation

   (c) meet quality criteria set in local legislation for fertilizer application (not limited to e.g. heavy metal content and microbiological quality as applicable)

   (d) be non-fossil fraction of an industrial waste[36]

### 3.3. Methodological Tool Approval Date

8. The date of adoption of this document shall be effective from XXXXX.

## 4. Project Boundaries

### 4.1 Spatial boundary:

9. For spatial boundaries, rules and requirements defined in the SOC Framework Methodology apply.

10. Under this activity module, organic soil improvers from pulp and paper mill sludges are applied to improve soil in agricultural systems, increasing soil organic carbon content.

11. Areas joining the project shall be areas within a farm (e.g. fields, parcels). Project area shall be stratified into modelling units (MU) by mineral soil type, climate zone, tillage practices, and level of vegetation cover.

**12.** Risk of project areas leaving a project within crediting period are low, however any areas leaving the project shall be considered as full reversal following SOC framework methodology. At project level long-term storage of organic inputs will be ensured by a) using reduced or minimum tillage, where residue, mulch, or sod is left on the soil surface to protect soil and conserve moisture, and/or b) avoiding bare fallow periods by increasing vegetation cover.

### 4.2 Temporal boundary:

13. The maximum crediting period is 20 years. All project applying this module shall follow 5-year certification cycle.

### 4.3 Carbon Pools:

**14.** This tool focuses on the soil organic carbon pool. The project shall account for carbon pools for assessment in line with the SOC Framework Methodology.

14. the SOC Framework Methodology.

15. This tool focuses on the soil organic carbon pool. The project shall account for carbon pools for assessment in line with the SOC Framework Methodology.

| Scenario | Carbon Pools | $CO_2$ | $CH_4$ | $N_2O$ |
|---|---|---|---|---|
| Baseline Scenario | | | | |
| The amount of soil carbon storage (SOC content) | | Yes | No | No |
| Project scenario | | | | |
| The amount of soil carbon storage (SOC content) | | Yes | No | No |
| GHG emissions from the processing, transport and application of organic soil improvers from pulp and paper mill sludges | | Yes | Yes[37] | Yes |

Justification for inclusion of other carbon pools than SOChan SOC

- GHG emissions from the processing, transport and application of organic soil improvers from pulp and paper mill sludges shall be included in the project scenario. The processing, transport and application of organic biomass residue generate GHG emissions from fuel/energy including electricity use.
- Optional: GHG emissions from waste handling. Emissions of $CO_2$ and $CH_4$ in the baseline scenario from the use of pulp and paper mill sludges for incineration/energy purposes, or landfilling in case it is not incinerated. The incineration and landfilling generate GHG emissions that are avoided in the project scenario; hence it is conservative. However, if the project activity intends to claim avoided/reduced GHGs emission from waste disposal and/or treatment, the project shall take all relevant emission as per applicable methodology.
- Optional: The storage of C in landfills or other areas (in case all the sludge is not incinerated). Some C could be stored in landfill. Justification for the inclusion or

[37] In the processing of organic residues from pulp and paper mill sludges, $CH_4$ and $N_2 0$ emissions are negligible, and can be considered zero in the calculations.

exclusion of this optional pool is determined at the project level depending on the situational context and applicable methodology.

## 5. Emission Reduction Quantification Approach

16. Calculations for overall benefits follow the equations set out in Section: Emissions Reduction Quantification Approaches of the SOC Framework Methodology. Sections below specify approaches and calculations specific to this SOC activity module.

17. This SOC Activity Module focuses on the soil carbon pool. Calculations for overall benefits follow the equations set out in Section 7 of the SOC. Calculations for overall benefits follow the equations set out in Section 7 of the SOC Framework Methodology. Sections below specify approaches and calculations specific to this SOC activity module.[38]

18. To accommodate that soil measurements are not always available to projects, especially for small community-based activities, this SOC Activity Module allows selection of any of the three approaches to baseline and project activity quantification as described in the SOC Framework Methodology. This activity module recommends approach 1, whereas approach 2 may be applied following SOC framework methodology requirements. If a different approach is used for baseline and project scenarios in a stratum, conservativeness and comparability have to be ensured.

    (a) Approach 1: On-site, direct measurement of Baseline Scenario and Project Scenario SOC stocks and emissions reductions from the application of fertilizers and other soil improvers.

    (b) Approach 2: Peer-reviewed publications to quantify Baseline Scenario SOC stocks, other emissions reductions, and Project Scenario impact. Non-obligated entities shall

demonstrate that the research results are conservative and applicable to the proposed Project Area and management practice.

19. See the specific quantification approaches for baseline and project activity emissions in sections 7 and 8.

# 6. Approaches for baseline and project scenario quantification

20. As per the SOC Framework Methodology, the relevant baseline scenario is that as a continuation of historical cropping practices, in the absence of the project activity, no organic soil improvers from pulp and paper mill sludges are used in the baseline scenario.

21. Organic soil improvers from pulp and paper mill sludges are mainly incinerated on site or in some cases also landfilled. To determine optional waste-related reduction in emissions, the non-obligated entity shall use one of the applicable ICM methodologies taking into account the possible restrictions for transboundary movement of waste. In areas where the pulp and paper mill sludges are landfilled as the baseline, the non-obligated entity should add whether the energy is captured or not, and what the effects of it are. However, the energy value is expected to be small or negligible also in landfills.

22. To determine the baseline of the eligible project area the land shall be stratified into modelling units (MU) according to:

    - Mineral soil type
    - Climate zone
    - Tillage practices
    - Level of vegetation cover

23. For each stratum (MU), SOC measurements shall be performed (Approach 1) and/or model parameters identified and verified (Approach 2) in the beginning of the project.

## 6.1 Calculation of Baseline Scenario

24. Quantification for soil organic carbon in the baseline ($SOC_{BL,y}$) shall follow the rules, approaches, calculations, and parameters set out in Section: Baseline Scenario of the SOC Framework Methodology.

## 6.2 Calculation of Project Scenario

25. Under the project scenario, organic soil improvers from pulp and paper mill sludges are applied in the project area to increase SOC and retain soil moisture. Other inputs are used and monitored similarly to baseline, and possible exceptions are reported. Expected emissions reductions would be achieved by both creating a soil carbon sink.

26. As with the baseline, the eligible project area shall be stratified into modelling units (MU) according to

    - Mineral soil type

- Climate zone
- Tillage practices
- Level of vegetation cover

27. For each stratum (MU), SOC measurements have to be performed (Approach 1) and/or model parameters identified and verified (Approach 2) during the project at least every 5 years (following certification cycle).

28. In cases where a cropping change between the Baseline Scenario and Project Scenario occurs, the impact on SOC stocks in the medium term (following 5-year certification cycle) must be assessed. This shall be done considering the following:

    - Agrochemical inputs, e.g. fertilizer or other nutrient inputs, pesticides, other additives;
    - Change in hydrology, e.g. irrigation, draining, and seasonal shift in crop coverage;
    - Change in crop-related inputs, including plant residue and N-fixation;
    - Change in crop management (e.g. tillage, machine use, treatments, and harvest); and
    - Seasonal management activities (e.g. harvesting, fallow periods, seasonal no cover).

29. Quantification for soil organic carbon in the project scenario ($SOC_{t,y}$) shall follow the rules, approaches, calculations, and parameters set out in Section 7: Project scenario of the SOC Framework Methodology.

30. Project activity emissions consist of $CO_2$ emissions from the transport emissions of pulp and paper mill sludges from the industrial facility to the processing facility, the processing emissions of organic soil improvers and the possible application emissions of organic soil improvers from pulp and paper mill sludges to the farms.

31. Project emissions are determined as follows:

$$PE_y = \sum_i Q_{amend}\, x\, (EF_{CO2e,trans1} + EF_{CO2e,process} + EF_{CO2e,trans2} + EF_{CO2e,applic}) + \sum_i Q_{ingredient}\, x\, (EF_{CO2e,ingredient} + EF_{CO2e,trans3})$$

Where[39]

| Parameter | Description |
|---|---|
| $PE_y$ | total project emissions in year y (t CO2e) |
| $Q_{amend}$ | Amount of organic soil improver product applied in the modelling unit i (t of organic soil improver product / ha of land) |
| $EF_{CO2e,trans1}$ | Emission factor for transport of the pulp and paper mill sludges to the processing site (tCO2e/t pulp and paper mill sludges) |
| $EF_{CO2e,process}$ | Emission factor for processing organic soil improvers from pulp and paper mill sludges at the processing site (tCO2e/t organic soil improver product) |
| $EF_{CO2e,trans2}$ | Emission factor for transport of the organic soil improver product to the farm i (tCO2e/t organic soil improver product) |
| $EF_{CO2e,applic}$ | Emission factor for the application of the organic soil improver product on the farm i (tCO2e/t organic soil improver product). The emission factor is calculated based on the fuel or electricity use (fuel consumption × area/distance travelled) of the application machinery of the soil improver product. If there are any other significant emissions from application, e.g. due to environmental conditions of the project, these should be taken into account in the emission factor. |
| $Q_{ingredient}$ | Amount of any other ingredients (e.g. lime) used for processing of organic soil improver products (ton of ingredient; sum each ingredient separately) |
| $EF_{CO2e,ingredient}$ | Emission factor (tCO2e/t) for production of any other ingredients used for processing of organic soil improver products (each ingredient separately). The emission factor should also include $CH_4$ and $N_2O$ emissions, if applicable. (tCO2e/t) |
| $EF_{CO2e,trans3}$ | Emission factor for transportation of any other ingredients used for processing of organic soil improver products (each ingredient separately) (tCO2e/t ingredient) |

### 6.3 Uncertainty

32. Calculation of uncertainty shall follow the rules and equations set out in the SOC Framework Methodology.

33. Allocation of emissions reductions to a Buffer shall follow the rules set out in the SOC Framework Methodology.

[39] Sources for estimation of emission factors are defined in Monitoring section

### 6.4 Calculation of Other Emissions

**34.** Significant additional greenhouse gas emissions (>5% total) due to the project activity need to be accounted for. For this SOC Activity Module, this explicitly includes emissions from fossil fuel combustion, and other agrochemical emissions. Input of N fertilizer is not included since additional N fertilization is not necessary and in some cases N fertilization can even be reduced[40]. Calculation thereof shall follow the rules and equations set out in the SOC Framework Methodology. These emissions need to be subtracted from the soil organic carbon in the project scenario ($SOC_{t,y}$).

## 7. Estimation of leakage emission

35. Calculation of leakage shall follow the rules and equations set out in the SOC Framework Methodology. If a reduction in yield is detected in a performance certification, it is assumed that the lost production capacity will have to be made up for on land outside the project area. Emissions caused by such a shift shall be accounted for as leakage according to the equation listed in the SOC Framework Methodology.

36. As the Project Area is being actively maintained for commodity production during the project crediting period, yield-related leakage risks are expected to be relatively small. Crop producers are commonly risk averse and are unlikely to intentionally suffer reduced crop yields. Moreover, according to SOC Framework Methodology, projects shall not lead to a decrease in agricultural productivity. All projects shall be designed to maintain or increase crop yield. In principle, $LK_{t-0}$ is thus considered to be 0.[41]

37. Risk of reducing yield is low when enough soluble N is available for the plants. Baseline N fertilization levels are adequate and can even be reduced in some cases[42]. When applied in the autumn, organic soil improvers from pulp and paper sludges can immobilize excess soluble N and prevent nutrient leaching. Immobilized N may be mineralized to plant available N next growing seasons and then increase yields[43] . After application of organic soil an improver from pulp and paper sludges into the field, the next crop is typically sowed in the spring

38. Risks to soil health are linked to quality of organic soil improvers from pulp and paper sludges, main risks being heavy metals and microbiological contaminations. However, these risks are taken into account in local fertilizer legislations and are mitigated using approved soil improver products only. Monitoring of soil health using heavy metals markers as indicators (parameter ID 15) is required at least once at each certification renewal (i.e. every fifth year of the project).

---

[40] https://journal.fi/afs/article/view/95600

[41] Section 11. Leakage in SOC Framework Methodology

[42] Kinnula et al. (2020 Effects of mixed pulp mill sludges on crop yields and quality. Agricultural and Food Science, 29(4), 276–286. https://doi.org/10.23986/afsci.95600, https://journal.fi/afs/article/view/95600

[43] Camberato et al. (2006) Pulp and paper mill by-products as soil amendments and plant nutrient sources. Canadian Journal of Soil Science . DOI: 10.4141/S05-120

# 8. Monitoring

39. Monitoring approach and parameters shall be followed as set out in the SOC Framework Methodology. Also, the non-obligated entity shall collect and document evidence that this SOC activity module's applicability conditions are met at all times.

40. In addition to the data and parameters listed in the SOC Framework Methodology, the following parameters need to be monitored and recorded.

## 6.1 Data and Parameters Not Monitored

| | |
|---|---|
| **Data/parameter** | $EF_{FFC,MT}$ |
| **Data unit:** | t CO2e / liter of fuel |
| **Description:** | The emission factor for diesel / gasoline / light fuel oil used in transport vehicles and agricultural machinery (per machine type) |
| **Source of data:** | 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 2 Chapter 3 Table 3.3.1 |
| **Monitoring Frequency:** | - |
| **QA/QC procedures** | - |
| **Any comment:** | - |

## 6.2 Data and Parameters Monitored

| | |
|---|---|
| **Data/parameter** | $Q_{SOC,prod}$ |
| **Data unit:** | % |
| **Description:** | The content of organic carbon (C) in the processed organic biomass residue product |
| **Source of data:** | Direct C-content measurements at the organic soil improvers production company |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures** | Direct analyses according to SFS-EN 13039 standard performed by an independent laboratory |
| **Any comment:** | The level of uncertainty as per test results determined by independent laboratory shall be taken into account. |

| | |
|---|---|
| **Data/parameter** | $Q_{amend}$ |
| **Data unit:** | t of organic biomass residue product / ha of land |
| **Description:** | The amount of organic biomass residue product applied to each modelling unit in the project activity |
| **Source of data:** | Amount applied by organic soil improvers producer to the farmers in the modelling units (book-keeping of organic soil improvers producer company) |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures** | Verifying data entry and analysis techniques, ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter** | $D_{trans,industry}$ |
| **Data unit:** | km |
| **Description:** | The distance from each industrial facility to the organic soil improvers processing facility, multiplied by the amount of transportation trips. Amount of transportation trips is calculated by dividing the total amount of pulp and paper mill sludges delivered by the average load of residues per truck. This parameter is used in the calculation of the emission factor $EF_{CO2e,trans1}$ |
| **Source of data:** | Organic soil improvers processing company. The PD should ensure, that data sharing agreement among all parties involved is in place where applicable. |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures** | Verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter** | D *trans,farm* |
| **Data unit:** | km |
| **Description:** | The distance from the organic soil improvers processing facility to each farm in the project activity scenario, multiplied by the amount of transportation trips. Amount of transportation trips is calculated by dividing the total amount of organic soil improvers delivered by the average load of residues per truck. This parameter is used in the calculation of the emission factor $EF_{CO2e,trans2}$ |
| **Source of data:** | Organic soil improvers processing company |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures** | Verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter** | FFC *trans, wp,j,i,t* |
| **Data unit:** | liters |
| **Description:** | Consumption of fossil fuel type j in the project for transport i in year t. This parameter is used in the calculation of all of the transport-related emission factors mentioned in section 7.2.3. |
| **Source of data:** | Fuel efficiency factors can be obtained from the 2019 Refinement to the IPCC 2006 Volume 2 Chapter 3. Alternatively national or literature studies can be used[44] |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures** | Verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

[44] e.g. in Finland http://lipasto.vtt.fi/yksikkopaastot/indexe.htm

| | |
|---|---|
| **Data/parameter** | FFC $_{applic,\ wp,j,i,t}$ |
| **Data unit:** | liters |
| **Description:** | Consumption of fossil fuel type j in the project for application at farm i in year t. This parameter is used in the calculation of the emission factor $EF_{CO2e,applic}$ |
| **Source of data:** | Fuel efficiency factors can be obtained from the 2019 Refinement to the IPCC 2006 Volume 2 Chapter 3 |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures** | Verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter:** | FFC $_{process}$ |
| **Data unit:** | liter of fuel |
| **Description:** | The amount of fuel used for processing of the organic soil improvers from pulp and paper mill sludges. Record also the type of fuel used. This parameter is used in the calculation of the emission factor $EF_{CO2e,process}$ |
| **Source of data:** | Organic soil improvers processing company |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures:** | Verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter:** | EF $_{CO2e,ingredient}$ |
| **Data unit:** | t CO2e /t of ingredient |
| **Description:** | Emission factor for production of other ingredients used in the processing of organic biomass residue products. Record emission factors for any other ingredients needed in processing of the organic biomass residue products. In case of recycled residues, the value is considered to be zero. |
| **Source of data:** | Organic soil improvers processing company |

| | |
|---|---|
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures:** | Verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter:** | $Q_{ingredient}$ |
| **Data unit:** | t of ingredient |
| **Description:** | Amount of other ingredients used in the processing of the organic soil improver products. Specify each ingredient separately |
| **Source of data:** | Organic soil improvers processing company |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures:** | Verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter:** | $FFC_{ingredient}$ |
| **Data unit:** | liter of fuel |
| **Description:** | The amount of fossil fuel in transportation of any other ingredient (used for processing organic soil improver products). Record consumption of fossil fuel needed for transportation of any other ingredients needed in processing of the organic soil improver products. Record also the type of fuel used. This parameter is used in the calculation of $EF_{CO2e,trans3.}$ |
| **Source of data:** | Transportation company |
| **Monitoring Frequency:** | Annually |
| **QA/QC procedures:** | Verifying methods used to collect data, verifying data entry, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

| | |
|---|---|
| **Data/parameter:** | $SH_{HM,MC}$ |
| **Data unit:** | mg / kg solids |
| **Description:** | Soil health is monitored using changes in content of heavy metal marker compounds as the indicator. Non-obligated entities shall assess possible heavy metal related risks posed by organic soil improvers products to be used in the project. Relevant marker compound or compounds (e.g. Pb, Cd, Hg etc.) are identified and selected based on the assessment and monitored by testing content of marker compounds in soil samples per modelling unit. Content of heavy metal marker compounds should not rise significantly in the soil samples during the project. |
| **Source of data:** | Laboratory test results |
| **Monitoring Frequency:** | At least once at each certification renewal (i.e. every fifth year of the project) |
| **QA/QC procedures:** | Verifying selection of heavy metal marker compounds, verifying methods used to test heavy metal marker compounds in soil samples, verifying data entry into project book-keeping, and ensuring data maintenance and archiving for project time (20 years) |
| **Any comment:** | |

विद्युत मंत्रालय
MINISTRY OF
POWER

**METHODOLOGICAL TOOL**

BM-T-0XXX

# Soil Organic Carbon Activity Tool for Zero Tillage

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INDIAN
Carbon
MARKET

Publication Date: XXXX

Version 1.0

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# 1. Introduction

1. This tool provides an approach for quantification of soil organic carbon (SOC) increase which are listed in the SOC Framework Methodology. The tool focuses on zero tillage/no-till practice, an agricultural technique for growing crops or pasture without mechanically disturbing the soil through tillage.

# 2. Definitions

2. This tool uses the following specific definition:

   i. **Conservation Tillage -** Conservation tillage includes any form of minimum or reduced tillage in which residue, mulch, or sod is left on the soil surface to protect soil and conserve moisture. After planting, at least 30% of the soil surface remains covered by residue to reduce soil erosion by water.

   ii. **Conventional Tillage** - Seedbed preparation using cultivation instruments such as harrows, mouldboard ploughs, offset harrows, subsoilers, and rippers. Conventional tillage methods, involving extensive seedbed preparation, cause the greatest soil disturbance and leave little plant residue on the surface.

   iii. **Zero Tillage/No-Till -** Zero tillage/no-till is an agricultural technique for growing crops or pasture without mechanically disturbing the soil through tillage (including disturbance from non-turning tillage such as rippers and disc harrows). Crop is sown directly into soil that has not been tilled since the harvest of the previous crop. This technique has more than 30% coverage of the soil surface with residues.

# 3. Scope & applicability

## 3.1. Scope

3. This tool is applicable to project activities that introduces zero tillage/no-till practice, an agricultural technique for growing crops or pasture without mechanically disturbing the soil through tillage.

## 3.2. Applicability

4. Projects applying this tool shall comply with the applicability conditions that are specified below and are within the SOC Framework Methodology. In addition, projects shall comply with applicable requirements specified in the detailed procedure for offset mechanism.

5. Applicability conditions specific to this SOC tool:

   i. Geographic location – The tool is applicable for projects implemented in India

   ii. Project area –

      a) Generally, entire farms or selected fields/areas of a farm are eligible (see Section 4.1: Spatial Boundary).

b) The methodology shall not be applicable to farms which have been applying zero tillage techniques in more than one full rotation cycle in the baseline period prior to project start, even if such application was on fields not included in the project. For single crop systems, baseline practices shall not include more than two subsequent years of zero tillage practices. In the baseline period, at least one tillage event (soil disturbance) shall be no less than two years prior to project start.

iii. Zero tillage refers to practices on arable land for which no tillage is applied between harvest and sowing. Crop is sown directly into soil that has not been tilled since the harvest of the previous crop. At least 30% of the soil surface remains covered by residue to reduce soil erosion by water.

iv. Soil type –

a) Proposed projects on sites with organic soils (Histosols), as defined by the World Reference Base for Soil Resources (FAO 2015), are ineligible. While no-till practices can reduce C losses from organic soils, they shall not commonly lead to an increase in soil carbon as targeted by this tool. Thus, only mineral soil types are eligible.

v. In line with the SOC Framework Methodology, any reduction in crop yield which can be attributed to the project activity shall be avoided. To date, the majority of studies report little or no difference in yield between the zero and conventional tillage managements (Shakoor et al. 2021, Huang et al. 2018). However, it has been reported that yields in the first one to two years following zero tillage implementation can decline (Pittelkow et al. 2015b). This SOC tool thus requires monitoring whether the project activity impacts crop yields, calculated as the five-year average. Reductions of more than 5% in five-year average yield during the crediting period shall be accounted for as leakage (see Section 11). An exception to this may be allowed for reductions in the first three years after project start to allow refinement of practices, subject to review at verification.

vi. In line with the SOC Framework Methodology, this tool only accounts for benefits in the SOC pool. No other pools or emission reductions (e.g. in $N_2O$ due to reduced fertilizer need) can be accounted for as benefits.

vii. No reversal to conventional or reduced tillage practices is allowed on the project area fields. This explicitly prohibits tillage rotation approaches applying periodic tillage runs between or after several years of zero tillage during the crediting period.

viii. Organic carbon inputs to the field (other than on-field crop residue) shall not be changed as part of the project activity by more than 5% from baseline period average.

Exception: If this tool is stacked with an additional ICM methodology or tool which uses the same quantification approach in the project (measurement or compatible SOC model allowing distinction of impacts between zero tillage and organic inputs) and considers respective leakage effects, this applicability condition may be omitted.

ix. If project activity includes change of other agricultural practices impacting SOC (e.g. cover crops, organic inputs), the same quantification approach

(measurements and/or models) shall be used for all activities and shall be fully aligned to ensure that overall benefits are quantified correctly and not double-counted.

x. Application of herbicides to prepare fields and/or terminate crops shall be minimized, and negative environmental impacts shall be prevented. Use of herbicides related to project activities, specifically for crop termination and weed control, shall not be increased by more than 5% from baseline period average.

### 3.3. Methodological Tool Approval Date

6. The date of adoption of this document shall be effective from XX/XX 2026.

## 4. Project Boundaries

### 4.1. Spatial boundary

7. For spatial boundaries, the rules and requirements defined in the SOC Framework Methodology apply.

8. Spatial boundary description specific to this SOC tool:

i. Generally, entire farms or selected fields/areas of a farm are eligible. If project activities are only implemented on selected fields, all applicability conditions, and requirements from the SOC Framework Methodology and from this tool shall be applied to these specific fields. These fields shall remain in the project during the entire crediting period (i.e. no change of participating fields within a farm during the crediting period is allowed).

ii. This tool requires GHG sources, sinks, and reservoirs (SSR) as identified in Figure 05 to be considered, with a focus on their impact on SOC and increase of emissions from project activities. The boundary thus includes all impacted field activities (machine use, tillage, and application of fertilizer and pesticides). Since SOC increase from zero tillage is also driven through inputs of biomass (root and residue), residue management for all crops is essential to assess SOC impact. For the same reason, other organic inputs (e.g. manure, compost) have to be monitored. Since no changes are expected to be made to farm infrastructure or handling of main crops, these activities are not included in the project boundary for quantification.

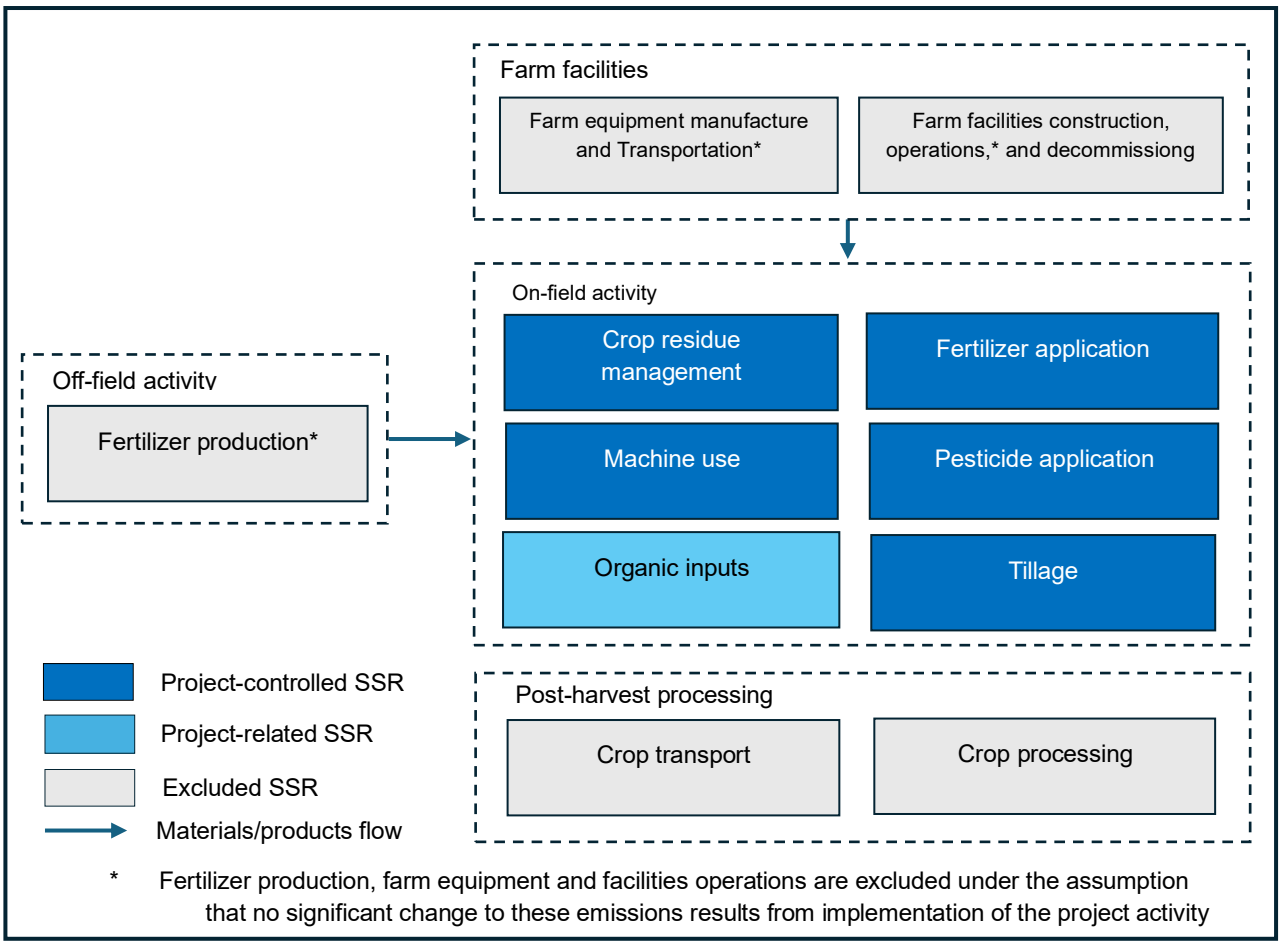



**Figure 05: Spatial project boundary**

## 4.2. Temporal boundary

9. The fixed crediting period allowed under this tool is 20 years The requirements listed in the detailed procedure for offset mechanism shall be followed for crediting period renewal.

## 4.3. Carbon pools

10. Carbon pool description:

    i. In line with the SOC Framework Methodology, this SOC tool only accounts for benefits in the SOC pool. Additional pools (e.g. biomass and litter) may be used for calculation of SOC change, but no GHG sequestration in pools other than soil carbon shall be accounted for as benefits.

    ii. No emission reduction from other GHG sources (e.g. fossil fuel use or fertilizer application) shall be reported for benefits under this methodology. To account for emission reductions from reduced fuel use or fertilizer application, a separate ICM methodology may be applied.

## 4.4. Greenhouse gases

11. Following the SOC Framework Methodology, all GHG sinks and sources affected by its activities shall be monitored.

12. For this SOC tool, this specifically includes any significant increases in $CO_2$, CH4, and $N_2O$ emissions from project activities.

# 5. Emissions Reduction Quantification Approach

13. Calculations for overall benefits follow the equations set out in the Emissions Reduction Quantification Approaches section of the SOC Framework Methodology. Sections below describe the approaches and calculations specific to this SOC tool.

## 5.1. Approaches for Baseline and Project Scenario Quantification

14. This SOC tool allows application of quantification Approaches 1 and 2, as described in the SOC Framework Methodology. It does not allow application of Approach 3 (default equations and parameters). The non-obligated entity shall select the most specific approach possible with the data available, as outlined in the decision tree in Figure 2 of the SOC Framework Methodology.

## 5.2. Approach 1 (On-Site Measurements of SOC)

15. Additional requirements:

    i. SOC measurements shall be taken at a suitable time of year, e.g. at the beginning of the season for main crops, with repeated measurements performed at the same time of year to avoid bias due to short-term variation in soil carbon pools. The project documentation shall specify the timing of soil sampling, i.e. the season and crop rotation phase, as applicable, to be implemented by all farms participating in the project or program.

    ii. To avoid overestimation of SOC stocks, measurements shall always be performed before any significant carbon inputs, such as organic or partially organic fertilizers, including compost and manure, as well as other carbon sources (e.g. in lime or other soil inputs).

    iii. In line with the SOC Framework Methodology, soil samples shall be collected to a depth of 50 cm.

    iv. Since sampling constitutes a disturbance in the zero tillage field, the project description shall include an approach to avoid repeated sampling at the same position (e.g. by adequately marking sampling sites after closure or recording exact GPS coordinates and prescribing a minimum distance for the next sampling round).

    v. As introduction of zero tillage practices can lead to soil compaction, bulk density must be considered in all quantification approaches. Quantification is to be done on an equivalent mass basis to properly quantify carbon impacts, i.e. if increased bulk density is measured in the project scenario, respective layer depth shall be reduced proportionally when calculating SOC change.

    vi. Special care shall be taken to ensure that superficial litter (crop residue) is not included in the SOC quantification.

16. Alternative SOC quantification approaches beyond physical sampling and chemical laboratory analysis, such as proximal sensing (in situ) or lab spectroscopy approaches

may be used in projects if such approaches are approved by the ICM. Application of such SOC quantification is subject to scientific proof that the results obtained under project conditions[45] do not show significant differences to the laboratory results to a soil depth of 50 cm. Revalidation of measurements in the project area shall be undertaken in accordance with the detailed procedure for offset mechanism requirements.

17. If a new technology has not been tested under comparable conditions, a project may establish project-specific datasets to test the accuracy of the technology. This testing shall include repeated SOC stock measurements in at least two points in time, with each sample quantified with the new technology and parallel laboratory analysis. If a system (re)calibration is needed, separate datasets shall be used for calibration of the technology and validation for the project site. Upon confirmation of accuracy for measurements with the new technology, the project shall request a design change and get approval from the ICM administrator using the alternative technology. After confirmation by the ICM administrator, the project may use the new technology for quantification.

18. In any case, uncertainty of the SOC quantification under project conditions shall have to be considered following an Uncertainty Assessment as outlined in the SOC Framework Methodology.

## 5.3. Approach 2 (Application of Models, Research data)

19. Additional requirements:

    i. The non-obligated entities shall provide evidence that calculation approaches, datasets, parameters, and models from peer-reviewed publications to estimate baseline and project SOC stocks in the project are conservative and applicable to the project site conditions and management practices.[46]

    ii. Models and datasets applied must be calibrated locally, i.e. using project specific data. Global models without such local parametrization and validation shall not be applied. Application of global defaults for soil respiration, e.g. Q10 values to quantify temperature effects on respiration, is not allowed.

    iii. Project documentation shall include evidence that the sources used are applicable to main crops and levels of yield (production intensity) as well as management practices (including tillage practice) in the project area for baseline and project scenarios. Evidence shall also be provided that datasets and models have sufficient temporal resolution (i.e. multiple data points per year) to account for impacts of seasonal variations in practices and other factors such as changing weather conditions.

    iv. Parameters for process-based models shall be calibrated and verified following best practice, using at least five years of data for calibration and three years of independent data for verification. Both calibration and verification datasets shall match project conditions as outlined above. Any gaps in datasets and procedures how gaps were addressed for the model calibration and verification shall be reported at validation.

[45] Refer to stratification criteria (Section 7.1.4) for factors determining applicability to project conditions. All factors listed for stratification shall be assessed to confirm applicability.

[46] Refer to stratification criteria (Section 7.1.4) for factors determining applicability to project conditions. All factors listed for stratification shall be assessed to confirm applicability.

v. SOC measurements for a representative sample in each stratum of the project area shall be made at project start and at least once every five years, applying proven methodology (see Section 5.2) and used to statistically assess model results or research data used for quantification. If significant differences from modelled results or research data are found, the models shall be refined.

# 6. Baseline Scenario

## 6.1. Baseline Calculations

20. Quantification for SOC in the baseline ($SOC_{BL,y}$) shall follow the rules, approaches, calculations, and parameters set out in the Baseline Scenario section of the SOC Framework Methodology.

21. Baseline data is required for five years as required by the SOC Framework Methodology. If five-year baseline data is not fully available for a field within a stratum (e.g. as it was leased less than five years prior to project start), evidence shall be provided that average baseline conditions for the stratum apply to the field lacking data, referencing the stratification criteria described below.

22. Baseline period may be extended to cover full rotation periods; respectively avoid including fragments of rotations in the baseline quantification (e.g. in a three-year crop rotation system, a six-year baseline period shall deliver more consistent results, including all relevant practices, inputs and crop yields for two full rotation periods).

23. To determine the baseline of the eligible project area, the land shall be stratified into modelling units (MUs) according to criteria set out in Section 6 of the SOC Framework Methodology . In addition, the following criteria shall be considered in stratification:

    - Mineral soil type
    - Tillage practices (e.g. tillage depth, frequency, tillage equipment)
    - Specific crops (or crop rotations) and production periods

    For each stratum, SOC measurements shall be performed (Approach 1) and/or model parameters identified and verified (Approach 2).

24. If a model is used to quantify baseline SOC (Approach 2), full transparency shall be provided on input data, parametrization, and quantification approach. Application of “black box” models is not allowed.

# 7. Project Scenario

## 7.1. Project Calculations

25. Quantification for SOC in the project scenario ($SOC_{t,y}$) shall follow the rules, approaches, calculations, and parameters set out in the Project Scenario section of the SOC Framework Methodology.

26. This SOC tool addresses change to zero tillage practices. Quantifications shall consider all changes in practices (including changes to crop management, fertilization, and

application of other agrochemicals). For model application (Approach 2), calculation and parameter must be adapted to the specific project situation.

27. If other agricultural activities besides zero tillage are introduced in a project (e.g. cover crops, organic inputs), interactions shall be considered by applying the same SOC quantification approaches (Approach 1 or 2). If models are applied (Approach 2), they shall specifically cover all activities in the project scenario.

28. For the project scenario, the eligible project area shall be stratified into MUs according to criteria set out in Section 7 of the SOC Framework Methodology. In addition, the following criteria shall be considered in stratification.

- Mineral soil type
- Tillage practices (zero tillage)
- Specific crops (or crop rotations) and production periods

For each stratum, SOC measurements shall be performed (Approach 1) and/or model parameters identified and verified (Approach 2).

29. If a model is used to quantify project SOC (Approach 2), full transparency shall be provided on input data, parametrization, and quantification approach. Application of "black box" models is not allowed.

# 8. Procedures for calculation approach change

30. Any approach change between baseline and project scenario calculations shall be in compliance with requirements of section 8: Procedure for approach change of SOC Framework Methodology.

# 9. Uncertainty

31. Calculation of uncertainty shall follow the rules and equations set out in the SOC Framework Methodology.

32. If no information on Standard deviation (SD) or Standard error (SE) is known for a parameter, SE of 50% of the parameter value shall be assumed. For the calculations of the upper and lower confidence intervals, a t-value of 3 shall be applied. Exceptions to this rule are accepted default values considered constant (e.g. physical conversion rates, global warming potentials).

# 10. Other emissions

## 10.1. Project emissions

33. Significant additional GHG emissions (>5% total) due to the project activity need to be accounted for. For this SOC tool, this explicitly includes increased fertilizer input, fuel and electricity use, and other agrochemical emissions (*AE*).

34. As such emissions may differ between strata, calculation for this SOC tool shall follow the rules and Equations 1 through 7 that are described below, which replace the area-wide

calculation set out in Equations 12 through 18 of the SOC Framework Methodology. They follow the same approach but allow application of different emission factors and quantities for different strata.

$$PE_{t-0} = \Delta FE_{t-0} + \Delta FU_{t-0} + \Delta AE_{t-0} \quad (1)$$

Where:

$PE_{t-0}$ = emissions from project activities in the calculation period [tCO2e]

$\Delta FE_{t-0}$ = emissions from increased fertilizer use in the calculation period [$tCO_2e$]

$\Delta FU_{t-0}$ = emissions from increased fuel and electricity use in the calculation period [$tCO_2e$]

$\Delta AE_{t-0}$ = other agrochemical emissions in the calculation period [$tCO_2e$]

## 10.2. Increased Nitrogen Fertilizer Input

35. Emissions from increased nitrogen (N) fertilizer input in project scenario as compared to the baseline scenario are calculated as follows. No differentiation is made between synthetic and organic N fertilizer. This equation shall not be applied for reductions in N fertilizer input, in which case $\Delta FE_{t-0}$ is considered 0. To account for reductions in fertilizer input and the respective GHG emissions reductions, a separate ICM methodology shall be applied.

$$\Delta FE_{t-0} = \sum y\left[EF_{FE,y} \times \sum_{a=1}^{T} \left(FE_{PR,y,a} - FE_{BL,y}\right)\right] \quad (2)$$

Where:

$\Delta FE_{t-0}$ = emissions from increased fertilizer use in the calculation period [tCO2e]. Must be ≥ 0 in this methodology (i.e. no accounting of reductions).

$FE_{PR,y,a}$ = N fertilizer input in stratum y under the project scenario in year a of the calculation period [kgN]

$FE_{BL,y}$ = mean annual N fertilizer input in stratum y under the baseline scenario [kgN]

$T$ = number of years in the calculation period [yr]

$EF_{FE,y}$ = Conversion factor for emissions from N fertilizer in stratum y [tCO2e kgN-1 ]. IPCC 2019 aggregated default value[47] for EFFE is 0.01. Disaggregated default values in IPCC 2019 Table 11.1 may be used if fertilizer inputs are known per fertilizer type.

$FE_{PR}$ and $FE_{BL}$ shall be documented by the non-obligated entity. For $FE_{BL}$, mean annual input is calculated based on respective management records for five years prior to project start. If no adequate documentation can be provided, $FE_{BL}$ shall be no more than 50% of $FE_{PR}$.

[47] IPCC 2019, Vol 4 AFOLU, Table 11.1 (Aggregated default value)

## 10.3. Increased Combustion of Fossil Fuels and Electricity Use

36. Additional CO2 emissions from use of fossil fuel and electricity in project activities (e.g. fuel used by farm machines due to needs for stronger tractors or additional passes to close/treat the surface or fuel/electricity for irrigation pumps) need to be accounted for unless the non-obligated entity can demonstrate that the fossil fuel/electricity used in the project scenario is less than or does not differ significantly from fossil fuel/electricity used in the baseline, in which case $\Delta FU_{t-0}$ is considered 0.

$$\Delta FU_{t-0} = \sum y\left[\sum_{a=1}^{T}\left(FU_{PR,y,a} - FU_{BL,y}\right) + \left(EU_{PR,y,a} - EU_{BL,y}\right)\right] \quad (3)$$

Where:

$\Delta FU_{t-0}$ = emissions from increased fossil fuel and electricity use in the calculation period [tCO2e]

$FU_{PR,y,a}$ = emissions from use of fossil fuels in stratum y under the project scenario in year a of the calculation period [tCO2e]

$FU_{BL,y}$ = mean annual emissions from use of fossil fuels in stratum y under the baseline scenario [tCO2e]

$EU_{PR,y,a}$ = emissions from use of electricity in stratum y under the project scenario in year a of the calculation period [tCO2e]

$EU_{BL,y}$ = mean annual emissions from use of electricity in stratum y under the baseline scenario [tCO2e]

$T$ = number of years in the calculation period [yr]

37. $FU_{PR}$ and $FU_{BL}$ shall be documented by the non-obligated entity and generally calculated with the equation below, based on fuel consumption by machine type and fuel emission factor.

$$FU_{i,y,a} = \sum MT\, FUL_{i,y,MT,a} \times FEF_{i,y,MT} \quad (4)$$

Where:

$FU_{i,y,a}$ = emissions from use of fossil fuels in stratum y in year a [tCO2e ha-1 ]

$FUL_{i,y,MT,a}$ = fuel consumption in stratum y by the machinery type MT used in year a [litres]

$FEF_{i,y,MT}$ = emissions factor for the fuel used in stratum y in machinery MT [tCO2e litres-1 ]

$MT$ = machinery type (gasoline two-stroke, gasoline four-stroke, diesel)

*i* = formula used for baseline (i=BL) as well as project scenario (i=PR)

For *FUBL*, mean annual emissions are calculated based on respective management records for five years prior to project start. If this is not available, the amount of fuel combusted can be estimated using fuel efficiency (for example l/100 kilometres (km), l/tonne-km, l/hour) of the vehicle and the appropriate unit of use for the selected fuel

efficiency (e.g. km driven if efficiency is given in l/100 km). If no adequate documentation can be provided, *FUBL* shall be no more than 50% of *FUPR*.

Non-CO2 GHG emissions caused by the use of fossil fuel from project activities (management operations, machinery, etc.) are insignificant and may thus be neglected.

38. *EUPR* and E*UBL* shall be documented by the non-obligated entity and generally calculated with the equation below, based on electricity consumption by appliance and respective emission factor. If electricity is generated on-site using fossil fuels (e.g. in diesel generators for irrigation pumps), emissions from fuel combustion shall be calculated instead, following the approach described above.

$$EU_{i,y,a} = \sum SE\ EUW_{i,y,SE,a} \times EEF_{i,y,SE} \quad (5)$$

Where:

$EU_{i,y,a}$ = emissions from electricity consumption in stratum y in year a [tCO2e ha-1]

$EUW_{i,y,SE,a}$ = electricity consumption in stratum y from source SE in year a [kWh]

$EEF_{i,y,SE}$ = emissions factor for the electricity used in stratum y in source SE [tCO2e kWh-1]

$SE$ = electricity source type (grid, fossil fuel generator, etc)

*i* = formula used for baseline (i=BL) as well as project scenario (i=PR)

For *EUBL*, mean annual emissions are calculated based on respective management records for five years prior to project start. If no adequate documentation can be provided, *EUBL* shall be no more than 50% of *EUPR*.

## 10.4. Other Agrochemical Emissions

39. Additional AE related to the project activities from increased use of agrochemicals, especially pesticides or non-N fertilizers, need to be accounted for unless the non-obligated entity can demonstrate that agrochemicals used in the project scenario are less than or do not differ significantly from agrochemicals used in the baseline, in which case $\Delta AE_{t-0}$ is considered 0.

40. If use of agrochemicals (herbicides, pesticides) or non-N fertilizer is significantly higher in the project than in the baseline scenario, the non-obligated entity shall calculate respective emissions by using specific amounts and emission factors. Emission factors applied shall be based on manufacturer information or scientific sources.

$$\Delta AE_{t-0} = \sum y\left[\sum_{a=1}^{T} \left(AE_{PR,y,a} - AE_{BL,y}\right)\right] \quad (6)$$

Where:

$\Delta AE_{t-0}$ = additional emissions from project activity in the calculation period [tCO2e]

$AE_{PR,y,a}$ = other emissions in stratum y under the project scenario in year a of the calculation period [tCO2e]

$AE_{BL,y}$ = other emissions (annual mean) in stratum y under the baseline scenario [tCO2e]

$T$ = number of years in the calculation period [yr]

41. $AE_{PR}$ and $AE_{BL}$ shall be documented for each emitter type (agrochemical) by the non-obligated entity and calculated with the equation below, based on emission type, underlying quantity, and respective emission factor.

$$AE_{i,y,a} = \sum ET\ AQ_{i,y,ET,a} \times AEF_{i,y,ET} \quad (7)$$

Where:

$AE_{i,y,a}$ = emissions from use of other agrochemicals in stratum y in year a [tCO2e ha-1 ]

$AQ_{i,y,ET,a}$ = quantity of agrochemicals in stratum y for emitter type ET applied in year a [kg]

$AEF_{i,y,ET}$ = emissions factor of the agrochemical used in stratum y (for emitter type ET) [tCO2e kg-1 ]

$ET$ = emitter type (specific pesticide, fertilizer, or other agrochemical)

$i$ = formula used for baseline (i=BL) as well as project scenario (i=PR)

42. For $AE_{BL}$, mean annual emissions are calculated based on respective management records for five years prior to project start. If no adequate documentation can be provided, $AE_{BL}$ shall be no more than 50% of $AE_{PR}$.

# 11. Leakage

43. Calculation of leakage shall follow the rules and equations set out in the SOC Framework Methodology. If a reduction in yield is detected in a performance certification, it is assumed that the lost production capacity has to be made up for on land outside the project area. Emissions caused by such a shift shall be accounted for as leakage according to the equation listed in the SOC Framework Methodology. Note that accounting of positive leakage is not allowed according to the SOC Framework Methodology.

44. Additional requirements: Under this SOC tool, leakage shall be assessed at stratum level. Instead of Equation 19 in the SOC Framework Methodology, leakage shall thus be calculated for each stratum separately, using Equation 8:

$$LK_{t-0} = \sum y \left[ max \left[ \frac{CY_{min,y} - CY_{t,y}}{CY_{BL,y}}; 0 \right] \times \bar{A}_{y,t} \right] \times \left( \Delta BC_{LA} + \Delta SOC_{LA,t-0} + \Delta FE_{LA,t-0} + \Delta FU_{LA,t-0} \right) \quad (7)$$

Where:

| | | |
|---|---|---|
| $LK_{t-0}$ | = | emissions due to shift of production to non-project lands (leakage area) [tCO2e] |
| $CY_{t,y}$ | = | crop yield in stratum y at time t (five-year average*) [kg ha$^{-1}$] |
| $CY_{min,y}$ | = | lowest crop yield (five-year average**) in stratum y in any calculation period since project start [kg ha$^{-1}$] |
| $CY_{BL,y}$ | = | crop yield in stratum y under the baseline scenario (five-year average*) [kg ha$^{-1}$] |
| $\bar{A}_{y,t}$ | = | eligible project area in stratum y (five-year average*) [ha] |
| $\Delta BC_{LA}$ | = | emissions from change in biomass carbon stocks in leakage area [tCO2e ha$^{-1}$] |
| $\Delta SOC_{LA,t-0}$ | = | emissions from change in SOC stocks in leakage area [tCO2e ha$^{-1}$] |
| $\Delta FE_{LA,t-0}$ | = | change in emissions from use of fertilizer in leakage area [tCO2e ha$^{-1}$] |
| $\Delta FU_{LA,t-0}$ | = | change in emissions from fuel use in leakage area [tCO2e ha$^{-1}$] |

* If the baseline period is extended to cover full rotation cycles, crop yield and stratum area shall be calculated as the average across an equal length period. In the project scenario, the five-year average shall be calculated as a running average from year t-4 to year t (for longer baseline periods, averaging period shall be increased accordingly).

** Baseline years shall be in included in $CY_{min}$ for the project period (i.e. in project year 1, $CY_{min} = CY_{BL}$).

# 12. Monitoring

45. Monitoring approach and parameters shall be followed as set out in the SOC Framework Methodology. Also, the non-obligated entity shall collect and document evidence that this SOC tool's applicability conditions are met at all times and that project activity has been implemented as stated, specifically that no tillage events have occurred. Acceptable evidence are detailed field reports listing key interventions, dates and machinery used, remote sensing analysis of vegetation, and crop residue cover or timestamped photographs of fields before, during, and after the cropping season, including fallow periods as applicable.

46. Additional requirements: The following parameters shall be monitored in this SOC tool, depending on the quantification approach selected. Where the parameters listed below overlap with the SOC Framework Methodology Monitoring section, the below descriptions shall be applied under this SOC tool to ensure stratum-level quantification and appropriate data sources.

### I. Data and parameters fixed:

47. Data and parameters collected for baseline calculation, when project areas (farms/land parcels) are being added, and at renewable of crediting period if required:

**Data / Parameter table 5.**

| **Data / Parameter:** | $CY_{BL,y}$ |
|---|---|
| Unit: | kg ha-1 |
| Description: | Average annual crop yield per hectare (ha) in stratum y in the project area during the baseline period (five-year average*) |
| Source of data: | Farm records, e.g. field records, sales receipts |
| Value(s) applied | - |
| Measurement procedures | Yield is recorded for each crop season and accumulated annually (as sum of each crop type). For the baseline, an average annual yield is calculated across the five-year baseline period.*<br><br>* If the baseline period is extended to cover full rotation cycles, crop yield shall be calculated as the average across an equal length period. In the project scenario, the five-year average shall be calculated as a running average from year t-4 to year t (for longer baseline periods, averaging period shall be increased accordingly). |
| Monitoring frequency | Project start |
| QA/QC procedures | - |
| Additional comments | Used in Equation 8. This parameter replaces parameter $CY_{BL}$ in the SOC Framework Methodology (Equation 19). |

**Data / Parameter table 6.**

| **Data / Parameter:** | $EF_{FE,y}$ |
|---|---|
| Unit: | tCO2e kgN-1 |
| Description: | Conversion factor for emissions from N fertilizer in stratum y |
| Source of data: | IPCC 2019 |
| Value(s) applied | IPCC 2019 aggregated default value for $EF_{FE}$ is 0.01. Disaggregated default values in IPCC 2019 Table 11.1 may be used if fertilizer inputs are known per fertilizer type. |
| Measurement procedures | - |
| Monitoring frequency | Project start |
| QA/QC procedures | - |
| Additional comments | Used in Equation 2 |

**Data / Parameter table 7.**

| **Data / Parameter:** | $FE_{BL,y}$ |
|---|---|
| Unit: | kgN |
| Description: | Mean annual N fertilizer input in stratum y under the baseline scenario |
| Source of data: | Farm records (field level) |
| Value(s) applied | Average annual N input in stratum y, calculated across baseline period (five years) |
| Measurement procedures | |
| Monitoring frequency | Project start |
| QA/QC procedures | Used in Equation 2 |
| Additional comments | |

## II. Data and parameters monitored

**Data / Parameter table 4.**

| **Data / Parameter:** | $\bar{A}y,t$ |
|---|---|
| Unit: | ha |
| Description: | Eligible project area in stratum y at time t (five-year average*) |
| Source of data: | Farm records (GPS data, GIS files) |
| Value(s) applied: | |
| Measurement procedures: | Documentation allowing unambiguous identification and documenting area size at field level<br><br>* If the baseline period is extended to cover full rotation cycles, area shall be calculated as the average across a period of equal length. |
| Monitoring frequency: | Annually (for calculation of five-year average*) |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 8 |

**Data / Parameter table 5.**

| **Data / Parameter:** | $AQ_{i,y,ET,a}$ |
|---|---|
| Unit: | km |

| | |
|---|---|
| Description: | Quantity of agrochemicals in stratum y for emitter type (ET) applied in year a |
| Source of data: | Farm records |
| Value(s) applied: | |
| Measurement procedures: | Documentation of quantity and type of non–N fertilizer agrochemicals used for zero tillage (e.g. for crop termination, weed control) |
| Monitoring frequency: | Annually |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 7 |

**Data / Parameter table 6.**

| **Data / Parameter:** | $AEF_{i,y,ET}$ |
|---|---|
| Unit: | tCO2e kg-1 |
| Description: | Emissions factor of the agrochemical used in stratum y (for emitter type [ET]) |
| Source of data: | Supplier information |
| Value(s) applied: | |
| Measurement procedures: | Supplier's life cycle assessment (LCA) information for production and transport. If no supplier information is available, national, or international default values may be applied. |
| Monitoring frequency: | Annually |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 7 |

**Data / Parameter table 7.**

| **Data / Parameter:** | $CY_{t,y}$ |
|---|---|
| Unit: | kg ha-1 |
| Description: | Average annual crop yield per ha in stratum y in the project area (five-year average*) |
| Source of data: | Farm records, e.g. field records, sales receipts |
| Value(s) applied: | |

| | |
|---|---|
| Measurement procedures: | Yield is recorded for each crop season and cumulated annually (as sum of each crop type). Average annual yield is calculated for each stratum across the five-year reporting period.*<br><br>* If the baseline period is extended to cover full rotation cycles, crop yield shall be calculated as the average across a period of equal length. |
| Monitoring frequency: | Annually (for calculation of five-year average*) |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 8. This parameter replaces parameter $CY_t$ in the SOC Framework Methodology (Equation 19). |

**Data / Parameter table 8.**

| **Data / Parameter:** | $\boldsymbol{CY_{min,y}}$ |
|---|---|
| Unit: | Kilogram (kg) ha-1 |
| Description: | Minimum annual crop yield per ha in stratum y since project start |
| Source of data: | Farm records, e.g. field records, sales receipts |
| Value(s) applied: | |
| Measurement procedures: | Yield is recorded for each crop season and cumulated annually (as sum of each crop type). $CY_{min,y}$ represents lowest average annual yield per stratum ($CY_{t,y}$) since project start and serves as threshold to assess yield reduction.<br><br>Note: As the running average yield ($CY_{t,y}$) also includes baseline years, $CY_{min,y}$ for project year one is equal to $CY_{BL}$. |
| Monitoring frequency: | At verification |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 8. This parameter replaces parameter $CY_{min}$ in the SOC Framework Methodology (Equation 19). |

**Data / Parameter table 9.**

| **Data / Parameter:** | $\Delta BC_{LA}$ |
|---|---|
| Unit: | tCO2e ha-1 |
| Description: | Emissions from change in biomass carbon stocks in leakage area |
| Source of data: | Remote sensing analysis or public data on leakage area |
| Value(s) applied: | |

| | |
|---|---|
| Measurement procedures: | Emissions from change in biomass are calculated as the expected difference between pre-leakage carbon stocks and post-leakage carbon stocks (e.g. deforestation or removal of woody biomass due to intensification) in the leakage area. Loss of carbon shall be accounted for as tons of CO2 equivalent released.<br><br>In line with the requirement in the SOC Framework Methodology that no positive leakage must be accounted, no increase in biomass in the leakage area shall be accounted for. |
| Monitoring frequency: | At verification |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 8 |

**Data / Parameter table 10.**

| **Data / Parameter:** | $\Delta SOC_{LA,t-0}$ |
|---|---|
| Unit: | tCO2e ha-1 |
| Description: | Emissions from change in SOC stocks in leakage area |
| Source of data: | Remote sensing analysis or public data on leakage area |
| Value(s) applied: | |
| Measurement procedures: | Emissions from change in SOC stocks are calculated as the expected difference between pre-leakage SOC stocks and post leakage SOC stocks (e.g. loss in SOC due to tillage or other agricultural practices) in the leakage area. Loss of SOC shall be accounted for as tons of CO2 equivalent released.<br><br>In line with the requirement in the SOC Framework Methodology that no positive leakage must be accounted, no increase in SOC in the leakage area shall be accounted for. |
| Monitoring frequency: | At verification |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 8 |

**Data / Parameter table 11.**

| **Data / Parameter:** | $EEF_{i,y,SE}$ |
|---|---|
| Unit: | $tCO_2e\ kWh^{-1}$ |
| Description: | Emissions factor for the electricity used in stratum y in electricity source type |

| | |
|---|---|
| Source of data: | a) If the electricity supply source is a regional/national electric grid, apply the provisions of the latest version of TOOL05.<br>b) If the electricity supply source is a mini grid, apply the provisions of the latest version of "AMS-I.F.: Renewable electricity generation for captive use and mini-grid".<br>c) If the electricity supply source is a renewable energy captive power plant, a value of 0 tCO2/MWh shall be applied.<br>d) For fossil-based captive sources, use national GHG inventory sources.<br>e) If no national data is available, default emission factors in IPCC 2006 Volume 2, Chapter 3, Table 3.3.1 may be applied (taking into account listed uncertainties). |
| Value(s) applied: | |
| Measurement procedures: | |
| Monitoring frequency: | Project start, annual review for national updates |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 5 |

**Data / Parameter table 12.**

| **Data / Parameter:** | $EUWi,y,_{SE,a}$ |
|---|---|
| Unit: | kWh |
| Description: | Electricity consumption in stratum y from source SE in year a |
| Source of data: | Farm records |
| Value(s) applied: | |
| Measurement procedures: | 1) Document electric equipment and power source used for zero tillage practices in each stratum.<br>2) Quantify electricity used for zero tillage activity based on:<br>a. direct records (electricity documentation), or<br>b. duration of equipment use (hours) and power consumption per equipment type (kWatt). |
| Monitoring frequency: | Annually: use-based documentation, aggregated (sum) per year |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 5 |

**Data / Parameter table 13.**

| **Data / Parameter:** | $FE_{PR,y,a}$ |
|---|---|
| Unit: | kgN |
| Description: | N fertilizer input in stratum y under the project scenario in year a |
| Source of data: | Farm records (field level) |
| Value(s) applied: | Sum of N inputs in stratum y in year a |
| Measurement procedures: | |
| Monitoring frequency: | Use-based, aggregated annually |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 2 |

**Data / Parameter table 14.**

| **Data / Parameter:** | $FEF_{i,y,MT}$ |
|---|---|
| Unit: | $tCO_2e\ liter^{-1}$ |
| Description: | Emissions factor for the fuel used in stratum y in machinery type (MT) |
| Source of data: | a) National GHG inventory sources.<br>b) If no national data is available, default emission factors in IPCC 2006 Volume 2, Chapter 3, Table 3.3.1 may be applied (taking into account listed uncertainties). |
| Value(s) applied: | |
| Measurement procedures: | |
| Monitoring frequency: | Project start, annual review for national updates |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 4 |

**Data / Parameter table 15.**

| **Data / Parameter:** | $FUL_{i,y,MT,a}$ |
|---|---|
| Unit: | Liter |
| Description: | Fuel consumption in stratum y by the machinery type (MT) used in year a |

| | |
|---|---|
| Source of data: | Farm records |
| Value(s) applied: | |
| Measurement procedures: | 1) Document machines and fuel type for equipment used for zero tillage practices in each stratum.<br>2) Document amount of equipment use for zero tillage from:<br>a. direct records (fuel use), or<br>b. equipment usage (hours or distance) and fuel consumption (liters/hour or liters/km). |
| Monitoring frequency: | Baseline: project start<br>Project: use-based documentation, aggregated (sum) per year |
| QA/QC procedures: | |
| Additional comments: | Used in Equation 4 |

# 13. References

48. In addition to the methodologies, methodological tools, guidelines, and key sources listed in the SOC Framework Methodology, this SOC tool refers to the following key publications:


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